Changes in DNA

When it was recognized that changes (mutations) in genes occur spontaneously (T. H. Morgan, 1910) and can be induced by X-rays (H. J. Muller, 1927), the mutation theory of heredity became a cornerstone of early genetics. Genes were defined as mutable units, but the question what genes and mutations are remained. Today we know that mutations are changes in the structure of DNA and their functional consequences. The study of mutations is important for several reasons. Mutations cause diseases, including all forms of cancer. They can be induced by chemicals and by irradiation. Thus, they represent a link between heredity and environment. And without mutations, well-organized forms of life would not have evolved. The following two plates summarize the chemical nature of mutations.

A. Error in replication

The synthesis of a new strand of DNA occurs by semiconservative replication based on complementary base pairing (see DNA replication). Errors in replication occur at a rate of about 1 in 10^5. This rate is reduced to about 1 in 10^7 to 10^9 by proofreading mechanisms. When an error in replication occurs before the next cell division (here referred to as the first division after the mutation), e.g., a cytosine (C) might be incorporated instead of an adenine (A) at the fifth base pair as shown here, the resulting mismatch will be recognized and eliminated by mismatch repair (see DNA repair) in most cases. However, if the error is undetected and allowed to stand, the next (second) division will result in a mutant molecule containing a CG instead of an AT pair at this position. This mutation will be perpetuated in all daughter cells. Depending on its location within or outside of the coding region of a gene, functional consequences due to a change in a codon could result.

B. Mutagenic alteration of a nucleotide

A mutation may result when a structural change of a nucleotide affects its base-pairing capability. The altered nucleotide is usually present in one strand of the parent molecule. If this leads to incorporation of a wrong base, such as a C instead of a T in the fifth base pair as shown here, the next (second) round of replication will result in two mutant molecules.

C. Replication slippage

A different class of mutations does not involve an alteration of individual nucleotides, but results from incorrect alignment between allelic or nonallelic DNA sequences during replication. When the template strand contains short tandem repeats, e.g., CA repeats as in microsatellites (see DNA polymorphism and Part II, Genomics), the newly replicated strand and the template strand may shift their positions relative to each other. With replication or polymerase slippage, leading to incorrect pairing of repeats, some repeats are copied twice or not at all, depending on the direction of the shift. One can distinguish forward slippage (shown here) and backward slippage with respect to the newly replicated strand. If the newly synthesized DNA strand slips forward, a region of nonpairing remains in the parental strand. Forward slippage results in an insertion. Backward slippage of the new strand results in deletion.
Microsatellite instability is a characteristic feature of hereditary nonpolyposis cancer of the colon (HNPCC). HNPCC genes are localized on human chromosomes at 2p15 – 22 and 3p21.3. About 15% of all colorectal, gastric, and endometrial carcinomas show microsatellite instability. Replication slippage must be distinguished from unequal crossing-over during meiosis. This is the result of recombination between adjacent, but not allelic, sequences on nonsister chromatids of homologous chromosomes (Figures redrawn from Brown, 1999).

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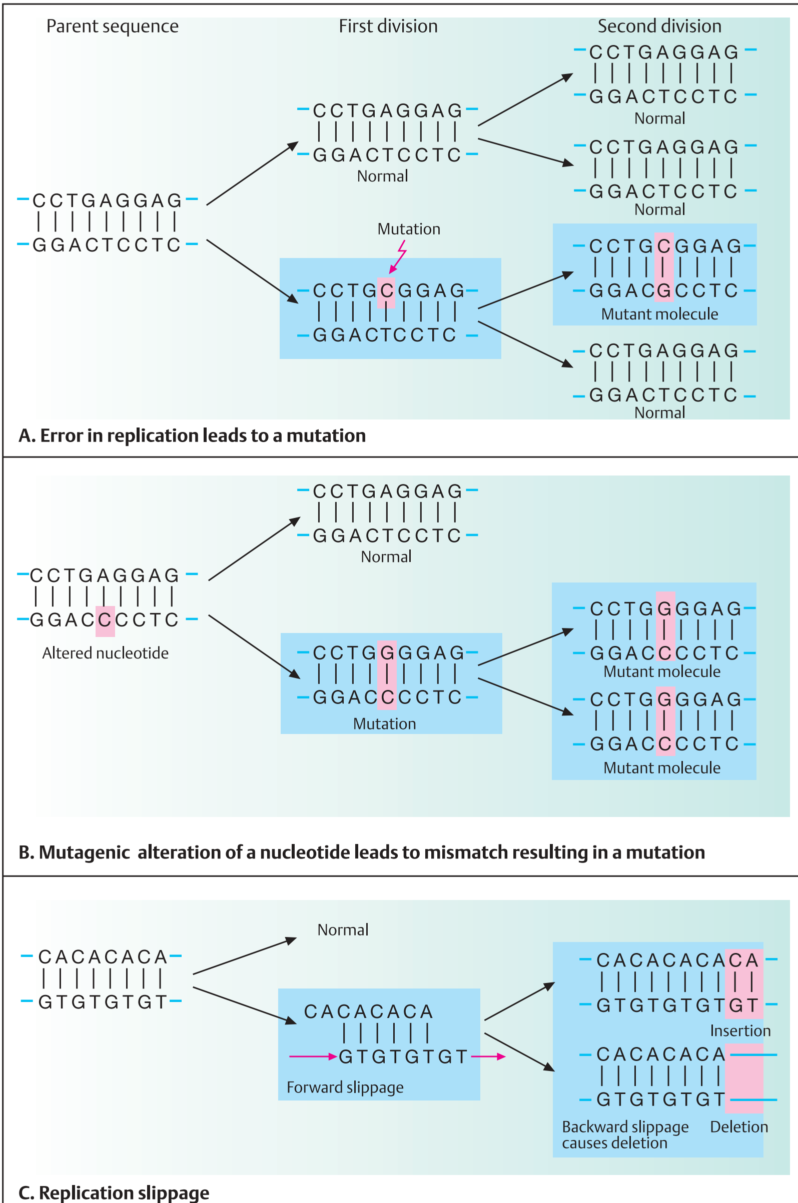

A. Error in replication leads to a mutation

B. Mutagenic alteration of a nucleotide leads to mismatch resulting in a mutation

C. Replication slippage

Mutation Due to Different Base Modifications

Mutations can result from chemical or physical events that lead to base modification. When they affect the base-pairing pattern, they interfere with replication or transcription. Chemical substances able to induce such changes are called mutagens. Mutagens cause mutations in different ways. Spontaneous oxidation, hydrolysis, uncontrolled methylation, alkylation, and ultraviolet irradiation result in alterations that modify nucleotide bases. DNA-reactive chemicals change nucleotide bases into different chemical structures or remove a base.

A. Deamination and methylation

Cytosine, adenine, and guanine contain an amino group. When this is removed (deamination), a modified base with a different base-pairing pattern results. Nitrous acid typically removes the amino group. This also occurs spontaneously at a rate of 100 bases per genome per day (Alberts et al., 1994, p. 245). Deamination of cytosine removes the amino group in position 4 (1). The resulting molecule is uracil (2). This pairs with adenine rather than guanine. Normally this change is efficiently repaired by uracil-DNA glycosylase. Deamination at the RNA level occurs in RNA editing (see Expression of genes). Methylation of the carbon atom in position 5 of cytosine results in 5-methylcytosine, containing a methyl group in position 5 (3). Deamination of 5-methylcytosine will result in a change to thymine, containing an oxygen in position 4 instead of an amino group (4). This mutation will not be corrected because thymine is a natural base. Adenine (5) can be deaminated in position 6 to form hypoxanthine, which contains an oxygen in this position instead of an amino group (6), and which pairs with cytosine instead of thymine. The resulting change after DNA replication is a cytosine instead of a thymine in the mutant strand.

B. Depurination

About 5000 purine bases (adenine and guanine) are lost per day from DNA in each cell (depurination) owing to thermal fluctuations. Depurination of DNA involves hydrolytic cleavage of the *N*-glycosyl linkage of deoxyribose to the guanine nitrogen in position 9. This leaves a depurinated sugar. The loss of a base pair will lead to a deletion after the next replication if not repaired in time (see DNA repair).

C. Alkylation

Alkylation is the introduction of a methyl or an ethyl group into a molecule. The alkylation of guanine involves the replacement of the hydrogen bond to the oxygen atom in position 6 by a methyl group, to form 6-methylguanine. This can no longer pair with cytosine. Instead, it will pair with thymine. Thus, after the next replication the opposite cytosine (C) is replaced by a thymine (T) in the mutant daughter molecule. Important alkylating agents are ethylnitrosourea (ENU), ethylmethane sulfonate (EMS), dimethylnitrosamine, and *N*-methyl-*N*-nitro-*N*-nitrosoguanidine.

D. Nucleotide base analogue

Base analogs are purines or pyrimidines that are similar enough to the regular nucleotide DNA bases to be incorporated into the new strand during replication. 5-Bromodeoxyuridine (5-BrdU) is an analog of thymine. It contains a bromine atom instead of the methyl group in position 5. Thus, it can be incorporated into the new DNA strand during replication. However, the presence of the bromine atom causes ambiguous and often wrong base pairing.

E. UV-light-induced thymine dimers

Ultraviolet irradiation at 260 nm wavelength induces covalent bonds between adjacent thymine residues at carbon positions 5 and 6. If located within a gene, this will interfere with replication and transcription unless repaired. Another important type of UV-induced change is a photoproduct consisting of a covalent bond between the carbons in positions 4 and 6 of two adjacent nucleotides, the 4–6 photoproduct (not shown). (Figures redrawn from Lewin, 2000).

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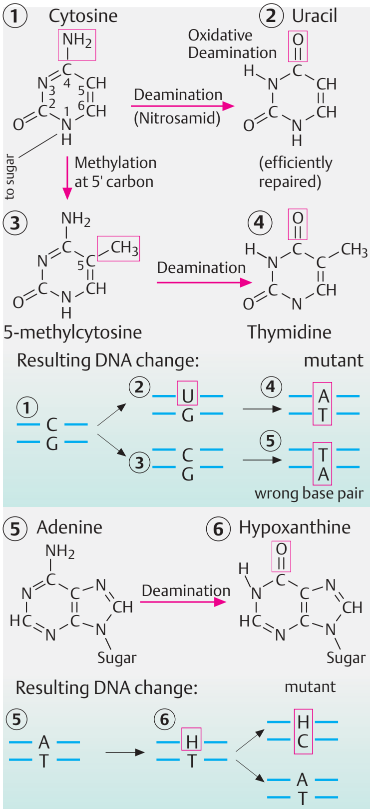
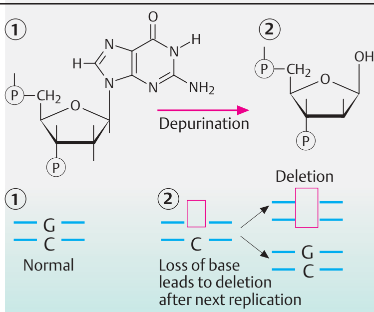
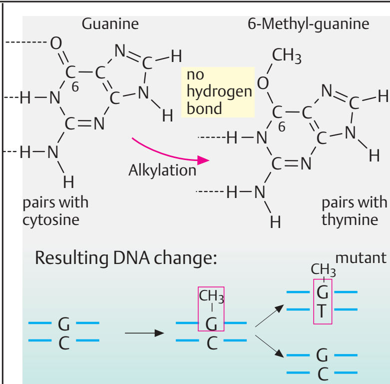


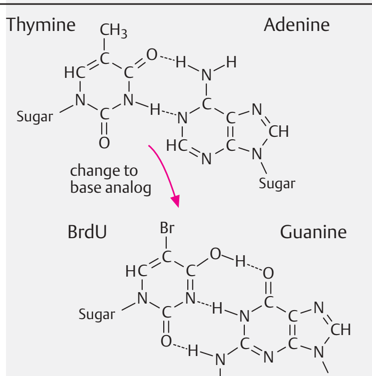
A. Deamination and methylation



B. Depurination

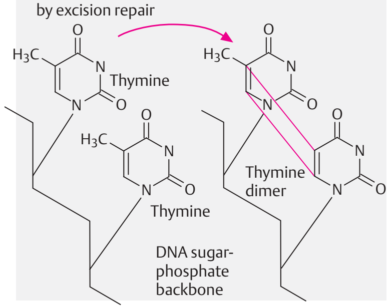


C. Alkylation of guanine



D. Base analog

Ultraviolet irradiation forms thymine dimers with convalent bonds, distorts DNA. Corrected by excision repair



E. UV-light-induced thymine dimers

DNA Polymorphism

Genetic polymorphism is the existence of variants with respect to a gene locus (alleles), a chromosome structure (e.g., size of centromeric heterochromatin), a gene product (variants in enzymatic activity or binding affinity), or a phenotype. The term DNA polymorphism refers to a wide range of variations in nucleotide base composition, length of nucleotide repeats, or single nucleotide variants. DNA polymorphisms are important as genetic markers to identify and distinguish alleles at a gene locus and to determine their parental origin.

A. Single nucleotide polymorphism (SNP)

These allelic variants differ in a single nucleotide at a specific position. At least one in a thousand DNA bases differs among individuals (1). The detection of SNPs does not require gel electrophoresis. This facilitates large-scale detection. A SNP can be visualized in a Southern blot as a restriction fragment length polymorphism (RFLP) if the difference in the two alleles corresponds to a difference in the recognition site of a restriction enzyme (see Southern blot, p. 62).

B. Simple sequence length polymorphism (SSLP)

These allelic variants differ in the number of tandemly repeated short nucleotide sequences in noncoding DNA. Short tandem repeats (STRs) consist of units of 1, 2, 3, or 4 base pairs repeated from 3 to about 10 times. Typical short tandem repeats are CA repeats in the 5′ to 3′ strand, i.e., alternating CG and AT base pairs in the double strand. Each allele is defined by the number of CA repeats, e.g., 3 and 5, as shown (1). These are also called microsatellites. The size differences due to the number of repeats are determined by PCR. Variable number of tandem repeats (VNTR), also called minisatellites, consist of repeat units of 20–200 base pairs (2).

C. Detection of SNP by oligonucleotide hybridization analysis

Oligonucleotides, short stretches of about 20 nucleotides with a complementary sequence to the single-stranded DNA to be examined, will hybridize completely only if perfectly matched. If there is a difference of even one base, such as due to an SNP, the resulting mismatch can be detected because the DNA hybrid is unstable and gives no signal.

D. Detection of STRs by PCR

Short tandem repeats (STRs) can be detected by the polymerase chain reaction (PCR). The allelic regions of a stretch of DNA are amplified; the resulting DNA fragments of different sizes are subjected to electrophoresis; and their sizes are determined.

E. CEPH families

An important step in gene identification is the analysis of large families by linkage analysis of polymorphic marker loci on a specific chromosomal region near a locus of interest. Large families are of particular value. DNA from such families has been collected by the Centre pour l’Étude du Polymorphisme Humain (CEPH) in Paris, now called the Centre Jean Dausset, after the founder. Immortalized cell lines are stored from each family. A CEPH family consists of four grandparents, the two parents, and eight children. If four alleles are present at a given locus they are designated A, B, C, and D. Starting with the grandparents, the inheritance of each allele through the parents to the grandchildren can be traced (shown here as a schematic pattern in a Southern blot). Of the four grandparents shown, three are heterozygous (AB, CD, BC) and one is homozygous (CC). Since the parents are heterozygous for different alleles (AD the father and BC the mother), all eight children are heterozygous (BD, AB, AC, or CD).

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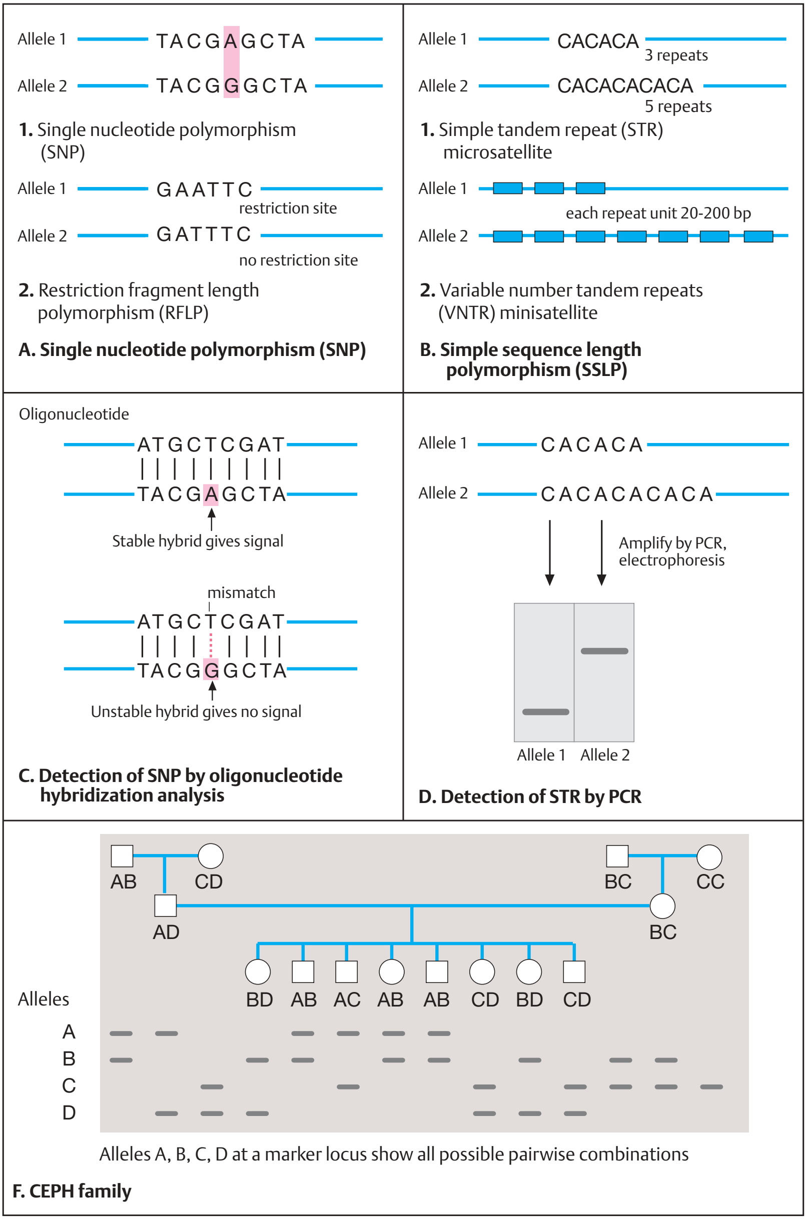

A. Single nucleotide polymorphism (SNP)

B. Simple sequence length polymorphism (SSLP)

C. Detection of SNP by oligonucleotide hybridization analysis

D. Detection of STR by PCR

F. CEPH family

Recombination

Recombination lends the genome flexibility. Without genetic recombination, the genes on each individual chromosome would remain fixed in their particular position. Changes could occur by mutation only, which would be hazardous. Recombination provides the means to achieve extensive restructuring, eliminate unfavorable mutation, maintain and spread favorable mutations, and endow each individual with a unique set of genetic information. This greatly enhances the evolutionary potential of the genome.

Recombination must occur between precisely corresponding sequences (homologous recombination) to ensure that not one base pair is lost or added. The newly combined (recombined) stretches of DNA must retain their original structure in order to function properly. Two types of recombination can be distinguished: (1) generalized or homologous recombination, which in eukaryotes occurs at meiosis (see p. 116) and (2) site-specific recombination. A third process, transposition, utilizes recombination to insert one DNA sequence into another without regard to sequence homology. Here we consider homologous recombination, a complex biochemical reaction between two duplexes of DNA. The necessary enzymes, which can involve any pair of homologous sequences, are not considered. Two general models can be distinguished, recombination initiated from a single-strand DNA break and recombination initiated from a double-strand break.

A. Recombination initiated by single-strand breaks

This model assumes that the process starts with breaks at corresponding positions of one of the strands of homologous DNA (same sequences of different parental origin) (1). A nick is made by a single-strand-breaking enzyme (endonuclease) in each molecule at the corresponding site (2), but see below. This allows the free ends of one nicked strand to join with the free ends of the other nicked strand, from the other molecule, to form single-strand exchanges between the two duplex molecules at the recombination joint (3). The recombination joint moves along the duplex (branch migration) (4). This is an important feature because it ensures that sufficient distance for the second nick is present in each of the other strands (5). After the two other strands have joined and gaps have been sealed (6), a reciprocal recombinant molecule is generated (7). Recombination involving DNA duplexes requires topological changes, i.e., either the molecules must be free to rotate or the restraint must be relieved in some other way.

This model has an unresolved difficulty: How is it assured that the single-strand nicks shown in step 2 occur at precisely the same position in the two double helix DNA molecules?

B. Recombination initiated by double-strand breaks

The current model for recombination is based on initial double-strand breaks in one of the two homologous DNA molecules (1). Both strands are cleaved by an endonuclease, and the break is enlarged to a gap by an exonuclease that removes the new 5′ ends of the strands at the break and leaves 3′ single-stranded ends (2). One free 3′ end recombines with a homologous strand of the other molecule (3). This generates a D loop consisting of a displaced strand from the "donor" duplex. The D loop is extended by repair synthesis until the entire gap of the recipient molecule is closed (4). This displaced strand anneals to the single-stranded complementary homologous sequences of the recipient strand and closes the gap (5). DNA repair synthesis from the other 3′ end closes the remaining gap (6). The integrity of the two molecules is restored by two rounds of single-strand repair synthesis. In contrast to the single-strand exchange model, the double-strand breaks result in heteroduplex DNA in the entire region that has undergone recombination. An apparent disadvantage is the temporary loss of information in the gaps after the initial cleavage. However, the ability to retrieve this information by resynthesis from the other duplex avoids permanent loss. (Figures redrawn from Lewin, 2000).

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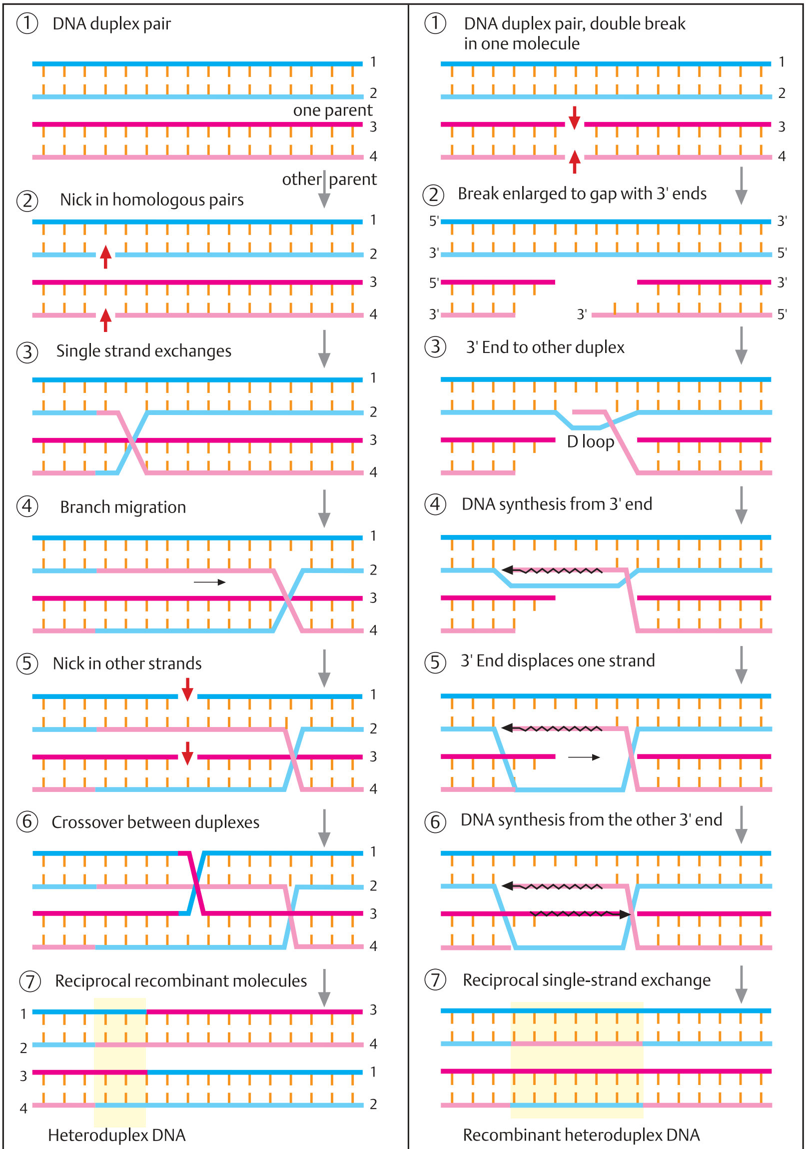

A. Recombination initiated by single-strand breaks

B. Recombination initiated by double-strand breaks

Transposition

Aside from homologous recombination, the overall stability of the genome is interrupted by mobile sequences called transposable elements or transposons. There are different classes of distinct DNA sequences that are able to transport themselves to other locations within the genome. This process utilizes recombination but does not result in an exchange. Rather, a transposon moves directly from one site of the genome to another without an intermediary such as plasmid or phage DNA (see section on prokaryotes). This results in rearrangements that create new sequences and change the functions of target sequences. Transposons may be a major source of evolutionary changes in the genome. In some cases they cause disease when inserted into a functioning gene. Three examples are presented below: insertion sequences (IS), transposons (Tn), and retroelements transposing via an RNA intermediate.

A. Insertion sequences (IS) and transposons (Tn)

A characteristic feature of IS transposition is the presence of a pair of short direct repeats of target DNA at either end. The IS itself carries inverted repeats of about 9–13 bp at both ends and depending on the particular class consists of about 750–1500 bp, which contain a single long coding region for transposase (the enzyme responsible for transposition of mobile sequences). Target selection is either random or at particular sites. The presence of inverted terminal repeats and the short direct repeats of host DNA result in a characteristic structure (1). Transposons carry in addition a central region with genetic markers unrelated to transposition, e.g., antibiotic resistance (2). They are flanked either by direct repeats (same direction) or by inverted repeats (opposite direction, 3).

B. Replicative and nonreplicative transposition

With replicative transposition (1) the original transposon remains in place and creates a new copy of itself, which inserts into a recipient site elsewhere. Thus, this mechanism leads to an increase in the number of copies of the transposon in the genome. This type involves two enzymatic activities: a transposase acting on the ends of the original transposon and resolvase acting on the duplicated copies.

In nonreplicative transposition (2) the transposing element itself moves as a physical entity directly to another site. The donor site is either repaired (in eukaryotes) or may be destroyed (in bacteria) if more than one copy of the chromosome is present.

C. Transposition of retroelements

Retrotransposition requires synthesis of an RNA copy of the inserted retroelement. Retroviruses including the human immunodeficiency virus and RNA tumor viruses are important retroelements (see p. 100 and p. 314). The first step in retrotransposition is the synthesis of an RNA copy of the inserted retroelement, followed by reverse transcription up to the polyadenylation sequence in the 3′ long terminal repeat (3′ LTR). Three important classes of mammalian transposons that undergo or have undergone retrotransposition through an RNA intermediary are shown. Endogenous retroviruses (1) are sequences that resemble retroviruses but cannot infect new cells and are restricted to one genome. Nonviral retrotransposons (2) lack LTRs and usually other parts of retroviruses. Both types contain reverse transcriptase and are therefore capable of independent transposition. Processed pseudogenes (3) or retropseudogenes lack reverse transcriptase and cannot transpose independently. They contain two groups: low copy number of processed pseudogenes transcribed by RNA polymerase II and high copy number of mammalian SINE sequences, such as human *Alu* and the mouse B1 repeat families.

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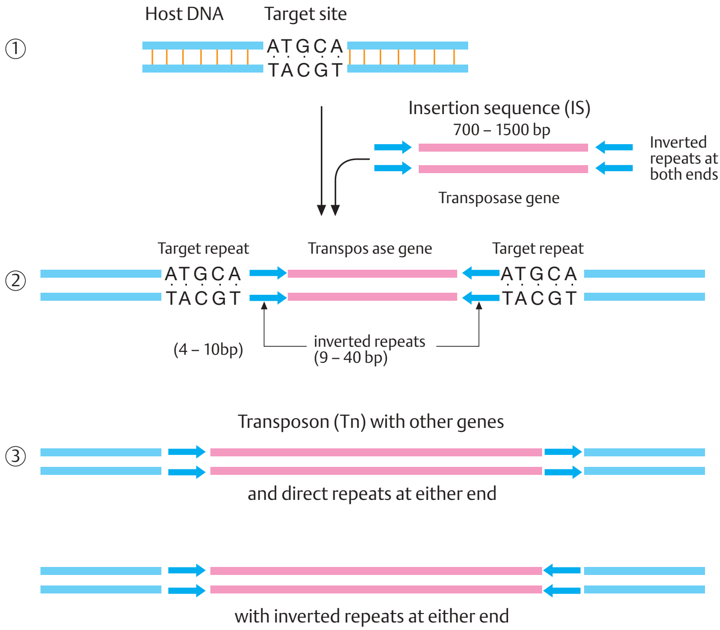
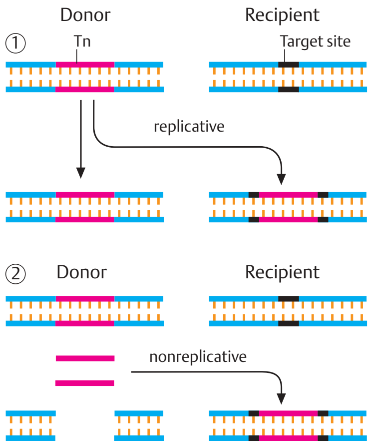
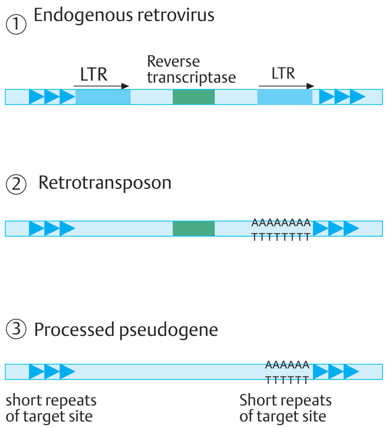


A. Insertion sequences (IS) and transposons (Tn)



B. Replicative and nonreplicative transposition



C. Transposition of retroelements

Trinucleotide Repeat Expansion

The human genome contains tandem repeats of trinucleotides. Normally they occur in groups of 5–35 repeats. When their number exceeds a certain threshold and they occur in a gene or close to it, they cause diseases. Once the normal, variable length has expanded, the increased number of repeats tends to increase even further when passed through the germline or during mitosis. Thus, trinucleotide expansions form a class of unstable mutations, to date observed in humans only.

A. Different types of trinucleotide repeats and their expansions

Trinucleotide repeats can be distinguished according to their localization with respect to a gene. Expansions are greater outside genes and more moderate within coding regions. In several severe neurological diseases, abnormally expanded CAG repeats are part of the gene. CAG repeats encode a series of glutamines (polyglutamine tracts). Within a normal number of repeats, which varies according to the gene involved, the gene functions normally (1). However, an expanded number of repeats leads to an abnormal gene product with altered function. Trinucleotide repeats also occur in noncoding regions of a gene (2). Fairly common types are CGG and GCC repeats. The increase in the number of these repeats can be drastic, up to 1000 or more repeats. The first stages of expansion usually do not lead to clinical signs of a disease, but they do predispose to increased expansion of the repeat in the offspring of a carrier (premutation).

B. Unstable trinucleotide repeats in different diseases

Disorders due to expansion of trinucleotide repeats can be distinguished according to the type of trinucleotide repeat, i.e., the sequence of the three nucleotides, their location with respect to the gene involved, and their clinical features. All involve the central or the peripheral nervous system. Type I trinucleotide diseases are characterized by CAG trinucleotide expansion within the coding region of different genes. The triplet CAG codes for glutamine. About 20 CAG repeats occur normally in these genes, so that about 20 glutamines occur in the gene product. In the disease state the number of glutamines is greatly increased in the protein. Hence, they are collectively referred to as polyglutamine disorders.

Type II trinucleotide diseases are characterized by expansion of CTG, GAA, GCC, or CGG trinucleotides within a noncoding region of the gene involved, either at the 5′ end (GCC in fragile X syndrome type A, FRAXA), at the 3′ end (CGG in FRAXE; CTG in myotonic dystrophy), or in an intron (GAA in Friedreich ataxia). A brief review of these disorders is given on p. 394.

C. Principle of laboratory diagnosis of unstable trinucleotide repeats

The laboratory diagnosis compares the sizes of the trinucleotide repeats in the two alleles of the gene examined. One can distinguish very large expansions of repeats outside coding sequences (50 to more than 1000 repeats) and moderate expansion within coding sequences (20 to 100–200). The figure shows 11 lanes, each representing one person: normal controls in lanes 1–3; confirmed patients in lanes 4–6; and a family with an affected father (lane 7), an affected son (lane 10), the unaffected mother (lane 11), and two unaffected children: a son (lane 8) and a daughter (lane 9). Size markers are shown at the left. Each lane represents a polyacrylamide gel and the $(CAG)_n$ repeat of the Huntington locus amplified by polymerase chain reaction shown as a band of defined size. Each person shows the two alleles. In the affected persons the band representing one allele lies above the threshold in the expanded region (in practice the bands are somewhat blurred because the exact repeat size varies in DNA from different cells).

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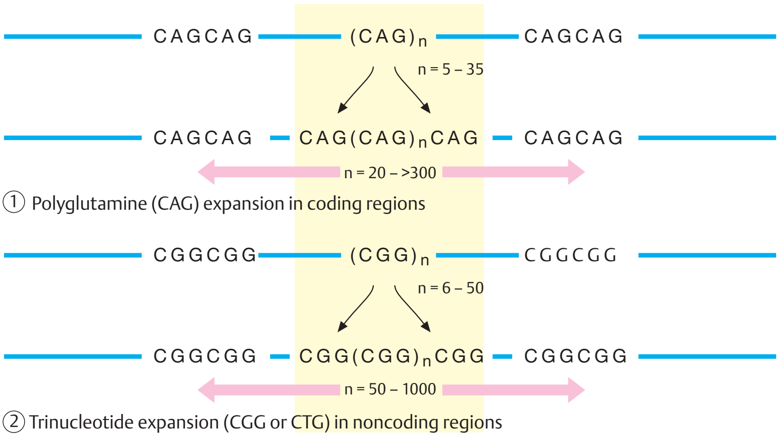
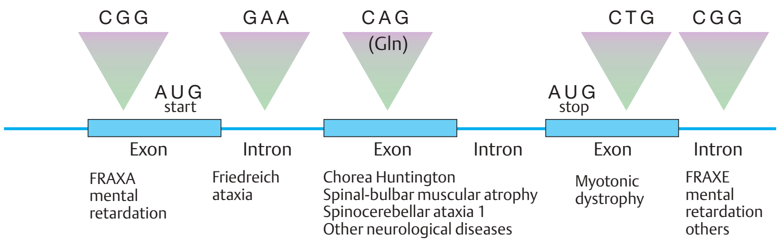
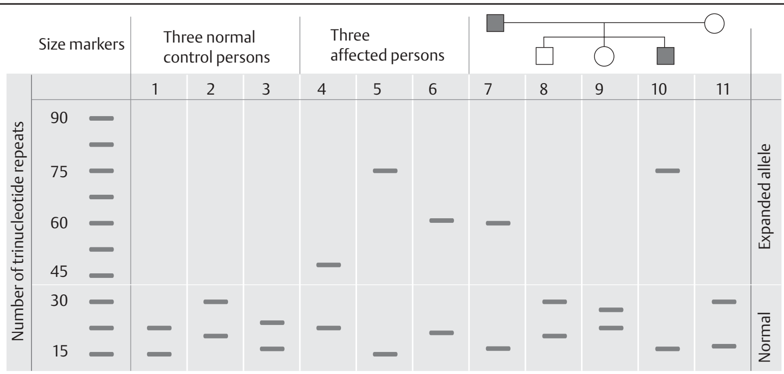


A. Different types of trinucleotide repeat expression



B. Unstable trinucleotide repeats in different diseases



C. Principle of laboratory diagnosis of unstable trinucleotide repeats leading to expansion

DNA Repair

Life would not be possible without the ability to repair damaged DNA. Since replication errors, including mismatch, and harmful exogenous factors are everyday problems for a living organism, a broad repertoire of repair genes has evolved in prokaryotes and eukaryotes. The following types of DNA repair can be distinguished by their basic mechanisms: (1) excision repair to remove a damaged DNA site, such as a strand with a thymine dimer; (2) mismatch repair to correct errors of replication by excising a stretch of single-stranded DNA containing the wrong base; (3) repair of UV-damaged DNA during replication; and (4) transcription-coupled repair in active genes.

A. Excision repair

The damaged strand of DNA is distorted and can be recognized by a set of three proteins, the UvrA, UvrB, and UvrC endonucleases in prokaryotes and XPA, XPB, and XPC in human cells. This DNA strand is cleaved on both sides of the damage by an exonuclease protein complex, and a stretch of about 12 or 13 nucleotides in prokaryotes and 27 to 29 nucleotides in eukaryotes is removed. DNA repair synthesis restores the missing stretch and a DNA ligase closes the gap.

B. Mismatch repair

Mismatch repair corrects errors of replication. However, the newly synthesized DNA strand containing the wrong base must be distinguished from the parent strand, and the site of a mismatch identified. The former is based on a difference in methylation in prokaryotes. The daughter strand is undermethylated at this stage. *E. coli* has three mismatch repair systems: long patch, short patch, and very short patch. The long patch system can replace 1 kb DNA and more. It requires three repair proteins, MutH, MutL, and MutS, which have the human homologues hMSH1, hMLH1, and hMSH2. Mutations in their respective genes lead to cancer due to defective mismatch repair.

C. Replication repair of UV-damaged DNA

DNA damage interferes with replication, especially in the leading strand. Large stretches remain unreplicated beyond the damaged site (in the 3′ direction of the new strand) unless swiftly repaired. The lagging strand is not affected as much because Okazaki fragments (about 100 nucleotides in length) of newly synthesized DNA are also formed beyond the damaged site. This leads to an asymmetric replication fork and single-stranded regions of the leading strand. Aside from repair by recombination, the damaged site can be bypassed.

D. Double-strand repair by homologous recombination

Double-strand damage is a common consequence of γ radiation. An important human pathway for mediating repair requires three proteins, encoded by the genes *ATM*, *BRCA1*, and *BRCA2*. Their names are derived from important diseases that result from mutations in these genes: ataxia telangiectasia (see p. 334) and hereditary predisposition to breast cancer (BRCA1 and BRCA2, see p. 328. ATM, a member of a protein kinase family, is activated in response to DNA damage (1). Its active form phosphorylates BRCA1 at specific sites (2). Phosphorylated BRCA1 induces homologous recombination in cooperation with BRCA2 and mRAD5, the mammalian homologue of *E. coli* RecA repair protein (3). This is required for efficient DNA double-break repair. Phosphorylated BRCA1 may also be involved in transcription and transcription-coupled DNA repair (4). (Figure redrawn from Ventikaraman, 1999).

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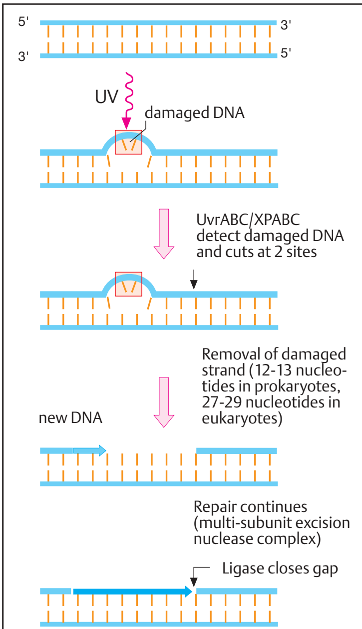
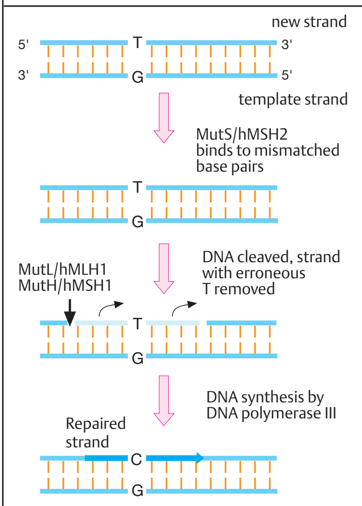
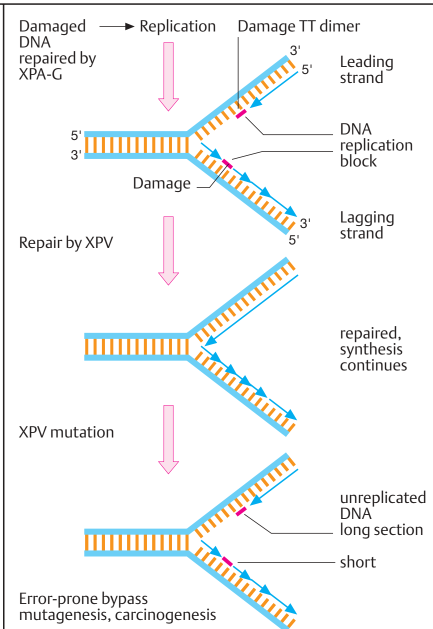


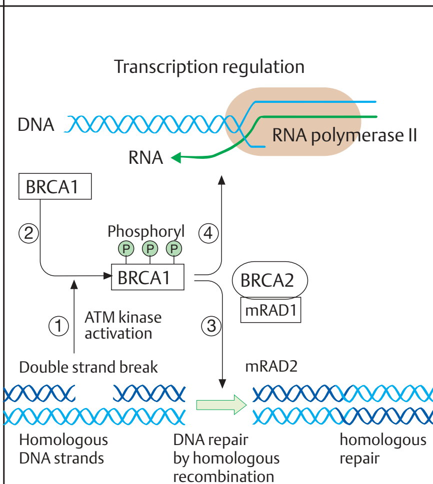
A. Excision repair (scheme)



B. Mismatch repair (scheme)



C. Replication repair of UV-damaged DNA



D. Double strand repair by homologous recombination

Xeroderma Pigmentosum

Xeroderma pigmentosum (XP) is a heterogeneous group of genetically determined skin disorders due to unusual sensitivity to ultraviolet light. They are manifested by dryness and pigmentation of the exposed regions of skin (xeroderma pigmentosum = "dry, pigmented skin"). The exposed areas of skin also show a tendency to develop tumors. The causes are different genetic defects of DNA repair.
Repair involves mechanisms similar to those involved in transcription and replication. The necessary enzymes are encoded by at least a dozen genes, which are highly conserved in bacteria, yeast, and mammals.

A. Clinical phenotype

The skin changes are limited to UV-exposed areas (1 and 2). Unexposed areas show no changes. Thus it is important to protect patients from UV light. An especially important feature is the tendency for multiple skin tumors to develop in the exposed areas (3). These may even occur in childhood or early adolescence. The types of tumors are the same as those occurring in healthy individuals after prolonged UV exposure.

B. Cellular phenotype

The UV sensitivity of cells can be demonstrated in vitro. When cultured fibroblasts from the skin of patients are exposed to UV light, the cells show a distinct dose-dependent decrease in survival rate compared with normal cells (1). Different degrees of UV sensitivity can be demonstrated. The short segment of new DNA normally formed during excision repair can be demonstrated by culturing cells in the presence of [^{3}H]thymidine and exposing them to UV light. The DNA synthesis induced for DNA repair can be made visible in autoradiographs. Since [^{3}H]thymidine is incorporated during DNA repair, these bases are visible as small dots caused by the isotope on the film (2). In contrast, xeroderma (XP) cells show markedly decreased or almost absent repair synthesis. (Photograph of Bootsma & Hoeijmakers, 1999).

C. Genetic complementation in cell hybrids

If skin cells (fibroblasts) from normal persons and from patients (XP) are fused (cell hybrids) in culture and exposed to UV light, the cellular XP phenotype will be corrected (1). Normal DNA repair occurs. Also, hybrid cells from two different forms of XP show normal DNA synthesis (2) because cells with different repair defects correct each other (genetic complementation). However, if the mutant cells have the same defect (3), they are not be able to correct each other (4) because they belong to the same complementation group. At present about ten complementation groups are known in xeroderma pigmentosum. They differ clinically in terms of severity and central nervous system involvement. Each complementation group is based on a mutation at a different gene locus. Several of these genes have been cloned and show homology with repair genes of other organisms, including yeast and bacteria.

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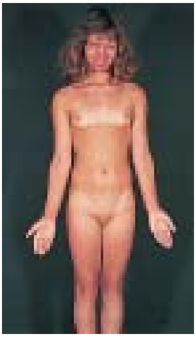
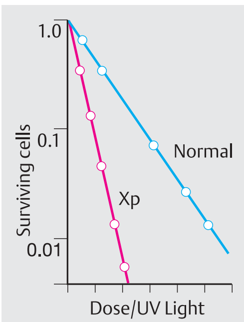


1.

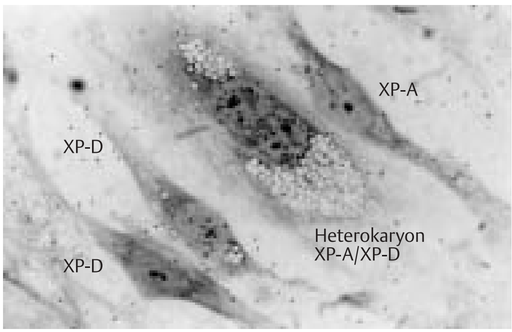
2.

3.

A. Clinical phenotype

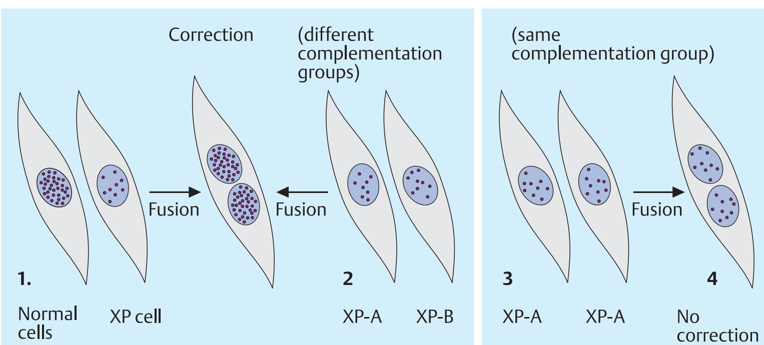


1. UV sensitivity



2. Complementation following fusion of XP-A and XP-D cells

B. Cellular phenotype



C. Genetic complementation in cell hybrids

Prokaryotic Cells and Viruses

Isolation of Mutant Bacteria

Important advances in genetics were made in the early 1950s through studies of bacteria. As prokaryotic organisms, bacteria have certain advantages over eukaryotic organisms because they are haploid and have an extremely short generation time. Mutant bacteria can be identified easily. The growth of some mutant bacteria depends on whether a certain substance is present in the medium (auxotrophism). Bacterial cultures are well suited for determining mutational events since an almost unlimited number of cells can be tested in a short time. Without great difficulty, it is possible to detect one mutant in 10^7 colonies. Efficiency to this degree is not possible in the genetic analysis of eukaryotic organisms.

A. Replica plating to recognize mutants

In 1952, Joshua and Esther Lederberg developed replica plating of bacterial cultures. With this method, individual colonies on an agar plate can be taken up with a stamp covered with velvet and placed onto other culture dishes with media of different compositions. Some mutant bacteria differ from nonmutants in their ability to grow. Here several colonies are shown in the Petri dish of the initial culture. Each of these colonies originated from a single cell. By means of replica plating, the colonies are transferred to two new cultures. One culture (right) contains an antibiotic in the culture medium; the other (left) does not. All colonies grow in normal medium, but only those colonies that are antibiotic resistant owing to a mutation grow in the antibiotic-containing medium. In this manner, mutant colonies can be readily identified.

B. Mutant bacteria identified through an auxotrophic medium

Here it is shown how different mutants can be distinguished, e.g., after exposure to a mutagenic substance. After a colony has been treated with a mutagenic substance, it is first cultivated in normal nutrient medium. Mutants can then be identified by replica plating. The culture with the normal medium serves as the control. In one culture with minimal medium, from which a number of substances are absent, two colonies do not grow (auxotrophic mutants). Initially, it is known for which of the substances the colonies are auxotrophic. If a different amino acid is added to each of two cultures with minimal medium, e.g., threonine (Thr) to one and arginine (Arg) to the other, it can be observed that one of the mutant colonies grows in the threonine-containing minimal medium, but the other does not. The former colony is dependent on the presence of threonine (Thr^-), i.e., it is an auxotroph for threonine. The other culture with minimal medium had arginine added. Only here can the other of the two mutant colonies, an auxotroph for arginine (Arg^-), grow. After the mutant colonies requiring specific conditions for growth have been identified, they can be further characterized. This procedure is relatively simple and makes rapid identification of mutants possible. Many mutant bacteria have been defined by auxotrophism. The wild-type cells that do not have special additional growth requirements are called prototrophs (Figures adapted from Stent & Calendar, 1978).

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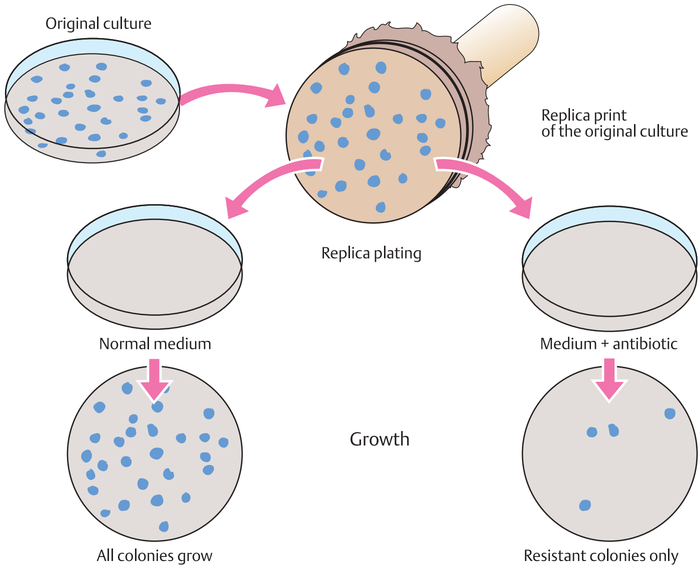
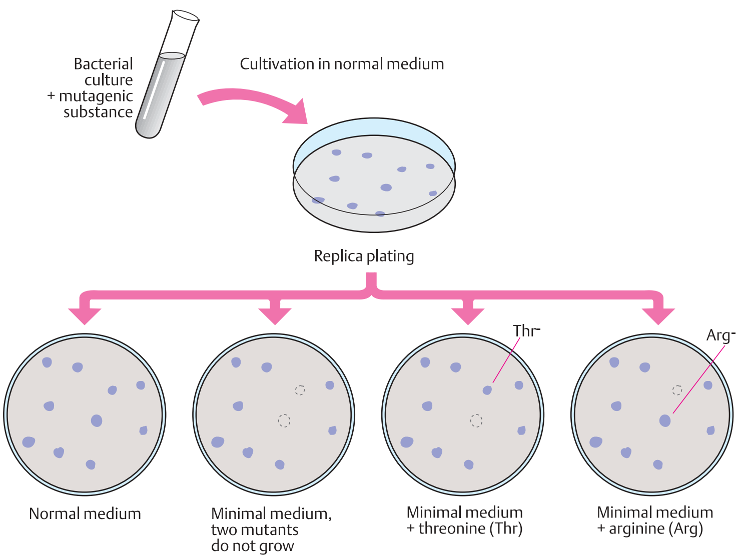


A. Replica plating for recognition of mutants



B. Mutant bacteria can be identified by auxotrophic medium

Recombination in Bacteria

In 1946, J. Lederberg and E. L. Tatum first demonstrated that genetic information can be exchanged between different mutant bacterial strains. This corresponds to a type of sexuality and leads to genetic recombination.

A. Genetic recombination in bacteria

In their classic experiment, Lederberg and Tatum used two different auxotrophic bacterial strains. One (A) was auxotrophic for methionine (Met^-) and biotin (Bio^-). This strain required methionine and biotin, but not threonine and leucine (Thr^+, Leu^+), to be added to the medium. The opposite was true for bacterial strain B, auxotrophic for threonine and leucine (Thr^-, Leu^-), but prototrophic for methionine and biotin (Met^+, Bio^+). When the cultures were mixed together without the addition of any of these four amino acids and then plated on an agar plate with minimal medium, a few single colonies (Met^+, Bio^+, Thr^+, Leu^+) unexpectedly appeared. Although this occurred rarely (about 1 in 10^7 plated cells), a few colonies with altered genetic properties usually appeared owing to the large number of plated bacteria. The interpretation: genetic recombination between strain A and strain B. The genetic properties of the parent cells complemented each other (genetic complementation). (Figure adapted from Stent & Calendar, 1978).

B. Conjugation in bacteria

Later, the genetic exchange between bacteria (conjugation) was demonstrated by light microscopy. Conjugation occurs with bacteria possessing a gene that enables frequent recombination. Bacterial DNA transfer occurs in one direction only. "Male" chromosomal material is introduced into a "female" cell. The so-called male and female cells of *E. coli* differ in the presence of a fertility factor (F). When F^+ and F^- cells are mixed together, conjugal pairs can form with attachment of a male (F^+) sex pilus to the surface of an F^- cell. (Photograph from Science **257**:1037, 1992).

C. Integration of the F factor into an Hfr^- chromosome

The F factor can be integrated into the bacterial chromosome by means of specific crossing-over. After the factor is integrated, the original bacterial chromosome with the sections a, b, and c contains additional genes, the F factor genes (e, d). Such a chromosome is called an Hfr chromosome (Hfr, high frequency of recombination) owing to its high rate of recombination with genes of other cells as a result of conjugation.

D. Transfer of F DNA from an F^+ to an F^- cell

Bacteria may contain the F factor (fertility) as an additional small chromosome, i.e., a small ring-shaped DNA molecule (F plasmid) of about 94 000 base pairs (not shown to scale). This corresponds to about 1/40th of the total genetic information of a bacterial chromosome. It occurs once per cell and can be transferred to other bacterial cells. About a third of the F^+ DNA consists of transfer genes, including genes for the formation of sex pili. The transfer of the F factor begins after a strand of the DNA double helix is opened. One strand is transferred to the acceptor cell. There it is replicated, so that it becomes double-stranded. The DNA strand remaining in the donor cell is likewise restored to a double strand by replication. Thus, DNA synthesis occurs in both the donor and the acceptor cell. When all is concluded, the acceptor cell is also an F^+ cell. (Figures in C and D adapted from Watson et al., 1987).

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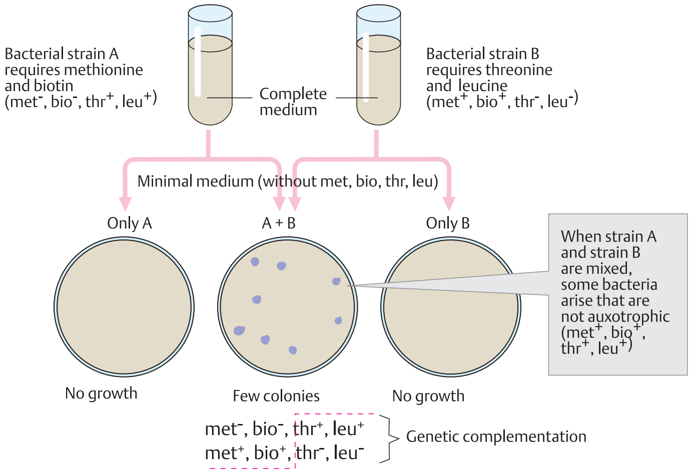
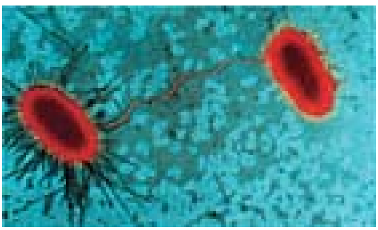
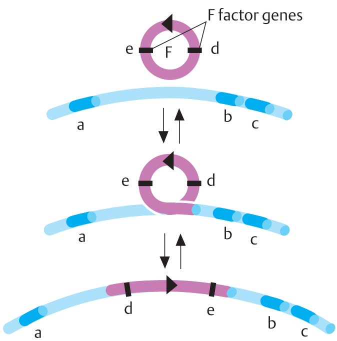


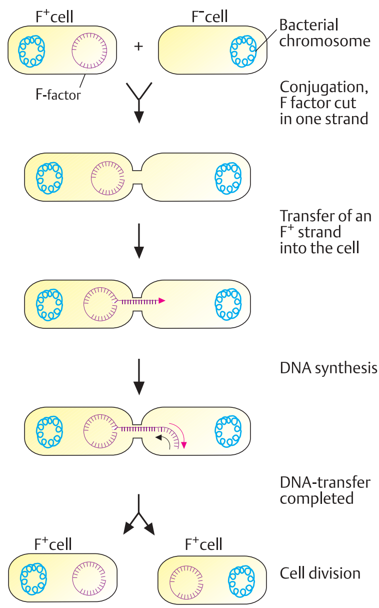
A. Genetic recombination in bacteria



B. Conjugation in bacteria



C. Integration of the F factor into an Hfr⁻ chromosome



D. Transfer of F DNA from an F⁺ into an F⁻ cell

Bacteriophages

The discovery of bacterial viruses (bacteriophages or phages) in 1941 opened a new era in the study of the genetics of prokaryotic organisms. Although they were disappointing in the original hope that they could be used to fight bacterial infections, phages served during the 1950s as vehicles for genetic analysis of bacteria. Unlike viruses that infect plant or animal cells, phages can relatively easily be analyzed in their host cells. Names associated with phage analysis are Max Delbrück, Salvador Luria, and Alfred D. Hershey (the "phage group," see: Cairns et al., 1966).

A. Attachment of a bacteriophage

Phages consist of DNA, a coat (coat protein) for protection, and a means of attachment (terminal filaments). Like other viruses, phages are basically nothing more than packaged DNA. One or more bacteriophages attach to a receptor on the surface of the outer cell membrane of a bacterium. The figure shows how an attached phage inserts its DNA into a bacterium. Numerous different phages are known, e.g., for *Escherichia coli* and *Salmonella* (phages T1, T2, P1, F1, lambda, T4, T7, phiX174 and others).

B. Lytic and lysogenic cycles of a bacteriophage

Phages do not reproduce by cell division like bacteria, but by intracellular formation and assembly of the different components. This begins with the attachment of a phage particle to a specific receptor on the surface of a sensitive bacterium. Different phages use different receptors, thus giving rise to specificity of interaction (restriction). The invading phage DNA contains the information for production of coat proteins for new phages and factors for DNA replication and transcription. Translation is provided for by cell enzymes. The phage DNA and phage protein synthesized in the cell are assembled into new phage particles. Finally, the cell disintegrates (lysis) and hundreds of phage particles are released. With attachment of a new phage to a new cell, the procedure is repeated (lytic cycle).

Phage reproduction does not always occur after invasion of the cell. Occasionally, phage DNA is integrated into the bacterial chromosome and replicated with it (lysogenic cycle). Phage DNA that has been integrated into the bacterial chromosome is designated a prophage. Bacteria containing prophages are designated lysogenic bacteria; the corresponding phages are termed lysogenic phages. The change from a lysogenic to a lytic cycle is rare. It requires induction by external influences and complex genetic mechanisms.

C. Insertion of a lambda phage into the bacterial chromosome by crossing-over

A phage can be inserted into a bacterial chromosome by different mechanisms. With the lambda phage (λ), insertion results from crossing-over between the *E. coli* chromosome and the lambda chromosome. First, the lambda chromosome forms a ring. Then it attaches to a homologous section of the bacterial chromosome. Both the bacterial and the lambda chromosome are opened by a break and attach to each other. Since the homologies between the two chromosomes are limited to very small regions, phage DNA is seldom integrated. The phage is released (and the lytic cycle is induced) by the reverse procedure. (Figures adapted from Watson et al., 1987).

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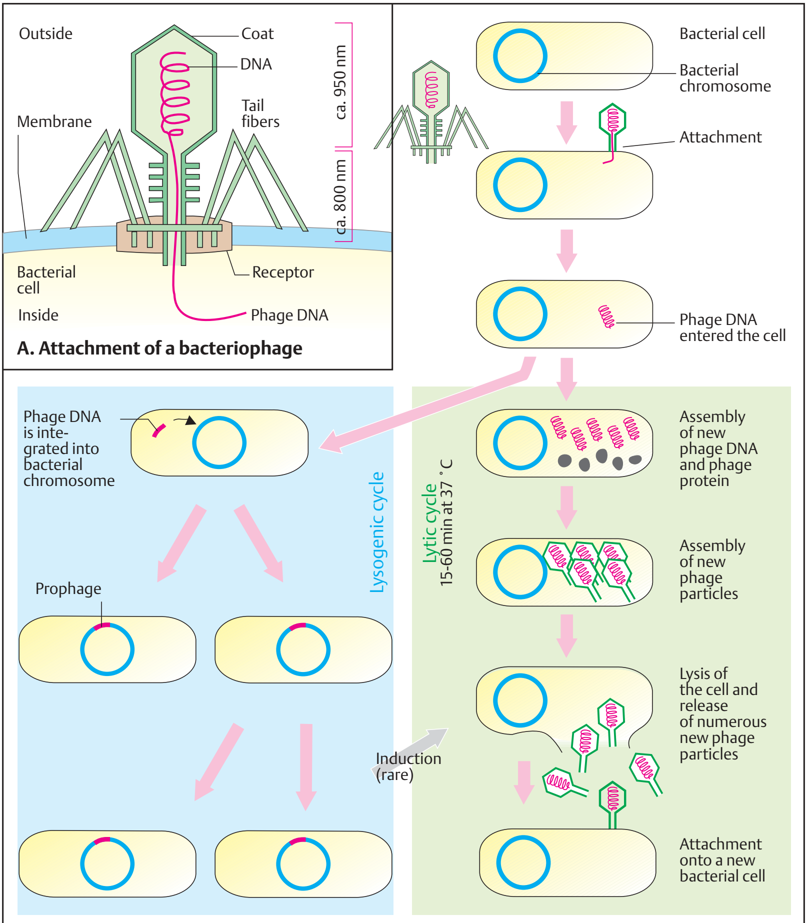
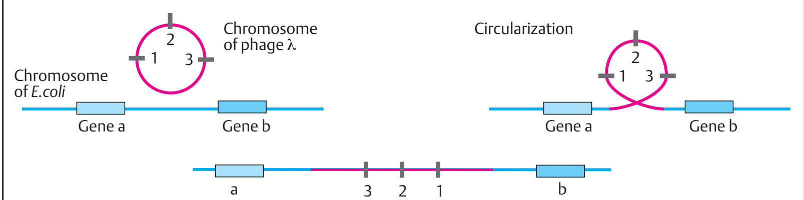


A. Attachment of a bacteriophage

B. Lytic and lysogenic cycle of a bacteriophage



C. Insertion of phage lambda into the bacterial chromosome by crossing-over

DNA Transfer between Cells

Transfer of DNA occurs not only by fusion of gametes in sexual reproduction but also between other cells of prokaryotic and eukaryotic organisms (conjugation of bacteria, transduction between bacteriophages and bacteria, transformation by plasmids in bacteria, transfection in cultures of eukaryotic cells). Cells altered genetically by taking up DNA are said to be transformed. The term transformation is used in different contexts and refers to the result, not the mechanism.

A. Transduction by viruses

In 1952, N. Zinder and J. Lederberg described a new type of recombination between two strains of bacteria. Bacteria previously unable to produce lactose (*lac*⁻) acquired the ability to produce lactose after being infected with phages that had replicated in bacteria containing a gene for producing lactose (*lac*⁺). A small segment of DNA from a bacterial chromosome had been transferred by a phage to another bacterium (transduction). General transduction (insertion of phage DNA into the bacterial genome at any unspecified location) is distinguished from special transduction (insertion at a particular location). Genes regularly transduced together (cotransduction) were used to determine the positions of neighboring genes on the bacterial chromosome (mapping of genes in bacteria).

B. Transformation by plasmids

Plasmids are small, autonomously replicating, circular DNA molecules separate from the chromosome in a bacterial cell. Since they often contain genes for antibiotic resistance (e.g., ampicillin), their incorporation into a sensitive cell renders the cell resistant to the antibiotic (transformation). Only these bacteria can grow in culture medium containing the antibiotic (selective medium).

C. Multiplication of a DNA segment in transformed bacteria

Plasmids are well suited as vectors for the transfer of DNA. A selective medium is used so that only those bacteria that have incorporated a recombinant plasmid containing the DNA to be investigated can grow.

D. Transfection by DNA

The transfer of DNA between eukaryotic cells in culture (transfection) can be used to examine the transmission of certain genetic traits (transfection assay). Left, a DNA transfer experiment is shown in a culture of mouse fibroblasts; right, in a culture of human tumor cells (Weinberg, 1985, 1987).
The mouse fibroblast culture (see p. 122) is altered by the chemical carcinogen methylcholanthrene (left). DNA from these cells is precipitated with calcium phosphate, extracted, and then taken up by a normal culture (transfection). About 2 weeks later, cells appear that have lost contact inhibition (transformed cells). When these cells are injected into mice that lack a functional immune system (naked mice), tumors develop. DNA from cultured human tumor cells (right) also can transform normal cells after several transfer cycles. The DNA segment must be of limited size (e.g., a gene), since long DNA segments do not remain intact after repeated cycles of extraction and precipitation. Detailed studies of cancer-causing genes (oncogenes) in eukaryotic cells were first carried out using transfection (see p. 90). (Figures in A–C adapted from Watson et al. 1987, D from Weinberg 1987).

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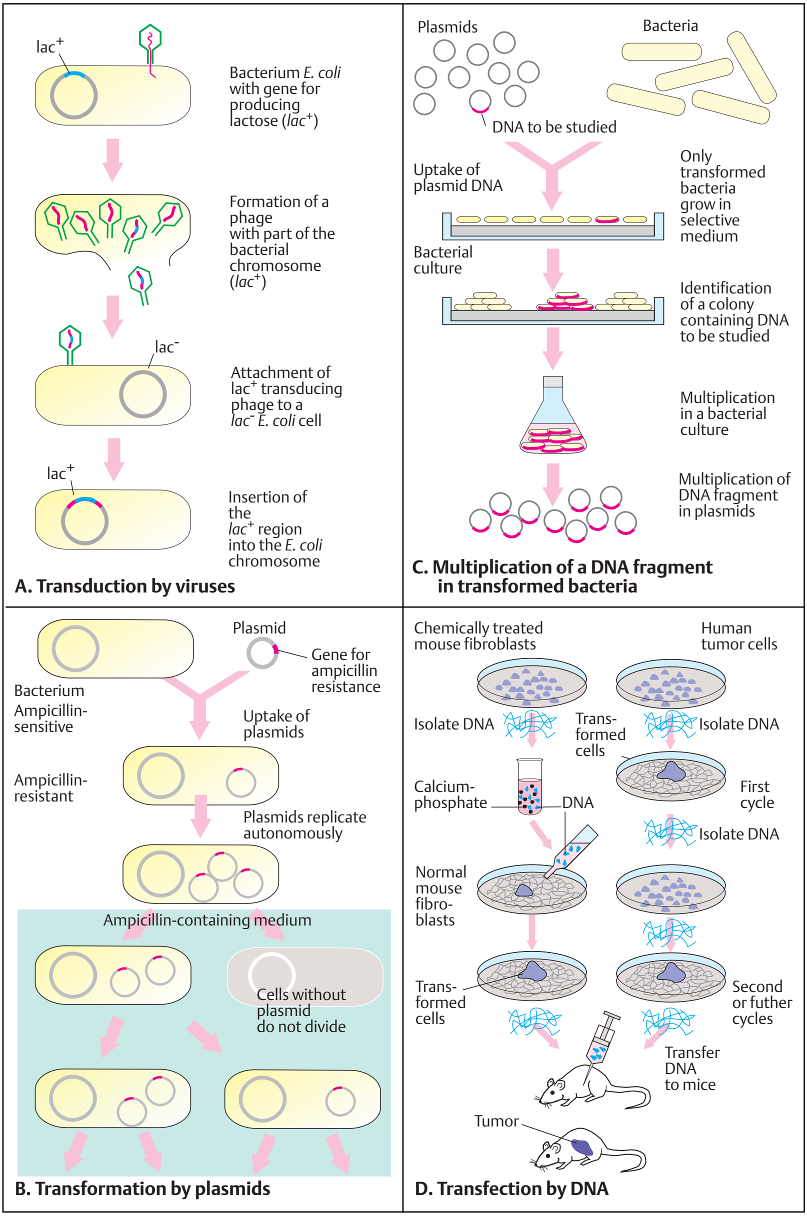
lac+
Bacterium E. coli with gene for producing lactose (lac+)
Formation of a phage with part of the bacterial chromosome (lac+)
lac-
Attachment of lac+ transducing phage to a lac- E. coli cell
lac+
Insertion of the lac+ region into the E. coli chromosome
A. Transduction by viruses
Plasmids
Bacteria
DNA to be studied
Uptake of plasmid DNA
Only transformed bacteria grow in selective medium
Bacterial culture
Identification of a colony containing DNA to be studied
Multiplication in a bacterial culture
Multiplication of DNA fragment in plasmids
C. Multiplication of a DNA fragment in transformed bacteria
Bacterium
Ampicillin-sensitive
Plasmid
Gene for ampicillin resistance
Uptake of plasmids
Ampicillin-resistant
Plasmids replicate autonomously
Ampicillin-containing medium
Cells without plasmid do not divide
B. Transformation by plasmids
Chemically treated mouse fibroblasts
Human tumor cells
Isolate DNA
Trans-formed cells
Isolate DNA
Calcium-phosphate
DNA
First cycle
Isolate DNA
Normal mouse fibro-blasts
Trans-formed cells
Second or futher cycles
Transfer DNA to mice
Tumor
D. Transfection by DNA

Viruses

Viruses are important pathogens in plants and animals, including man. The complete infectious viral particle is called a virion. Its genome carries a limited amount of genetic information, and it can replicate only in host cells. From analysis of the structure and expression of viral genes, fundamental biological processes such as DNA replication, transcription regulation, mRNA modification (RNA splicing, RNA capping, RNA polyadenylation), reverse transcription of RNA to DNA, viral genome integration into eukaryotic DNA, tumor induction by viruses, and cell surface proteins have been recognized and elucidated. The extracellular form of a virus particle includes a protein coat (capsid), which encloses the genome of DNA or RNA. The capsids contain multiple units of one or a few different protein molecules coded for by the virus genome. Capsids usually have an almost spherical, icosahedral (20 plane surfaces), or occasionally a helical structure. Some viral capsids are surrounded by a lipid membrane envelope.

A. Classification of viruses

Viruses can be classified on the basis of the structure of their viral coat, their type of genome, and their organ or tissue specificity. The genome of a virus may be enclosed simply in a virus-coded protein coat (capsid) or in the capsid plus an additional phospholipid membrane, which is of cellular origin. The genome of a virus may consist of single-stranded DNA (e.g., parvovirus), double-stranded DNA (e.g., papovavirus, adenovirus, herpesvirus, and poxvirus), single-stranded RNA (e.g., picornavirus, togavirus, myxovirus, rhabdovirus), or double-stranded RNA (e.g., reovirus). Viruses with genomes of single-stranded RNA are classified according to whether their genome is a positive (plus RNA) or negative (minus RNA) RNA strand. Only an RNA plus strand can serve as a template for translation (5′ to 3′ orientation).

B. Replication and transcription of viruses

Since viral genomes differ, the mechanisms for replicating their genetic material also differ. Viruses must pack all their genetic information into a small genome; thus, one transcription unit (gene) of a viral genome is often used to produce several mRNAs by alternative splicing, and each mRNA codes for a different protein. In some RNA viruses, initially large precursor proteins are formed from the mRNA and subsequently split into several smaller functional proteins. An RNA plus strand can be used directly for protein synthesis. An RNA minus strand cannot be used directly; an RNA plus strand must be formed from the RNA minus strand by a transcriptase before translation is possible. RNA viruses contain a transcriptase to replicate their RNA genomes. RNA viruses in which DNA is formed as an intermediate step (retroviruses) contain a reverse transcriptase. This can form DNA from RNA. The DNA intermediate step in the replication of retroviruses becomes integrated into the host cell. Several RNA viruses have segmented genomes. They consist of individual pieces of RNA genome, each of which codes for one or more proteins (e.g., influenza virus). The exchange of individual pieces of RNA genome of different viral serotypes plays an important role in the formation of new viral strains (e.g., influenza strains). (Figure after Watson et al., 1987).

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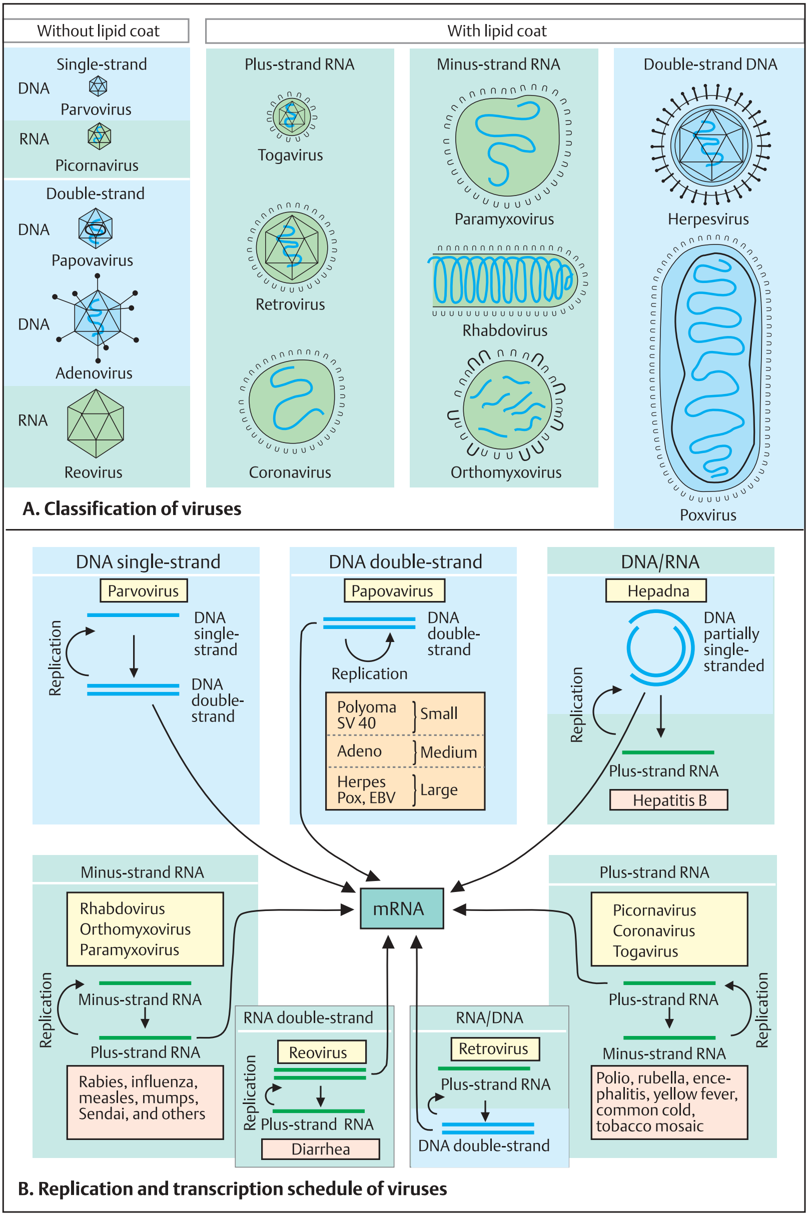

A. Classification of viruses

B. Replication and transcription schedule of viruses

Replication Cycle of Viruses

With all their different genomic structures, forms, and sizes, viruses basically have a relatively simple replication cycle. While only the genome of a bacteriophage enters a bacterium, the complete virus (genome and capsid) enters a eukaryotic cell.

A. General sequence of the replication cycle of a virus in a cell

The replication cycle of a virus consists of five principal consecutive steps: (1) entrance into the cell and release of the genome (uncoating), (2) transcription of the viral genes and (3) translation of the mRNAs to form viral proteins, (4) replication of the viral genome, (5) assembly of new viral particles in the cell and release of the complete virions from the host cell (6).

B. Uptake of a virus by endocytosis

Besides fusion of the lipid membrane of membrane-enclosed viruses with the cell membrane of the host cell, the most frequent mechanism for a virion to enter a cell is by a special form of endocytosis. The virus attaches to the cell membrane by using cell surface structures (receptors), which serve other important functions for the cell, e.g., for the uptake of macromolecules (see p. 352). Like these, the virus is taken into the cytoplasm by a special mechanism, receptor-mediated endocytosis (coated pits, coated vesicles). Within the cell, the virus-containing vesicle fuses with other cellular vesicles (e.g., primary lysosomes). The viral coat is extensively degraded in the endocytotic vesicle, and the viral core (genome, associated with virus-coded proteins) is released into the cytoplasm or nucleus, depending on the viral type. Replication and expression of the viral genome follow. Whether a cell can be infected by a virion depends on a specific interaction between the virus and a cellular receptor. Some viruses, such as the paramyxoviruses (e.g., mumps and Sendai virus), enter the cell by direct fusion of the viral and cellular membranes, mediated by a viral coat glycoprotein (F or fusion protein).

C. Transcription and replication of a virus

The first viral genes to be expressed after the virus has entered the cell are the early genes of the viral genome. Gene products of these early viral genes regulate transcription of the remaining viral genes and are involved in replicating the viral genome. Synthesis of the capsid proteins begins later (late genes), at the same time as genome replication, when new virions are formed from the genome and capsids (assembly). The virions (nucleocapsids = genome plus capsid) are then released from the cell by one of several mechanisms, depending on the type of virus.

D. Release of a virus by budding

The release of a virus coated by a lipid membrane occurs by budding. First, molecules of a viral-coded glycoprotein are built into the cell membrane, to which the virus capsid or virus core (containing the viral genome) attaches. Attachment of the genome leads to increased budding of that region of the cell membrane. Eventually, the entire virion is surrounded by a lipid membrane envelope of cellular origin containing viral proteins and is released. Virions can be expelled from the cell continuously and in great numbers without the death of the virus-producing cell. (Figures from J. D. Watson et al., 1987).

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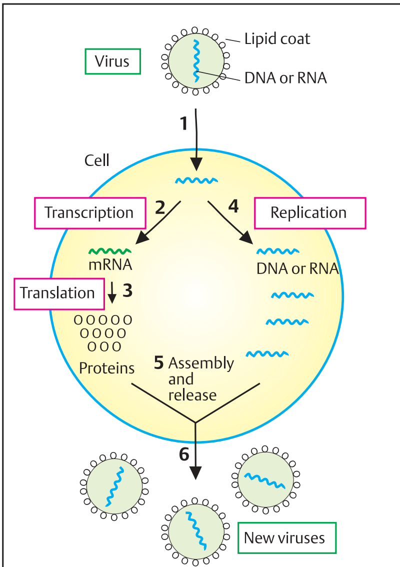
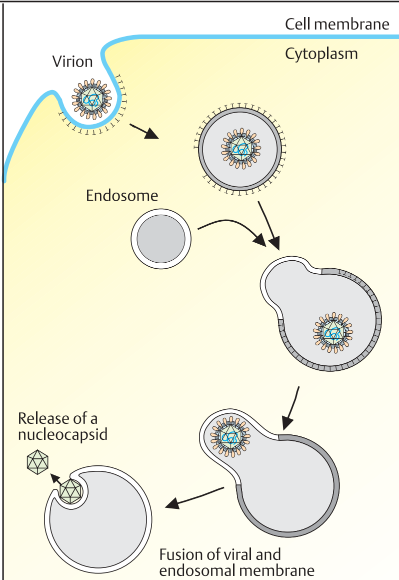
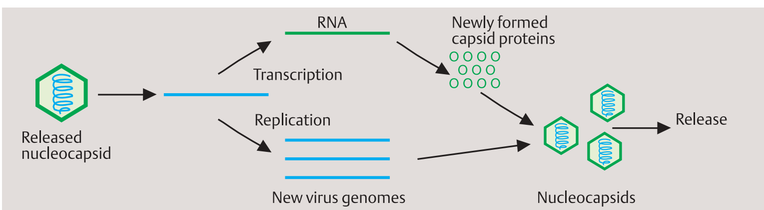


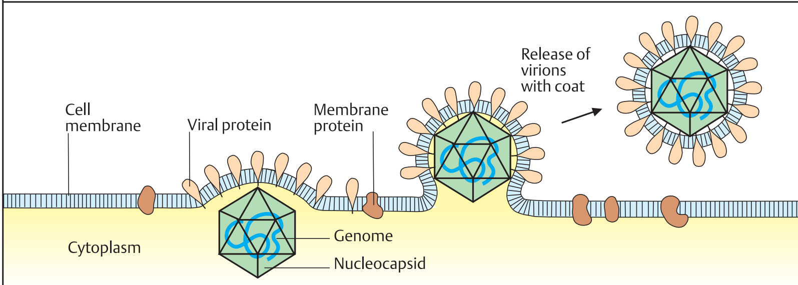
A. General sequence of the multiplication cycle of a virus in a cell



B. Uptake of a virus by endocytosis



C. Transcription and replication of a virus



D. Release of a virus by budding

RNA Viruses: Genome, Replication, Translation

The genomes of RNA viruses consist of double-stranded or single-stranded RNA. Single-stranded RNA can be of plus strands, which can be translated directly, or minus strands. The latter must first be transcribed into plus-strand RNA (sense RNA). Since eukaryotic cells do not have enzymes for copying RNA, RNA viruses contain enzymes necessary to transcribe RNA into DNA (transcriptase). In most of these RNA viruses, transcriptase molecules are included in the virus particle.

A. Genome of the poliovirus and its translation products

The genome of the poliovirus, a member of the enterovirus family, is one of the smallest genomes. It consists of about 7400 base pairs of plus-strand RNA of known nucleotide base sequence. It codes for a large precursor protein from which smaller, functional proteins are formed by proteolytic cleavage (VPO, 1, 2, 3, and 4), about 60 copies of each being present per virion. VP1 is responsible for attachment to the cellular receptor, which is found only on the epithelial cells, fibroblasts, and nerve cells of primates. Thus, the poliovirus can infect only a very limited range of host cells. The poliovirus is one of the fastest-replicating animal viruses: Within 6 – 8 hours, an infected cell can release about 10 000 new virions.

B. Togavirus: replication and translation

Togaviruses (e.g., yellow fever, rubella, encephalitis viruses) have an RNA plus strand of about 12 000 base pairs as genome. It is enclosed in a capsid and a lipid membrane. Only the replicase protein can be translated from the genomic RNA, since start codons (AUG) for initiating translation lie 3′ and are not recognized before replication. Thus, the capsid proteins are synthesized late after infection, i.e., after replication. This results in a differentiated regulation of viral gene expression.

C. Influenzavirus

The genome of the influenzavirus consists of eight segments of minus-strand RNA. Each codes for at least one, and some for more than one protein. The lipid membrane contains two virus-coded glycoproteins: hemagglutinin (HA), which recognizes and binds to cell surface receptors, and neuraminidase (NA), a receptor-degrading enzyme. In addition, the virions contain a matrix protein (M) and a nucleocapsid protein (N). If a cell has become infected with two different influenza virus strains, a new type of influenza virus can arise by exchange of genome segments. The recombinant virus is either not at all or only slowly recognized by the immune system.

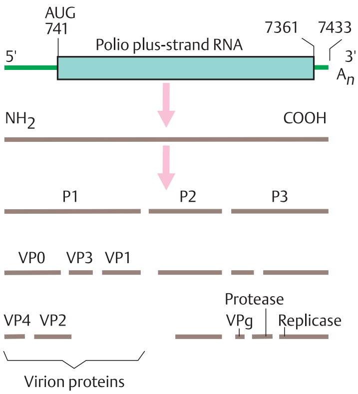
D. Rhabdovirus (vesicular stomatitis virus, VSV)

The RNA minus-strand genome of rhabdoviruses is enclosed in a characteristically formed (bullet-shaped) outer membrane; it codes for five proteins, the nucleocapsid protein (N) within the virion, the matrix protein (M) between capsid and outer membrane, a transmembrane viral glycoprotein (G) responsible for interacting with cellular receptors, and two enzymes for replication and mRNA synthesis: protein L (large) and NS (nonstructural). One of the best known types of rhabdoviruses is the rabies virus.

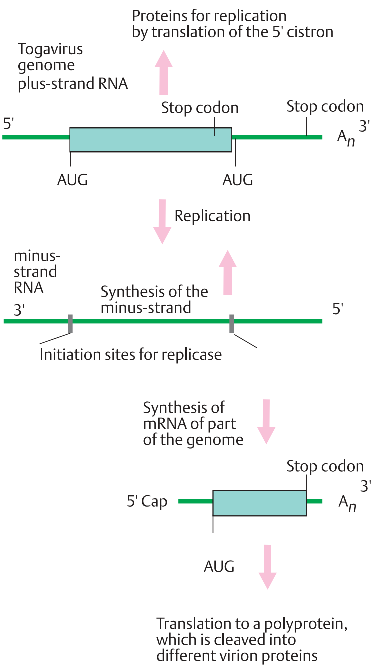
E. Transcription and translation of a minus-strand RNA virus

RNA viruses with minus-strand genomes (e.g., rhabdoviruses, myxoviruses) must first form an RNA plus strand by means of a virus-coded replicase contained in the viral particle. This serves as a template for the formation of a new genome (replication) and for mRNAs (transcription) for the synthesis of virus-coded proteins (translation). The new viral genomes are then packaged to form virions.

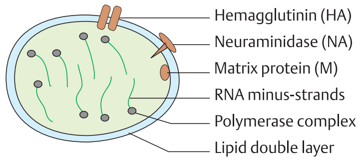
(Figures adapted from Watson et al., 1987).



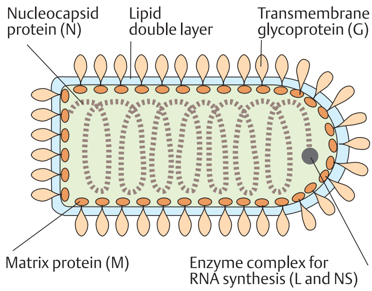
A. Genome of poliovirus and its translation products



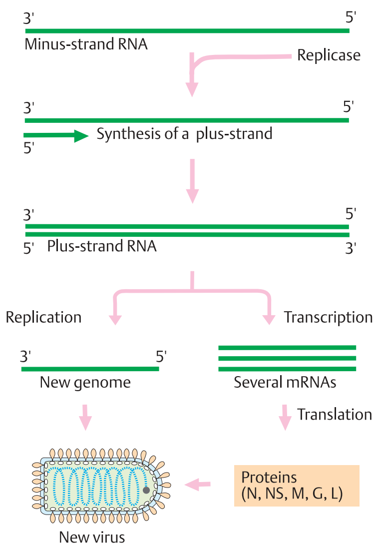
B. Togavirus: replication and translation



C. Influenzavirus



D. Rhabdovirus (vesicular stomatitis virus, VSV)



E. Transcription and translation of a minus-strand RNA virus

DNA Viruses

DNA viruses depend on host-cell DNA synthesis for their replication. Since the cellular proteins required for DNA synthesis are only present during the S phase of the cell cycle, most DNA viruses induce DNA synthesis in the host cell. A few contain genes for their own DNA polymerases and other proteins for DNA synthesis. Their genes are expressed in a well-defined chronological pattern. Important DNA viruses include herpesviruses and poxviruses.

A. SV40 virus

The SV40 (simian virus 40) virus has a double-stranded circular DNA genome (5243 base pairs) and belongs to the papovaviruses. The protein coat of the virion consists of three virus-coded proteins (VP1, VP2, VP3). An early and a late region of the viral genome can be distinguished according to their time of expression. Between them lies a regulative region. Two alternatively spliced mRNAs of the early region code for the large tumor antigen (T, ~ 90 kilodaltons or kDa) and the small tumor antigen (t, ~20 kDa). They regulate further transcription, the initiation of replication, and the expression of specific cellular genes. The late region, coding for the capsid proteins, is transcribed at the onset of viral DNA synthesis in the opposing DNA strand and in the opposite direction. The late proteins are translated from differently spliced, overlapping mRNAs. A fourth, late gene codes for a small protein (agno protein) of unknown function. Replication of the viral genome DNA begins at a defined point (OR, *o*rigin of *r*eplication).

B. Adenovirus

The adenovirus genome (1) consists of a linear double-stranded DNA of about 36 000 base pairs (36 kb). Both ends contain a repetitive nucleotide sequence (inverted terminal repeat), which is important for DNA replication. Early transcripts (E), which appear about 2 – 3 hours after infection and 6 – 8 hours before the onset of viral DNA replication, are transcribed at specific regions. The region E1a codes for proteins that initiate transcription of all other viral genes and that influence the expression of specific cellular genes. The E2 region codes for proteins directly involved in DNA replication, including a viral DNA polymerase (adenovirus-specific replication mechanism). A single promoter controls transcription of the late region (L), which is transcribed into a large primary RNA (2). From this, at least 20 different mRNAs are produced by alternative splicing. All mRNAs have the same 5′ terminus. Unlike the early transcripts, the late transcripts are encoded by only one DNA strand. The late genes code mainly for viral coat proteins.

C. Parvovirus

Parvoviruses are small, single-stranded DNA viruses that can replicate only in proliferating cells. Replication of the single DNA strand is initiated (self-priming) by the formation of a specific genomic structure (hairpin structure).

D. Herpesvirus genome

Herpesviruses and poxviruses are large viruses with DNA genomes of 80 – 200 kb that contain repetitive sequences. They code for many (50 – 200) different proteins, including their own DNA polymerase. Other proteins interfere with the regulation of the cell's nucleotide metabolism. The group of herpesviruses includes the Epstein – Barr virus, the varicellazoster virus, and the cytomegalovirus. (Figures after Watson et al., 1987).

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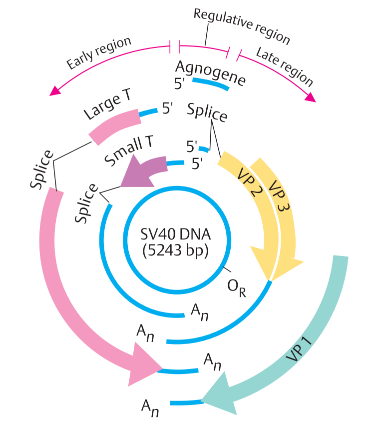
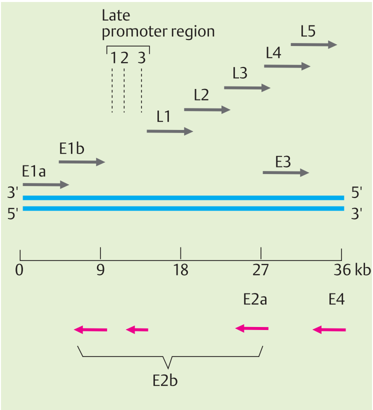
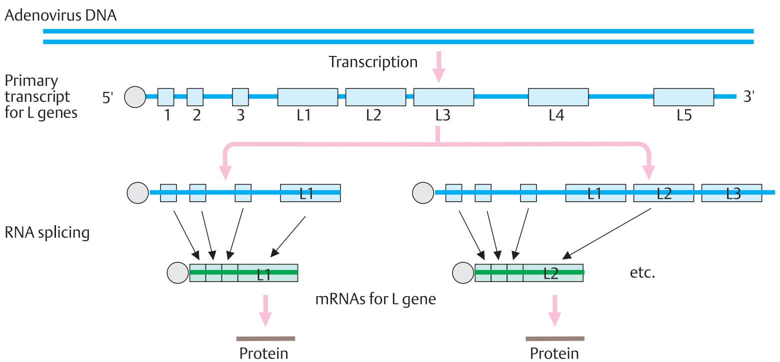
Watson, J.D. et al.: Molecular Biology of the Gene, 3rd ed. Benjamin/Cummings Publishing Co., Menlo Park, California, 1987.



A. SV40 genome and transcription

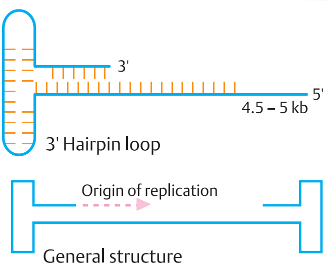


1. Adenovirus genome

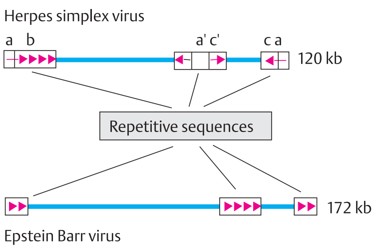


2. Adenovirus transcription and RNA splicing

B. Adenovirus



C. Parvovirus



D. Herpesvirus genome

Retroviruses

Retroviruses are RNA viruses with a developmental cycle in which double-stranded DNA is transcribed from the viral RNA genome and integrated into the genome of the host. Specific sequences of their genomes enable integrated proviruses to become integrated into new sites. When this happens, neighboring cellular sequences are occasionally transported along to a new region of the host genome or to the genome of another cell (retrotransposon). Important retroviruses are the AIDS virus (HIV I) and certain tumor viruses (see p. 314).

A. Retrovirus replication

After attaching to a cell surface receptor, the virion is taken into the cell. Immediately after entry into the cell, the viral RNA genome is transcribed into double-stranded DNA by an enzyme complex called reverse transcriptase, and the new DNA is integrated as a provirus into the DNA of the host cell. The RNAs transcribed from this DNA copy by cellular RNA polymerase II serve either as mRNA for the synthesis of virus proteins or as new virus genomes. Newly formed virions leave the cell by a specific process called exocytosis, without killing the cell.

B. Genomes of the retroviruses

Several retroviruses cause tumors in mice (mouse leukemia) or chickens (Rous sarcoma). The only retrovirus identified to cause a tumor in humans is HTLV (type 1 and 2) (human T cell leukemia/lymphoma virus). HTLV viruses have genetic similarities with the AIDS virus, which causes acquired immune deficiency syndrome (p. 314).

The RNA genome of a typical retrovirus contains short repetitive (R) and unique (U) sequences at both ends (RU5 at the 5′ end and U3R at the 3′ end). As a rule, three protein-coding genes lie between them: *gag* (group-specific antigen), *pol* (reverse transcriptase), and *env* (a gene that codes for a glycoprotein that is built into the lipid membrane coat of the virion). The 5′ end of the genome contains a nucleotide sequence that is complementary to the 3′ end of a host cell tRNA. This nucleotide sequence binds to tRNA, which serves as a primer for the synthesis by reverse transcriptase of viral DNA from the virus genome RNA. At the 3′ end of the genome, HTLV viruses carry several other genes (*px*, *lor*, *tat*, and others) that are involved in regulating the transcription of the viral genes. The AIDS virus (HIV I/II) contains further genes that do not occur in other retroviruses. These include *vif* (*v*irion *i*nfectivity *f*actor, formerly designated *sor*), *rev* (*r*egulator of *e*xpression of *v*irion proteins), and *nef* (*ne*gative *f*actor, formerly *orf*). The general structure of the genome is more complex than shown here.

C. DNA synthesis of a retrovirus

Reverse transcriptase transcribes the RNA genome into DNA by RNA-dependent DNA polymerase activity and catalyzes subsequent steps by means of DNA-dependent DNA polymerase activity. Furthermore, the reverse transcriptase has RNAase activity for degrading the RNA of the newly formed RNA/DNA hybrid molecule. The first step in replicating the retrovirus genome is initiated by a primer of host cell tRNA, which is hybridized to the 5′ end of the RNA genome (1). After synthesis of the first DNA strand and removal of the tRNA primer, synthesis of the second DNA strand (plus strand) begins at the RU5 region (2). Here, the previously formed minus DNA strand serves as the primer (3). As DNA synthesis is continued, the remaining RNA is degraded (4), and the DNA plus strand is synthesized to completion (5, 6). The double-stranded DNA copy of the virus contains long terminal repeats (LTR) at both ends. These enable the viral DNA intermediary step to be integrated into the host cell DNA, and they contain the necessary regulatory sequences for transcribing the provirus DNA (see p. 102). (Figures adapted from Watson et al., 1987).

Reference

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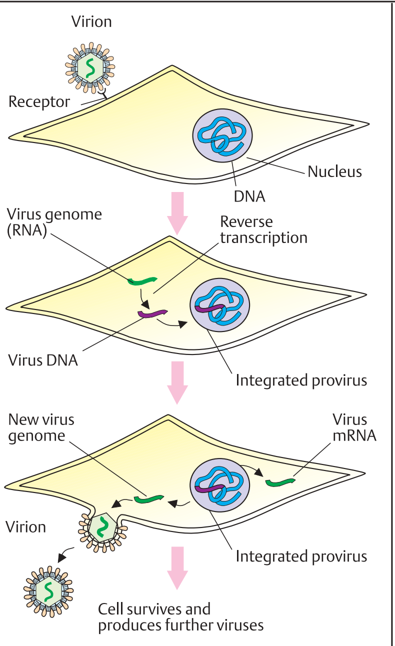
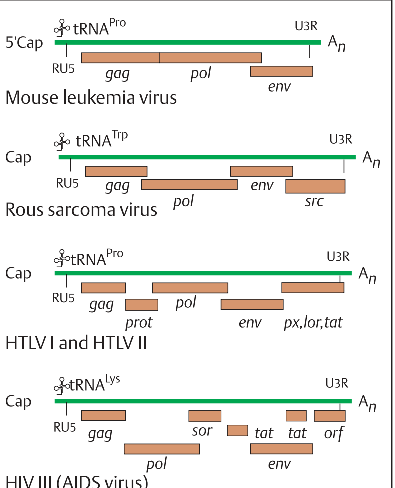
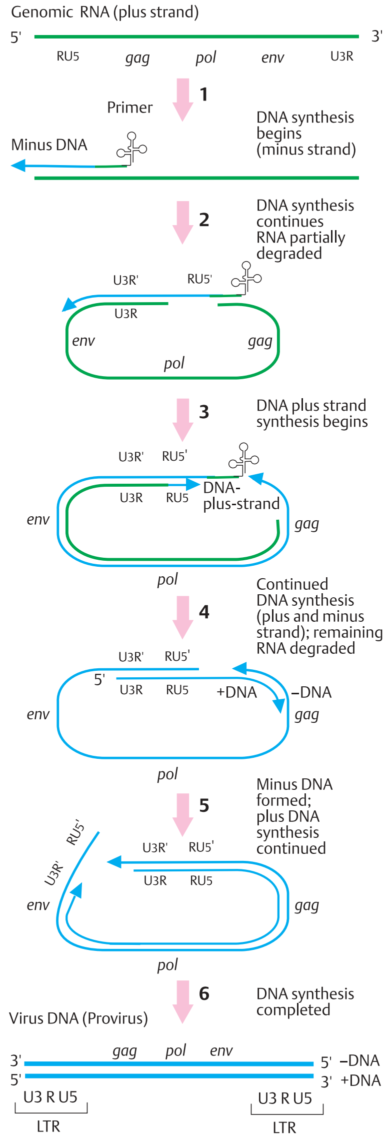


A. Retrovirus replication



B. Genomes of some retroviruses



C. DNA synthesis of a retrovirus by reverse transcriptase

Retrovirus Integration and Transcription

Integration of the DNA copy of a retrovirus into the host cell DNA occurs at a random location. This may alter cellular genes (insertion mutation). The viral genes of provirus DNA are transcribed by cellular RNA polymerase II. The resulting mRNA serves either for translation or for the production of new RNA genomes, which are packaged into the virion. Some retrovirus genomes may contain an additional viral oncogene (v-*onc*). Viral oncogenes are parts of cellular genes (c-*onc*) previously taken up by the virus. If they enter a cell with the virus, they may change (transform) the host cell so that its cell cycle is altered and the cell becomes the origin of a tumor (tumor virus) (see p. 320).

A. Retrovirus integration into cellular DNA

In the nucleus of the host cell, the double-stranded DNA (1, virus DNA) produced from virus genome RNA first forms a ring-shaped structure (2) by joining LTRs (long terminal repeats). This is possible because the LTRs contain complementary nucleotide sequences. Recognition sequences in the LTRs and in the cellular DNA (3) allow the circular viral DNA to be opened at a specific site (4) and the viral DNA to be integrated into the host DNA (5). Viral genes can then be transcribed from the integrated provirus (6). As a rule, the provirus remains in the genome of the host cell without disrupting the functions of the cell. One exception is the AIDS virus, which destroys a specific population of lymphocytes (helper T cells) (see p. 310). The genomes of vertebrates (including man) contain numerous DNA sequences that consist of endogenous proviruses. In mice, they may represent up to 0.5% of the total DNA. The genomes of higher organisms also contain LTR-like sequences that are very similar to those of an endogenous retrovirus. These sequences can change their location in the genome (mobile genetic elements or transposons). Since many of them have the fundamental structure of a retrovirus (LTR genes), they are designated retrotransposons (see pp. 76 and 252).

B. Control of retrovirus transcription

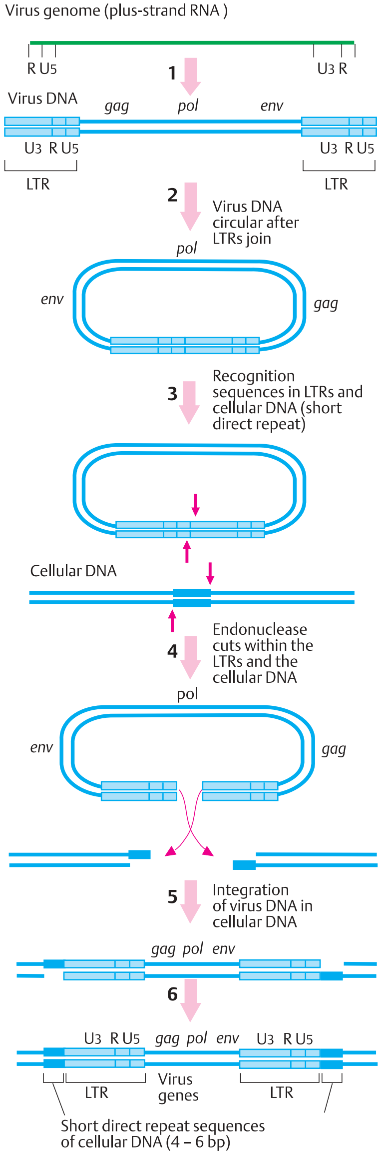
The LTRs are important not only for integration of the virus into cellular DNA, but also because they contain all regulatory signals necessary for efficient transcription of a viral gene. Typical transcription signals are the so-called “CCAAT” and “TATA” sequences of promoters, which are respectively located about 80 and 25 base pairs above the 5′ end of the sequence to be transcribed. Further upstream (in the 5′ direction) are nucleotide sequences that can increase the expression of viral genes (enhancer). Similar regulatory sequences are located at the 5′ ends of eukaryotic genes (see section on gene transcription). Newly synthesized viral RNA is structurally modified at the 5′ end (formation of a cap). Furthermore, numerous adenine residues are added at the 3′ end (polyadenylated, poly(A), see p. 214).

C. Virus protein synthesis by posttranscriptional modification of RNA

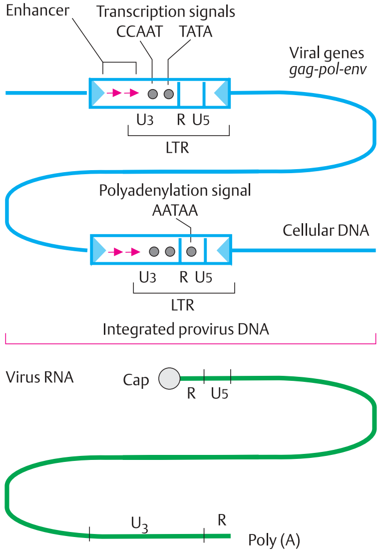
The RNA transcripts synthesized from the provirus by cellular RNA polymerase II serve either for the translation into virus-coded proteins (*gag*, *pol*) or as the genome for the formation of new virions. Some of the RNA is spliced to form new mRNAs that code for coat proteins (*env*). (Figures after Watson et al., 1987).

Reference

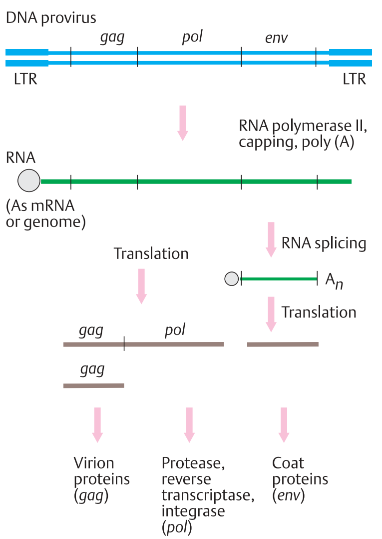
Watson, J.D. et al.: Molecular Biology of the Gene, 3rd ed. Benjamin/Cummings Publishing Co., Menlo Park, California, 1987.



A. Retrovirus integration into cellular DNA



B. Control of retrovirus transcription



C. Virus protein synthesis by post-transcriptional modification of RNA

Eukaryotic Cells

Yeast: Eukaryotic Cells with a Diploid and a Haploid Phase

Yeast is a single-celled eukaryotic fungus with a genome of individual linear chromosomes enclosed in a nucleus and with cytoplasmic organelles such as endoplasmic reticulum, Golgi apparatus, mitochondria, peroxisomes, and a vacuole analogous to a lysosome. About 40 different types of yeast are known. Baker's yeast, *Saccharomyces cerevisiae,* consists of cells of about 3 μm diameter that are able to divide every 90 minutes under good nutritional conditions. Nearly half of the proteins known to be defective in human heritable diseases show amino acid similarity to yeast proteins.

The haploid genome of *S. cerevisiae,* contains ca. 6200 genes in 1.4×10^7 DNA base pairs in 16 chromosomes, about 50% of the genes being without known function (Goffeau, 1996). For about 6200 proteins the following functions have been predicted: cell structure 250 (4%), DNA metabolism 175 (3%), transcription and translation 750 (13%), energy production and storage 175 (3%), biochemical metabolism 650 (11%), and transport 250 (4%). The *S. cerevisiae* genome is very compact compared to other eukaryotic genomes, with about one gene every 2 kb.

A. Yeast life cycle through a haploid and a diploid phase

Yeast can grow either as haploid or as diploid cells. Haploid cells of opposite types can fuse (mate) to form a diploid cell. Haploid cells are of one of two possible mating types, called **a** and α. The mating is mediated by a small secreted polypeptide called a pheromone or mating factor. A cell-surface receptor recognizes the pheromone secreted by cells of the opposite type, i.e., **a** cell receptors bind only α factor and α cell receptors bind only **a** factor. Mating and subsequent mitotic divisions occur under favorable conditions for growth. Under starvation conditions, a diploid yeast cell undergoes meiosis and forms four haploid spores (sporulation), two of type **a** and two of type α.

B. Switch of mating type

A normal haploid yeast cell switches its mating type each generation. The switch of mating type (mating-type conversion) is initiated by a double-strand break in the DNA at the MAT locus (recipient) and may involve the boundary to either of the flanking donor loci (HMR or HML). This is mediated by an HO endonuclease through site-specific DNA cleavage.

C. Cassette model for mating type switch

Mating type switch is regulated at three gene loci near the centromere (cen) of chromosome III of *S. cerevisiae.* The central locus is MAT (mating-type locus) which is flanked by loci HMLα (left) and HMRa (right). Only the MAT locus is active and transcribed into mRNA. Transcription factors regulate other genes responsible for the **a** or the α phenotype. The HMLα and HMRa loci are repressed (silenced). DNA sequences from either the HMLα or the HMLa locus are transferred into the MAT locus once during each cell generation by a specific recombination event called gene conversion. The presence of HMRa sequences at the MAT locus determines the **a** cell phenotype. When HMLα sequences are transferred (switch to an α cassette), the phenotype is switched to α. Any gene placed by recombinant DNA techniques near the yeast mating-type silencer is repressed, even a tRNA gene transcribed by RNA polymerase III, although it uses different transcription factors. Apparently the HML and HMR loci are permanently repressed because they are inaccessible to proteins (transcription factors and RNA polymerase) owing to the condensed chromatin structure near the centromere.

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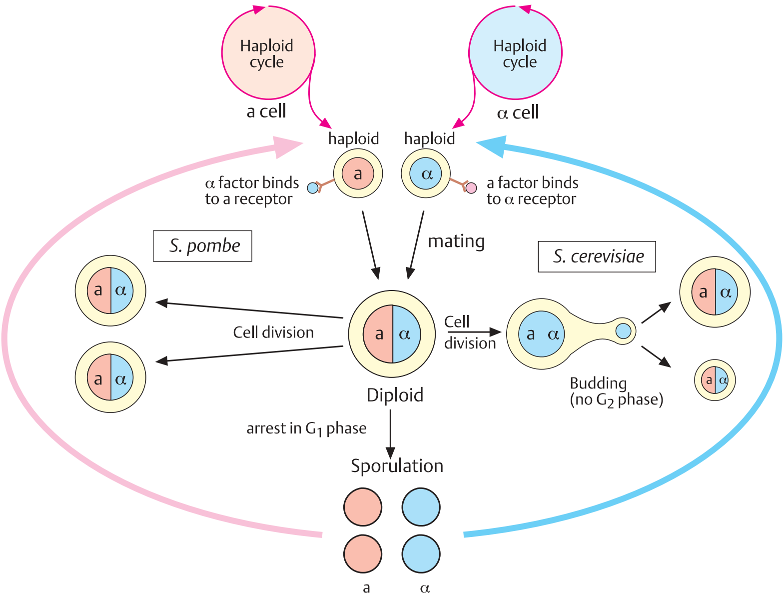
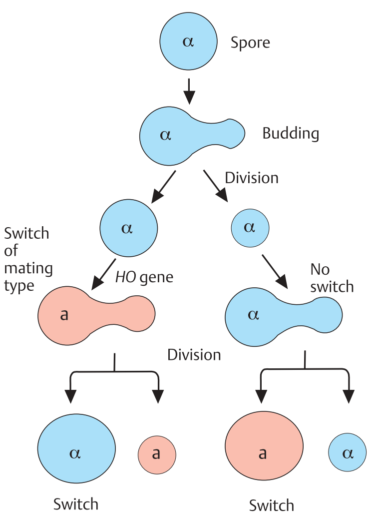
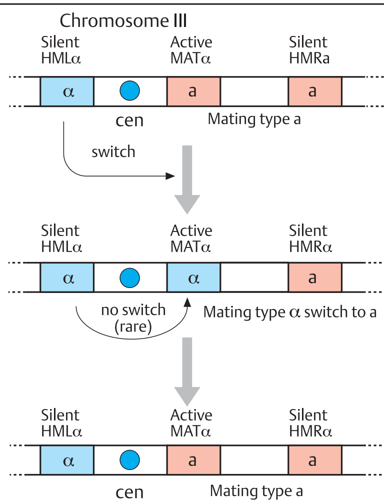


A. Yeast life cycle through a haploid and a diploid phase



B. Switch of mating type



C. Cassette model for mating type switch

Mating Type Determination in Yeast Cells and Yeast Two-Hybrid System

Yeast cells *(S. cerevisiae)* are unicellular eukaryotes with three different cell types: haploid **a** and α cells and diploid **a**/α cells. Owing to their relative simplicity compared with multicellular animals and plants, yeast serves as a model for understanding the underlying control mechanisms specifying cell types. The generation of many different cell types in different tissues of multicellular organisms probably evolved from mechanisms that determine cell fate in unicellular organisms such as yeast.

A. Regulation of cell-type specificity in yeast

Each of the three *S. cerevisiae* cell types expresses cell-specific genes. The resulting differences in combinations of DNA-binding proteins determine the cell-type specification. These regulatory proteins are encoded at the MAT locus in combination with a general transcription factor called Mcm1. Mcm1 is expressed in all three cell types. Cells of type **a** express **a**-specific genes, but not α-specific genes. In diploid (**a**/α) cells, diploid-specific genes are expressed. The cell-specific transcription factors are **a**1, α1, and α2, all encoded at the MAT locus. Mcm1 is a dimeric general transcription factor that binds to **a**-specific upstream regulatory sequences (URSs). This stimulates transcription of the **a**-specific genes, but it does not bind too efficiently to the α-specific URSs when α1 protein is absent. In α cells two specific transcription factors, α1 and α2, mediate transcriptional activity. α2-binding sequences associate with MCM helicase and block transcription of **a**-specific genes. α1-binding sequences form a complex with MCM and stimulate transcription of α-specific genes. In diploid cells (**a**/α) the haploid genes are repressed by α2-MCM1 and by α2/**a**1 complexes. In summary, each of the three yeast cell types is determined by a specific combination of transcription factors acting as activators or as repressors depending to which specific regulatory sites they bind.

B. Yeast two-hybrid system

The problem of determining the function of a newly isolated gene may be approached by determining whether its protein specifically reacts with another protein of known function. Yeast cells can be used in an assay for protein–protein interactions. The two-hybrid method rests on the observation of whether two different proteins, each hybridized to a different protein domain required for transcription factor activity, are able to interact and thereby reassemble the transcription factor. When this occurs, a reporter gene is activated. Neither of the two hybrid proteins alone is able to activate transcription. Hybrid 1 consists of protein X, the protein of interest (the "bait") attached to a transcription factor DNA-binding domain (BD). This fusion protein alone cannot activate the reporter gene because it lacks a transcription factor activation domain (AD). Hybrid 2, consisting of a transcription factor AD and an interacting protein, protein Y (the "prey"), lacks the BD. Therefore, hybrid 2 alone also cannot activate transcription of the reporter gene. Different ("prey") proteins expressed from cDNAs in vectors are tested. Fusion genes encoding either hybrid 1 or hybrid 2 are produced using standard recombinant DNA methods. Cells are cotransfected with the genes. Only cells producing the correct hybrids, i.e., those in which the X and Y proteins interact and thereby reconnect AD and BD to form an active transcription factor, can initiate transcription of the reporter gene. This can be observed as a color change or by growth in selective medium. (Figure adapted from Oliver, 2000, and Frank Kaiser, Essen, personal communication.)

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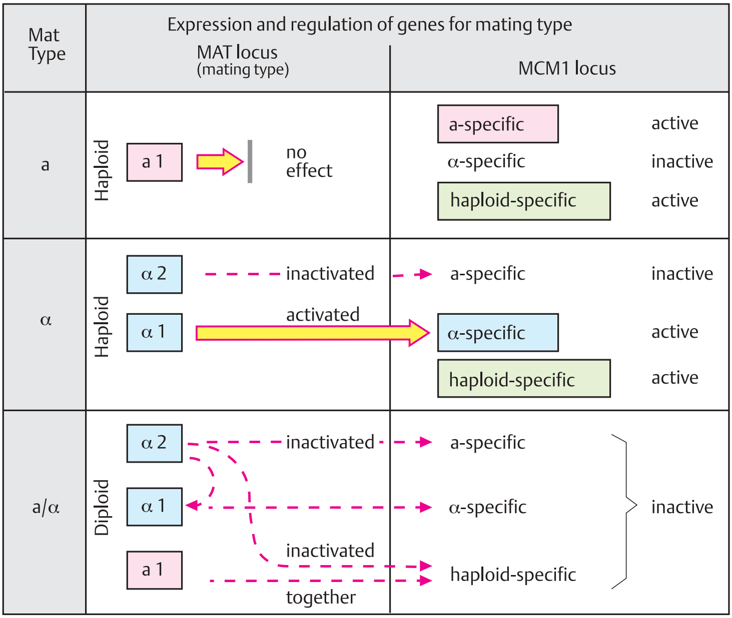
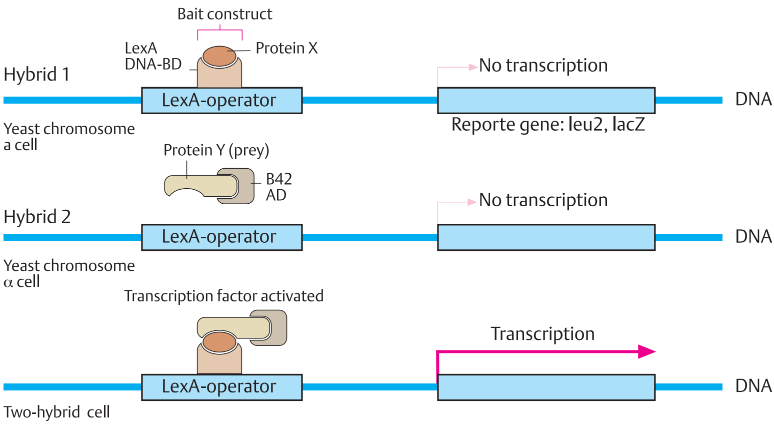


A. Regulation of cell-type specificity in yeast



B. Yeast two-hybrid sytem

Functional Elements in Yeast Chromosomes

The haploid genome of *S. cerevisiae* consists of 1.4×10^7 base pairs in 16 chromosomes. Several functional DNA sequences have been isolated and characterized from yeast chromosomes that are necessary for replication and normal distribution of the chromosomes during mitosis (mitotic segregation). If they lie on the same chromosome, they are designated *cis* elements (as opposed to *trans* on another chromosome). They are genetically stable and can be joined in vitro to DNA of any origin to form stable artificial chromosomes in yeast cells (see p. 104). Three types of functional DNA sequences are known: autonomous replicating sequences (ARS), centromere sequences (CEN), and telomere sequences (TEL). All eukaryotic chromosomes contain these functional elements, but they can be especially well demonstrated in yeast cell chromosomes.

A. Autonomous replicating sequences (ARS)

Replication proceeds in two directions (bidirectional). The ARS represent the starting points of replication in chromosomes. Their functional significance can be recognized in a transformation experiment: A mutant yeast cell (e.g., with the inability to produce leucine, Leu^-) can be transformed into a leucine-producing (Leu^+) cell (1) by means of transfection with cloned plasmids containing the gene for leucine formation *(Leu)*. These cells only are able to grow to medium lacking leucine. However, the daughter cells remain leucine-dependent because the plasmid cannot replicate. If the plasmid contains autonomously replicating sequences (ARS) along with the *Leu* gene, a small fraction of the daughter cells will be Leu^+ because plasmid replication has occurred (2). But most of the daughter cells remain Leu^- because mitotic distribution (segregation) is defective.

B. Centromere sequences (CEN)

If the plasmid DNA (1) contains sequences from the centromere (CEN) of the yeast chromosome along with the gene for leucine *(Leu)* and the autonomously replicating sequences (ARS), then normal mitotic segregation takes place (2). This demonstrates that the CEN sequences are necessary for normal distribution of the chromosomes at mitosis (see p. 114).

The centromere sequences are similar but not identical in different yeast chromosomes. They contain three elements with a total of about 220 base pairs (bp), which occur in all chromosomes. Element I is a conserved sequence of 8 bp; element II is AT rich and has about 80 bp; element III has about 25 bp. This segment is nuclease-protected and very important for mitotic stability.

C. Telomere sequences (TEL)

If the plasmid is linear instead of circular as in B, transformation will occur, but the plasmid does not replicate (2). However, if telomere sequences (TEL) are attached to both ends of the plasmid (3) before it is incorporated (4), then transformation of the yeast cell after incorporating the plasmid is followed by normal replication and mitosis (5).

These observations prove that autonomously replicating sequences (starting point of replication), centromere sequences, and telomere sequences (see p. 180) are necessary for normal chromosome function. These functional elements (ARS, CEN, TEL) can be utilized to produce artifical yeast chromosomes (YACs). (Figures adapted from Lodish et al., 2000, p. 330).

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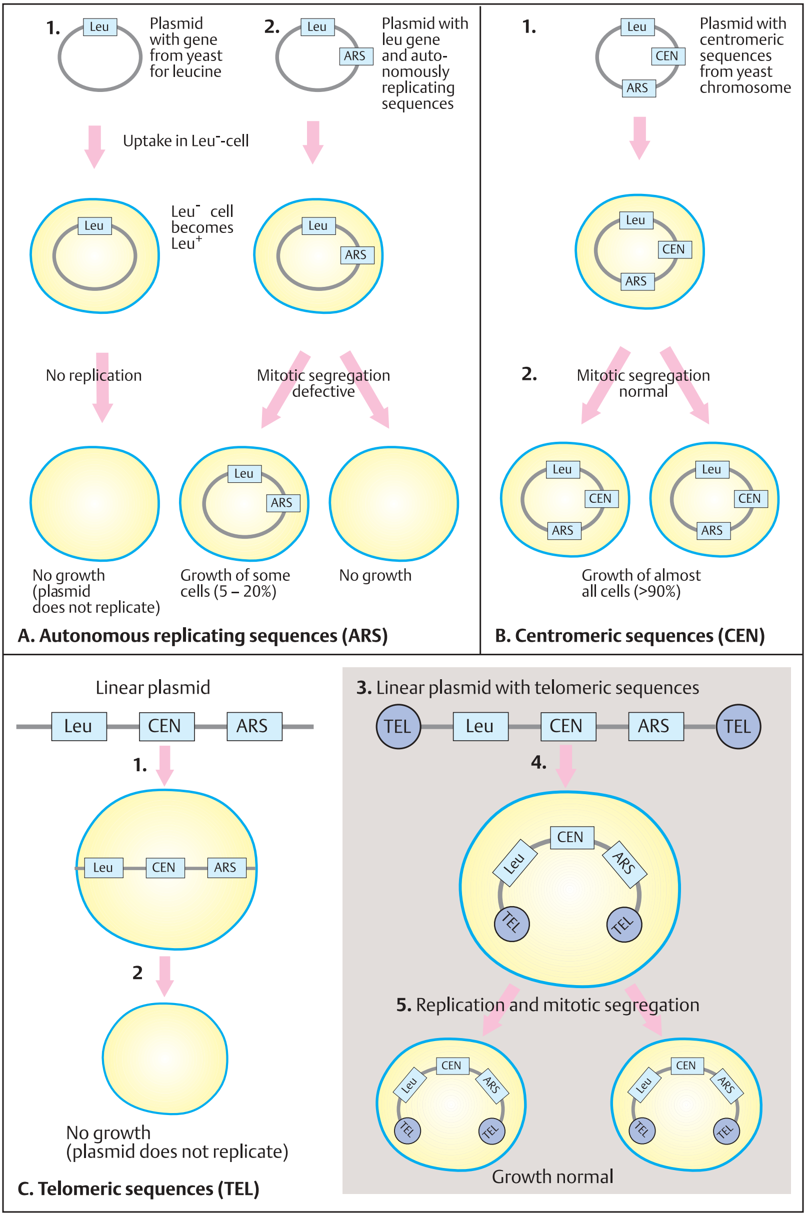

A. Autonomous replicating sequences (ARS)

B. Centromeric sequences (CEN)

C. Telomeric sequences (TEL)

Artificial Chromosomes for Analyzing Complex Genomes

The mapping of large and complex genomes (for instance in humans with three billion base pairs) would not be possible if cloned fragments of only a few dozen to a few thousand base pairs were available (with cosmids, up to about 50 kb). Large distances could not be bridged because nonclonable DNA segments would lie between the mapped segments. Since physical mapping of these segments is not possible, attempts were made to develop vectors for cloning large DNA fragments. Artificial chromosomes can be constructed from yeast chromosomes (YACs, yeast artificial chromosomes). These can incorporate DNA fragments of about 100–1000 kb). Since they are stable during growth, they represent an important instrument for gene mapping.

A. Construction of yeast artificial chromosomes (YACs) for cloning and mapping

The starting material is a linear vector. This consists of telomere sequences (TEL) and selectable markers, here leucine (LEU) and tryptophan (TRP), a starting point for replication (autonomously replicating sequences, ARS), and a centromere (CEN). The vector is incised and the DNA fragment to be replicated (foreign DNA) is inserted. The capacity of a YAC is about 750 kb. Subsequently, the incorporated DNA can be cloned in the artificial chromosome in yeast cells and amplified. In this way, a YAC library containing the entire genome of a complex organism can be produced in a manageable manner. From this, a physical map of large and overlapping fragments (contig) can be constructed. Increasingly, artificial mammalian chromosomes (mammalian artifical chromosomes, MACs) are also being developed. These consist of one arm of a YAC with a selectable marker, a replication starting point, and a telomere attached to a fragment of mammalian DNA with its own telomere.

B. YAC vector modification

Different methods can be used to modify YACs. Sequences can be incorporated so that, with their help, both ends of the YAC can be recovered as plasmids in *E. coli* (1). As a prerequisite, plasmid sequences with selectable markers (2), here for ampicillin resistance (Amp^R), neomycin resistance (Neo^R), and onset of plasmid replication (ORI), are included. Digestion with an appropriate restriction enzyme produces fragments of the incorporated foreign DNA and of both vector arms (3). The fragments are formed into a ring and incorporated into *E. coli*. The hybrid plasmids (4) can then be selected for (5) by means of the markers (Amp^R and Neo^R).

The use of YACs has considerably simplified the genetic analysis of large genomes. While a complete human-genome library requires about 500 000 clones of lambda phage vectors, a YAC library with 150 kb fragments reduces the number of clones to about 60 000. (Figures adapted from Schlessinger, 1990).

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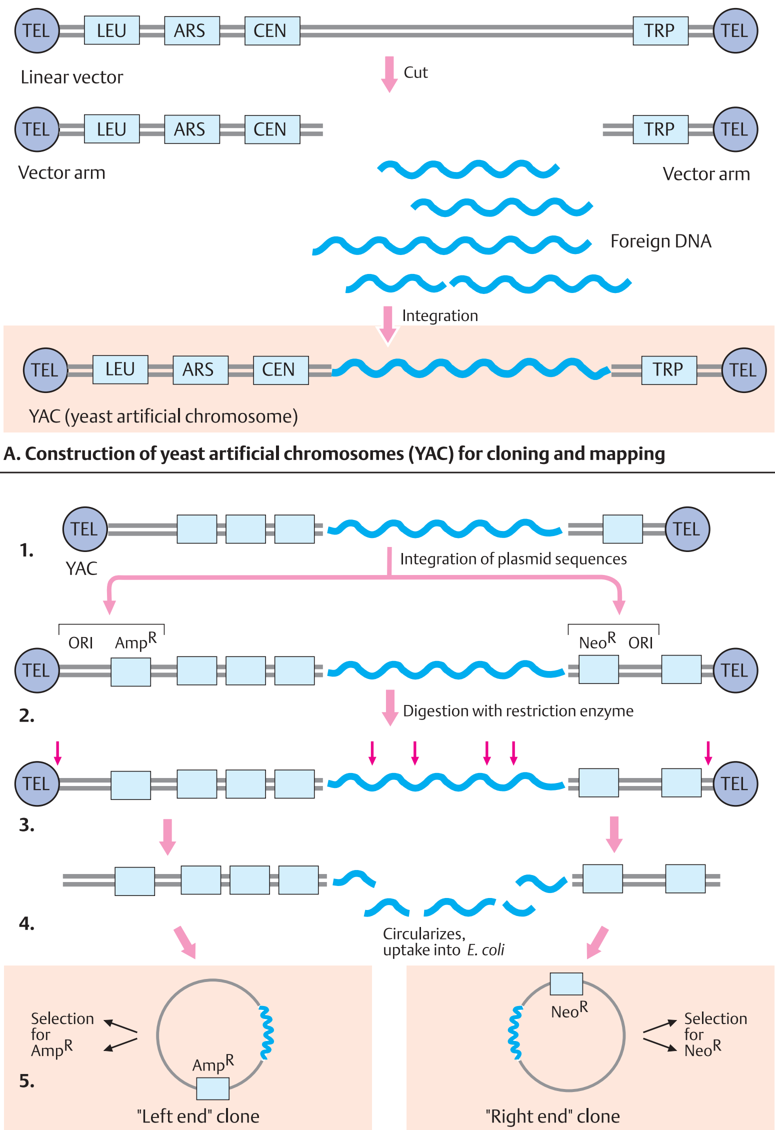

A. Construction of yeast artificial chromosomes (YAC) for cloning and mapping

B. YAC vector modification

Cell Cycle Control

The growth of multicellular organisms depends on precise replication of individual cells followed by cell division. During replication, eukaryotic cells go through an ordered series of cyclical events. The time from one cell division to the next is called a cell cycle. The cell cycle has two main phases, interphase and mitosis. Interphase is further divided into three distinct phases: G_1 (gap 1), S (DNA synthesis, lasting 6–8 hours in eukaryotic cells, at the end of which the chromosomes have been duplicated), and G_2 (gap 2, lasting about 4 hours). Mitosis (M, see p. 114) is the phase of actual division. Cell cycle control mechanisms, which include complex sets of interacting proteins, guide the cell through its cycle by regulating the sequential cyclical events. These are coordinated with extracellular signals and result in cell division at the right time.

A. Cell division cycle models in yeast

Budding yeast (baker's yeast) divides by mitotic budding to form one large and one small daughter cell. Since a microtubule mitotic spindle forms very early during the S phase, there is practically no G_2 phase (1). In contrast, fission yeast *(S. pombe)* forms a mitotic spindle inside the nucleus at the end of the G_2 phase, then proceeds to mitosis to form two daughter cells of equal size (2). Unlike in vertebrate cells, the nuclear envelope remains intact during mitosis. An important regulator is cdc2 (cell division cycle, *S. pombe*) (3). Absence of cdc2 activity ($cdc2^-$ mutant) results in cycle delay and prevents entry into mitosis. Thus, too large a cell with only one nucleus results. Increased activity of cdc2 (dominant mutant cdc^D) results in premature mitosis and cells that are too small (wee phenotype, from the Scottish word for small). Normally a yeast cell has three options: (a) halt the cell cycle if the cell is too small or nutrients are scarce, (b) mate (see p. 104), or (c) enter mitosis. (Figure adapted from Lodish et al., 2000).

B. Cell cycle control systems

The eukaryotic cell cycle is driven by cell cycle "engines", a set of interacting proteins, the cyclin-dependent kinases (Cdks). An important member of this family of proteins is cdc2 (also called Cdk1). Other proteins act as rate-limiting steps in cell cycle progression and are able to induce cell cycle arrest at defined stages (checkpoints). The cell is induced to progress through G_1 by growth factors (mitogens) acting through receptors that transmit signals to proceed towards the S phase. D-type cyclins (D1, D2, D3) are produced, which associate with and activate Cdks (4 and 6). Other proteins can induce G_1 arrest. If these proteins are inactive owing to mutations, cell proliferation becomes uncontrolled as in many forms of cancer. The detection of DNA damage and subsequent cell cycle arrest due to activated p53 is an important mechanism for preventing the cell from entering the S phase.

In early G_1 phase cdc2 is inactive. It is activated in late G_1 by association with G_1 cyclins, such as cyclin E. Once the cell has passed the G_1 restriction point, cyclin E is degraded and the cell enters the S phase. This is initiated, among many other activities, by cyclin A binding to Cdk2 and phosphorylation of the RB protein (retinoblastoma protein, see p. 330). The cell passes through the mitosis checkpoint only if no damage is present. Cdc2 (Cdk1) is activated by association with mitotic cyclins A and B to form the mitosis-promoting factor (MPF).

During mitosis, cyclins A and B are degraded, and an anaphase-promoting complex forms (details not shown). When mitosis is completed, cdc2 is inactivated by the S-phase inhibitor Sic1 in yeast. At the same time the retinoblastoma (RB) protein is dephosphorylated. Cells can progress to the next cell cycle stage only when feedback controls have ensured the integrity of the genome. (This figure is an overview only; it omits many important protein transactions).

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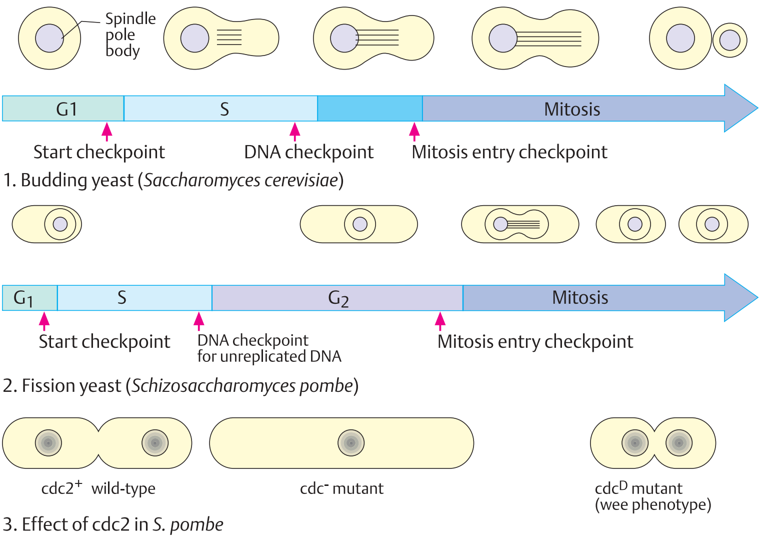
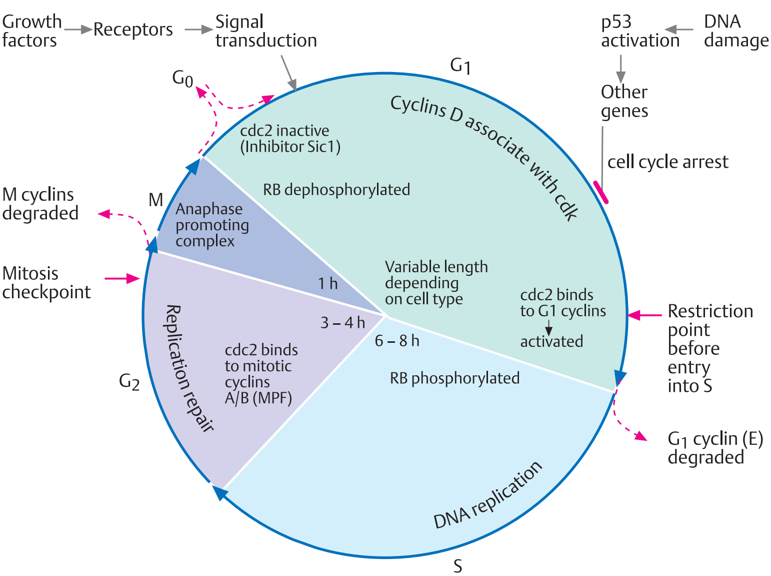


A. Cell division cycle models in yeast



B. Cell cycle control systems

Cell Division: Mitosis

Threadlike structures in dividing cells were first observed by Walter Flemming in 1879. He introduced the term mitosis for cell division. Flemming also observed the longitudinal division of chromosomes during mitosis. In 1884, Strasburger coined the terms prophase, metaphase, and anaphase for the different stages of cell division. A mitosis results in two genetically identical daughter cells.

A. Mitosis

During the transition from interphase to mitosis, the chromosomes become visible as elongated threads (prophase). In early prophase, each chromosome is attached to a specific site on the nuclear membrane and appears as a double structure (sister chromatids), the result of the foregoing DNA synthesis. The chromosomes contract during late prophase to become thicker and shorter (chromosomal condensation). In late prophase, the nuclear membrane disappears and metaphase begins. At this point, the mitotic spindle becomes visible as thin threads. It begins at two polelike structures (centrioles). The chromosomes become arranged on the equatorial plate, but homologous chromosomes do not pair. In late metaphase during the transition into anaphase, the chromosomes divide also at the centromere region. The two chromatids of each chromosome migrate to opposite poles, and telophase begins with the formation of a nuclear membrane. Finally the cytoplasm also divides (cytokinesis). In early interphase the individual chromosomal structures become invisible. Interphase chromosomes are called chromatin (Flemming 1879), i.e., nuclear structures stainable by basic dyes.

B. Metaphase chromosomes

Waldeyer (1888) coined the term chromosome for the stainable threadlike structures visible during mitosis. A metaphase chromosome consists of two chromatids (sister chromatids) and the centromere, which holds them together. The centromere may divide each of the chromatids into two chromosome arms. The regions at both ends of the chromosome are the telomeres. The point of attachment to the mitotic spindle fibers is the kinetochore. During metaphase and prometaphase, chromosomes can be visualized under the light microscope as discrete elongated structures, 3 – 7 μm long (see p. 182).

C. Cell cycle (mitosis)

Eukaryotic cells go through cell division cycles (cell cycle). In eukaryotic cells, each cell division begins with a phase of DNA synthesis, which lasts about 8 hours (S phase). This is followed by a phase of about 4 hours (G_2). During the G_2 phase (gap 2), the whole genome is double. Mitosis (M) in eukaryotic cells lasts about an hour (see A). This is followed by a phase, G_1 (interphase), of extremely varied duration. It corresponds to the normal functional phase. Cells that no longer divide are in the G_0 phase. Every cell must have a "memory" to tell it whether it is in G_1 or in G_2, since division of the chromosomes before they have been doubled would be lethal. All phases of the cell cycle are regulated by specific proteins encoded by numerous cell division cycle (cdc) genes. In particular, the transition from G_1 to S and from G_2 to M is regulated by specific cell-cycle proteins (see preceding plate).

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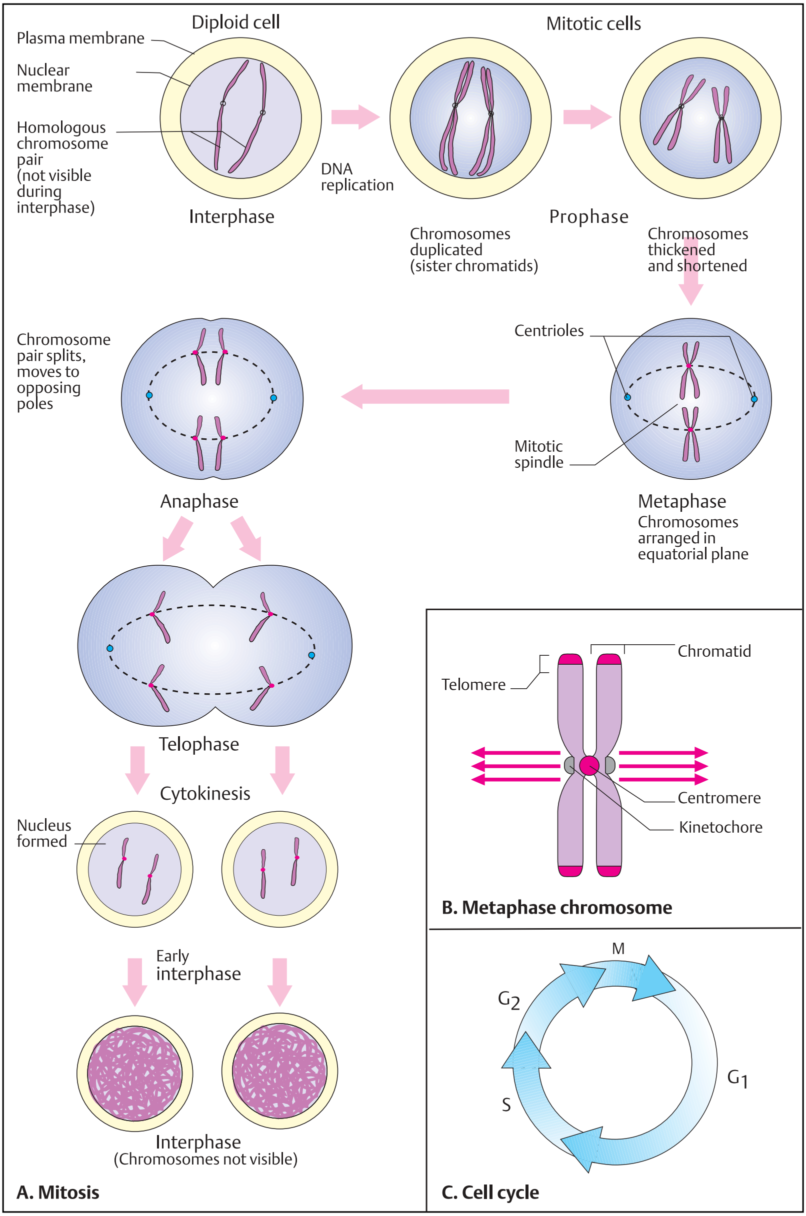

A. Mitosis

B. Metaphase chromosome

C. Cell cycle

Maturation Division (Meiosis)

Gametes are formed by a special type of cell division that differs from somatic cell division. Strasburger (1884) introduced the term meiosis (reduction division or maturation division) for this process.

Meiosis differs from mitosis in fundamental genetic and cytological respects. Firstly, homologous chromosomes pair up. Secondly, exchanges between homologous chromosomes occur regularly (crossing-over). This results in chromosome segments with new constitutions (genetic recombination). Thirdly, the chromosome complement is halved during the first cell division (meiosis I). Thus, the daughter cells resulting from this division are haploid (reduction division).

Meiosis is a complex cellular and biochemical process. The cytologically observable course of meiosis and the genetic consequences do not correspond exactly in time. A genetic process occurring in one phase is usually not cytologically manifest until a subsequent phase.

A. Meiosis I

A complete meiosis consists of two cell divisions, meiosis I and meiosis II. The relevant genetic events, genetic recombination by means of crossing-over and reduction to the haploid chromosome complement, occur in meiosis I.

Meiosis begins with chromosome replication. Initially the chromosomes in late interphase are visible only as threadlike structures, as in mitosis. At the beginning of prophase I, the chromosomes are doubled, but this is not visible until a later period of prophase I (see p. 118). Subsequently, the pairing of homologous chromosomes can also be visualized. The pairing makes an exchange between homologous chromosomes (crossing-over) possible by juxtapositioning homologous chromatids (chiasma formation). The result of crossing-over is an exchange of material between two chromatids of homologous chromosomes (genetic recombination). This exchange has occurred by the time the cell has entered metaphase I. After the homologous chromosomes migrate to opposite poles, anaphase I begins.

B. Meiosis II

Meiosis II consists of longitudinal division of the doubled chromosomes (chromatids) and a further cell division. Each daughter cell contains one chromosome of a chromosome pair and is therefore haploid. Because of the recombination that occurred in prophase I, the chromosomes of the resulting haploid cell differ from those of the original cells. Thus, unlike in mitosis, the chromosomes of the daughter cells are not genetically identical with those of the original cell. On each chromosome, recombinant and nonrecombinant sections can be identified. The genetic events relevant to these changes have occurred in the prophase of meiosis I (see p. 118).

The distribution of chromosomes during meiosis explains the segregation (separation or splitting) of traits according to Mendelian laws (1 : 1 segregation, cf. p. 134).

Recombination is the most striking event in meiosis. The molecular mechanisms of recombination are complex. Occasionally, recombination may occur during mitosis (mitotic recombination), e.g., during DNA repair. Gene conversion designates a unilateral, nonreciprocal exchange. With this, one allele is lost in favor of another.

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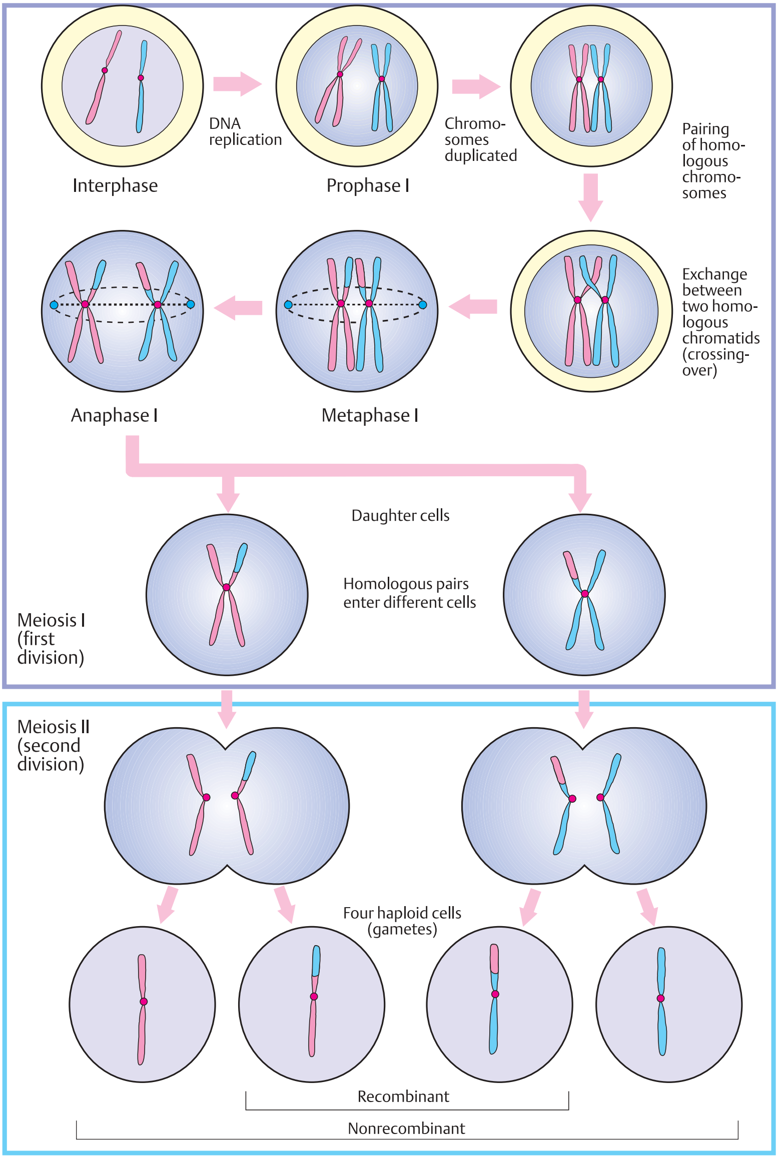

Interphase
DNA replication
Prophase I
Chromosomes duplicated
Pairing of homologous chromosomes
Exchange between two homologous chromatids (crossing-over)
Metaphase I
Anaphase I
Daughter cells
Homologous pairs enter different cells
Meiosis I (first division)
Meiosis II (second division)
Four haploid cells (gametes)
Recombinant
Nonrecombinant

A. Meiosis

Crossing-Over in Prophase I

Prophase of meiosis I is a complex period during which important cytological and genetic events occur. In this phase, exchanges between homologous chromosomes occur regularly by crossing-over. This results in new combinations of chromosome segments (genetic recombination).

A. Prophase of meiosis I

The prophase of meiosis I goes through a number of stages that can be differentiated schematically, although they proceed continuously. In the leptotene stage, the chromosomes are first visible as fine threadlike structures (in A only one chromosome pair is shown schematically). In zygotene the chromosomes are visible as paired structures. By this time, every chromosome has been doubled and consists of two chromatids, which are held together at the centromere (each chromatid contains a DNA double helix). Two homologous chromosomes that have paired are referred to as a bivalent. In the pachytene stage, the bivalents become thicker and shorter. In diplotene the two homologous chromosomes separate for the most part, but still remain attached to each other at a few points (chiasmata). In subsequent stages of diplotene, each of the chromosome pairs separates even more extensively (diakinesis), especially at the centromere region, but not yet at one or more distally located points of attachment (chiasmata). Each chiasma corresponds to a region in which crossover has taken place. In the last stage of prophase I, diakinesis, the chromosomes are widely separated, although still attached at their distal ends. The chiasmata have shifted distally (terminalization). At the end of diakinesis, the nuclear membrane disappears and the cell enters metaphase I.

B. Synaptonemal complex

Shortly before the onset of the pachytene stage, homologous chromosomes move very close together and form a synaptonemal complex. It consists of two chromatids (1 and 2) of maternal origin (mat) and two chromatids (3 and 4) of paternal origin (pat). This initiates chiasma formation and is the prerequisite for crossing-over and subsequent recombination. (Diagram after Watson et al., 1987).

C. Chiasmata

When a chiasma is formed, either of the two chromatids of one chromosome pairs with one of the chromatids of the homologus chromosome (e.g., 1 and 3, 2 and 4 and so on). Chiasma formation is the cytological prerequisite for crossing-over and is important in the definitive separation (segregation) of the chromosomes. The centromere (Cen) plays an important role in chromosome pairing.

D. Genetic recombination through crossing-over

Through crossing-over, new combinations of chromosome segments arise (recombination). As a result, recombinant and nonrecombinant chromosome segments can be differentiated. In the diagram, the areas A–E (shown in red) of one chromosome and the corresponding areas a–e (shown in blue) of the homologous chromosome become respectively a–b–C–D–E and A–B–c–d–e in the recombinant chromosomes.

E. Pachytene and diakinesis under the light microscope

In the micrograph, pachytene chromosomes are readily visualized as bivalents (a). An unusual structure in pachytene is formed by the X and the Y chromosomes. They appear to be joined end-to-end. Actually, short segments of the short arms in the regions with homologous sequences (pseudoautosomal region, see p. 390) have paired. In later stages (b), it can be seen that they have separated for the most part. (Photographs from Therman, 1986). Today, electron micrographs are usually used for meiotic studies.

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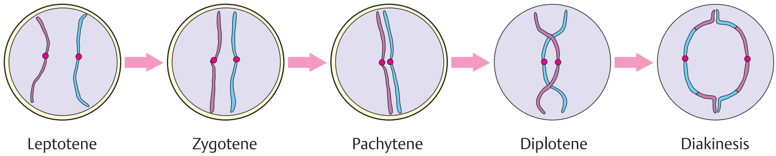
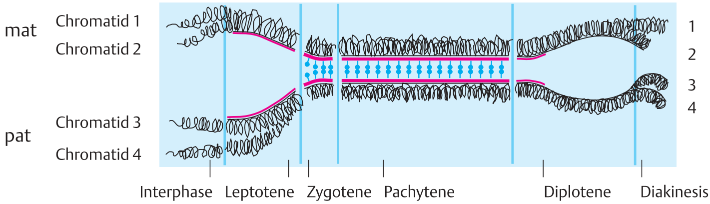
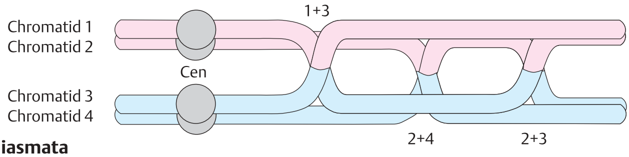


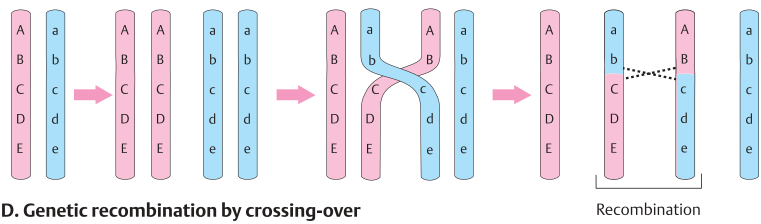
A. Prophase of meiosis I



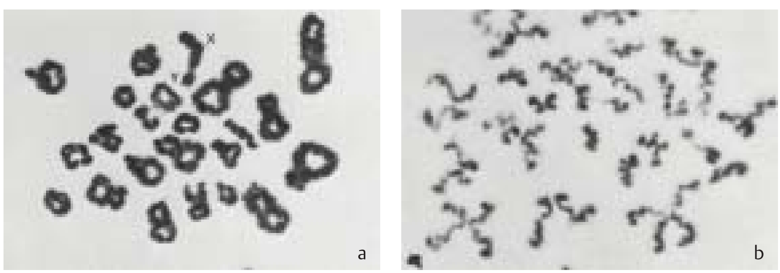
B. Synaptonemal complex



C. Chiasmata



D. Genetic recombination by crossing-over



E. Photographs of pachytene and diakinesis under the light microscope

Formation of Gametes

Germ cells (gametes) are produced in the gonads. In females the process is called oogenesis (formation of oocytes) and in males, spermatogenesis (formation of spermatozoa). The primordial germ cells, which migrate to the gonads during early fetal development, increase in number by mitotic division. The actual formation of germ cells (gametogenesis) begins with meiosis. Meiosis differs in duration and results between males and females.

A. Spermatogenesis

Diploid spermatogonia are formed by repeated mitotic cell divisions. At the onset of puberty, some of the cells begin to differentiate into primary spermatocytes. The first meiotic cell division occurs in these cells. At the completion of meiosis I, a primary spermatocyte has given rise to two secondary spermatocytes, each of which has a haploid set of duplicated chromosomes (recombination is not illustrated here). Each chromosome consists of two sister chromatids, which become separated during meiosis II. In meiosis II, each secondary spermatocyte divides to form two spermatids. Thus, one primary spermatocyte forms four spermatids, each with a haploid chromosome complement. The spermatids differentiate into mature spermatozoa. Male spermatogenesis is a continuous process. In human males, the time lapse between differentiation into a primary spermatocyte at the onset of meiosis I and the formation of mature spermatocytes is about 6 weeks.

B. Oogenesis

Oogenesis (formation of oocytes) differs from spermatogenesis in timing and in the result. At first the germ cells, which have migrated to the ovary, multiply by repeated mitosis (formation of oogonia). In human females, meiosis I begins about 4 weeks before birth. Primary oocytes are formed. However, meiosis I is arrested in a stage of prophase designated dictyotene. The primary oocyte persists in this stage until ovulation. Only then is meiosis I continued (recombination is not shown here).

In females, the cytoplasm divides asymmetrically in both meiosis I and meiosis II. The result each time is two cells of unequal size: a larger cell that will eventually form the egg and a small cell, called a polar body. When the primary oocyte divides, the haploid secondary oocyte and polar body I are formed. When the secondary oocyte divides, again unequally, the result is a mature oocyte and another polar body (polar body II). The polar bodies do not develop further, but degenerate. On rare occasions when this does not occur, a polar body may become fertilized. This can give rise to an incompletely developed twin.

In the secondary oocyte, each chromosome still exists as two sister chromatids. These do not separate until the next cell division (meiosis II), when they enter into two different cells. In most vertebrates, maturation of the secondary oocyte is arrested in meiosis II. At ovulation the secondary oocyte is released from the ovary, and if fertilization occurs, meiosis is then completed. Faulty distribution of the chromosomes (nondisjunction) may occur in meiosis I as well as in meiosis II (see p. 116).

The maximal number of germ cells in the ovary of the human fetus at about the 5th months is 6.8×10^6. By birth this has been reduced to 2×10^6, and by puberty to about 200,000. Of these, about 400 are ovulated (Connor & Ferguson-Smith, 1993).

The long period between meiosis I and ovulation is presumably a factor in the relatively frequent nondisjunction of homologous chromosomes in older mothers.

The difference in time in the formation of gametes during oogenesis and spermatogenesis is reflected in the difference in germline cell divisions. In the female there are 22 cell divisions before meiosis, resulting in a total of 23 chromosome replications. In contrast, 610 chromosome replications have taken place in the ancestral cells of spermatozoa produced in a male at the age of 40 (380 at the age of 30), yielding 25 times as many cell divisions during spermatogenesis (Crow, 2000). This probably accounts for the higher mutation rate in males, especially with increased paternal age.

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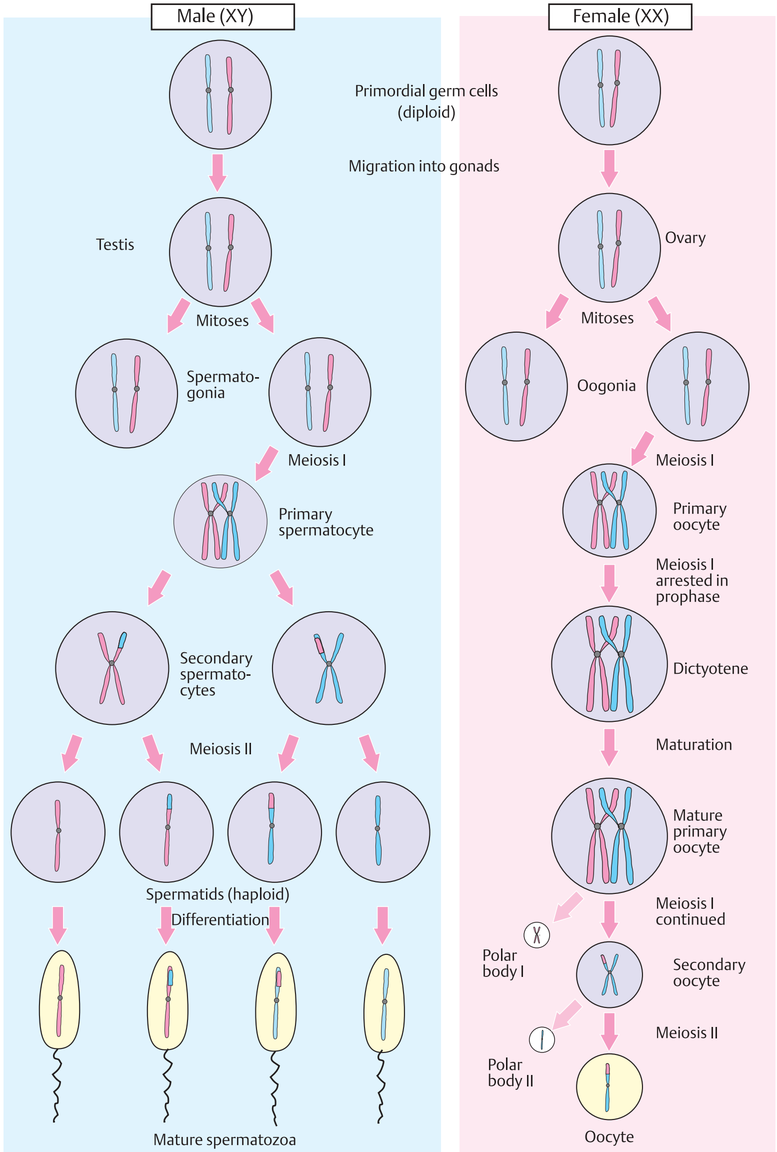

A. Spermatogenesis **B. Oogenesis**

Cell Culture

Cells of animals and plants can live and multiply in a tissue-culture dish (as a cell culture) at 37 °C in a medium containing vitamins, sugar, serum (containing numerous growth factors and hormones), the nine essential amino acids for vertebrate animals (His, Ile, Leu, Lys, Met, Phe, Thr, Tyr, Val), and usually also glutamine and cysteine. Cell cultures have been in wide use since 1965 and have become the basis for genetic studies not possible in the living mammals *(somatic cell genetics)*. A great variety of growth media are available for culturing mammalian cells and accommodating their requirements for growth.

The predominant cell type that grows from a piece of mammalian tissue in culture is the fibroblast. Although fibroblasts secrete proteins typical of fibrous connective tissue, in principle they retain the ability to differentiate into other cell types. Cultured skin fibroblasts have a finite life span (Hayflick, 1997). Human cells have a capacity for about 30 doublings until they reach a state called senescence. Cells derived from adult tissues have a shorter life span than those derived from fetal tissues.

Cultured cells are highly sensitive to increased temperature and do not survive above about 39 °C, whereas under special conditions they can be stored alive in vials kept in liquid nitrogen at –196 °C. They can be thawed after many years or even decades and cultured again.

A. Skin fibroblast culture

To initiate a culture, a small piece of skin (2×4 mm) is obtained under sterile conditions and cut into smaller pieces, which are placed into a culture dish. The pieces must attach to the bottom of the dish. After about 8 – 14 days, cells begin to grow out from each piece and begin to multiply. They grow and multiply only when adhering to the bottom of the culture vessel (adhesion culture due to anchorage dependency of the cells). When the bottom of the culture vessel is covered by a dense layer of cells, they stop dividing owing to contact inhibition (this is lost in tumor cells). When transferred into new culture vessels (subcultures), the cells will resume growing until they again become confluent. By a series of subcultures, several millions of cells can be obtained for a given study.

B. Hybrid cells for study

Cells in culture can be induced by polyethylene glycol or Sendai virus to fuse. If parental cells from different species are fused, interspecific (from different animal species) hybrid cells can be derived. The hybrid cells can be distinguished from the parental cells by using parental cells deficient in thymidine kinase (TK^-) or hypoxanthine phosphoribosyltransferase ($HPRT^-$). When cell cultures of parental type A (TK^-, 1) and type B ($HPRT^-$, 2) are cultured together, cells that did not fuse (3 and 5) will die in a selective medium containing hypoxanthine, aminopterin, and thymidine (HAT) (the TK^- cell cannot synthesize thymidine monophosphate; the $HPRT^-$ cell cannot synthesize purine nucleoside monophosphates), whereas fused cells of both parental types (4) containing the nuclei from both parents (heterokaryon, 6) will survive. Hybrid cells randomly lose chromosomes from one parent during further culturing (human/rodent cell hybrids lose the human chromosomes). When only one human chromosome is present, the genetic properties conferred by its genes to the cell can be studied (8).

C. Radiation hybrids

Radiation hybrids are rodent cells containing small fragments of human chromosomes. When human cells are irradiated with lethal roentgen doses of 3 – 8 Gy, the chromosomes break into small pieces (1) and the cells cannot divide in culture. However, if these cells are fused with nonirradiated rodent cells (2), some human chromosome fragments will be integrated into the rodent chromosomes (3). Cells containing human DNA can be identified by human chromosome-specific probes.

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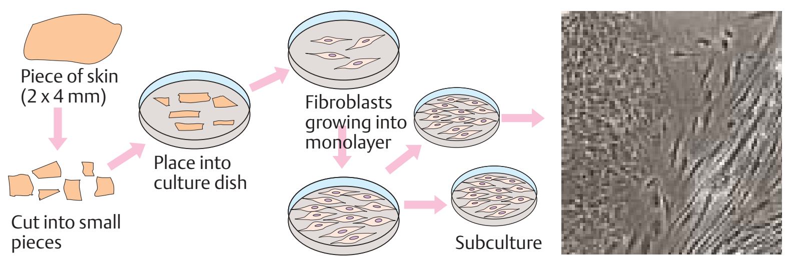
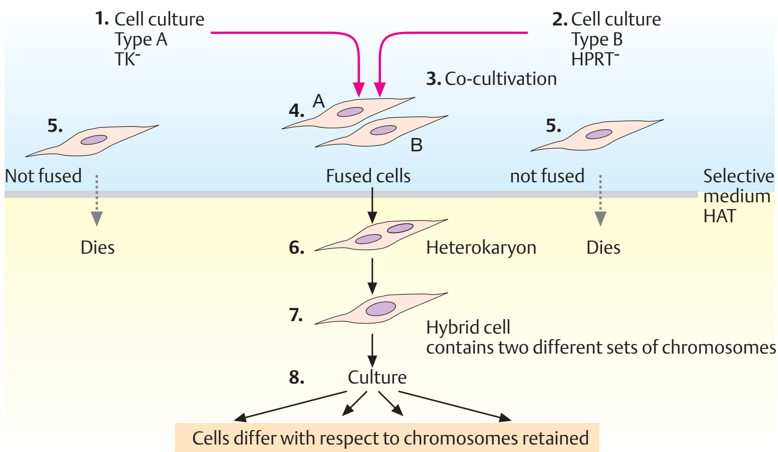
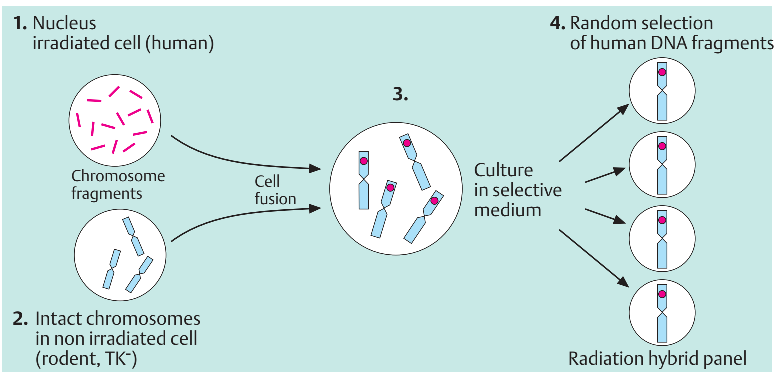


A. Skin fibroblast culture



B. Hybrid cells for study



C. Radiation hybrids

Mitochondrial Genetics

Genetically Controlled Energy-Delivering Processes in Mitochondria

Eukaryotic organisms contain essential genetic information separate from the nuclear DNA, in extrachromosomal genomes called mitochondria. The mitochondria of all eukaryotes and the chloroplasts of green plants and algae contain circular DNA molecules (mitochondrial DNA, mtDNA). Each eukaryotic cell contains 10^3 – 10^4 copies. Mitochondria and chloroplasts are the site of essential energy-delivering processes and photosynthesis. Each mitochondrion is surounded by two highly specialized membranes, the outer and inner membranes. The inner membrane is folded into numerous cristae and contains important molecules (see C).

The genes contained in mitochondrial DNA code for 13 proteins of the respiratory chain, subunits of the ATPase complex, and subunits of the NADH-dehydrogenase complex (ND); 22 other genes code for transfer RNA (tRNA) and two rRNAs. A number of diseases due to mutations and deletions in mtDNA are known in humans. Sequence homologies indicate evolutionary relationships. In particular, evolutionary transfer of DNA segments from chloroplasts to mitochondria, and from chloroplasts to nuclear DNA of eukaryotic organisms, has been demonstrated.

A. Principal events in mitochondria

The essential energy-conserving process in mitochondria is oxidative phosphorylation. Relatively simple energy carriers such as NADH and $FADH_2$ (nicotinamide–adenine dinucleotide in the reduced form and flavin adenine dinucleotide in the reduced form) are produced from the degradation of carbohydrates, fats, and other foodstuffs by oxidation. The important energy carrier adenosine triphosphate (ATP) is formed by oxidative phosphorylation of adenosine diphosphate (ADP) through a series of biochemical reactions in the inner membrane of mitochondria (respiratory chain).

B. Oxidative phosphorylation (OXPHOS) in mitochondria

Adenosine triphosphate (ATP) plays a central role in the exchange of energy in biological systems. ATP is a nucleotide consisting of adenine, a ribose, and a triphosphate unit. It is energy-rich because the triphosphate unit contains two phospho-anhydride bonds. Energy (free energy) is released when ATP is hydrolyzed to form ADP. The energy contained in ATP and bound to phosphate is released, for example, during muscle contraction.

C. Electron transfer in the inner mitochondrial membrane

The genomes of mitochondria and chloroplasts contain genes for the formation of the different components of the respiratory chain and oxidative phosphorylation. Three enzyme complexes regulate electron transfer: the NADH-dehydrogenase complex, the b–c_1 complex, and the cytochrome oxidase complex (C). Intermediaries are quinone (Q) derivatives such as ubiquinone and cytochrome *c*. Electron transport leads to the formation of protons (H^+). These lead to the conversion of ADP and P_i (inorganic phosphate) into ATP (oxidative phosphorylation). ATP represents a phosphate-bound reservoir of energy, which serves as an energy supplier for all biological systems. Thus it is understandable that genetic defects in mitochondria become manifest primarily as diseases with reduced muscle strength and other degenerative signs. (Figures adapted from Bruce Alberts et al., 1998).

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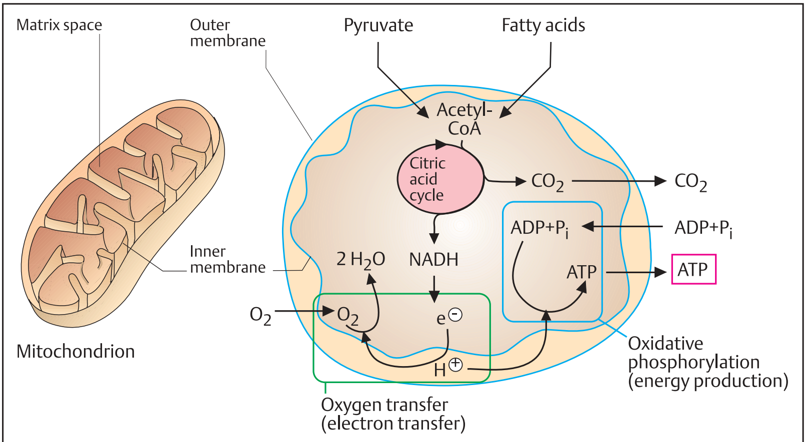
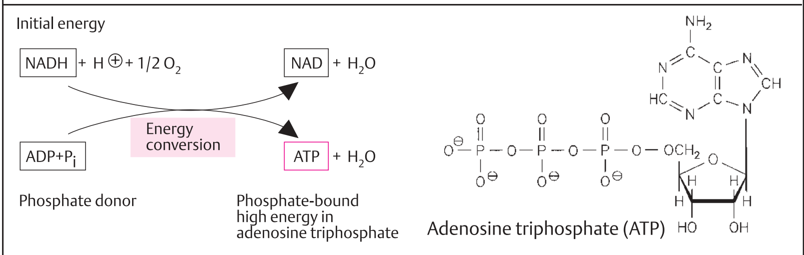
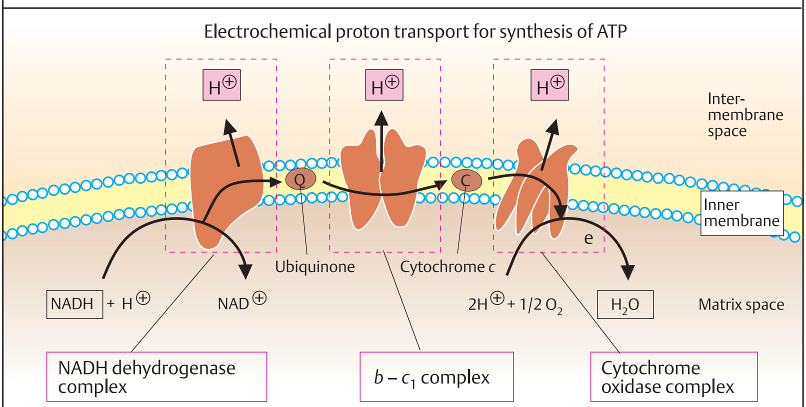


A. Principal events in mitochondria



B. Oxidative phosphorylation (OXPHOS) in mitochondria



C. Electron transfer in the inner mitochondrial membrane

The Genome in Chloroplasts and Mitochondria

Chloroplasts of higher plants and mitochondria of eukaryotic cells contain genomes of circular DNA. About 12000 base pairs (12 kb) of the genomes of chloroplasts and mitochondria are homologous. Furthermore, homologous regions are found in nuclear DNA. Thus it is assumed that the DNAs of chloroplasts, mitochondria, and nuclear DNA are evolutionary related.

A. Genes in the chloroplasts of a moss *(Marchantia polymorpha)*

The genome in chloroplasts (ctDNA) is large: 121 kb in the moss *M. polymorphia* and 155 kb in the tobacco plant. Yet the organization and number of their genes are similar. Protein synthesis shows certain similarities with that of bacteria. Many of the ribosomal proteins are homologous with those of *E. coli.* Genes in the chloroplast genomes are interrupted and contain introns. Each chloroplast contains about 20–40 copies of ctDNA and there about are 20–40 chloroplasts per cell. Among these are genes for two copies each of four ribosomal RNAs (16S rRNA, 23S rRNA, 4.5S rRNA, and 5S rRNA). The genes for ribosomal RNA are located in two DNA segments with opposite orientation (inverted repeats), which are characteristic of chloroplast genomes. An 18–19-kb segment with short single gene copies lies between the two inverted repeats. The genome of chloroplasts contains genetic information for about 30 tRNAs and about 50 proteins. The proteins belong to photosystem I (two genes), photosystem II (seven genes), the cytochrome system (three genes), and the H^+-ATPase system (six genes). The NADH dehydrogenase complex is coded for by six genes; ferredoxin by three genes; and ribulose by one gene. Twenty-nine genes have not been identified to data. (Data after Lewin, 2000).

B. Mitochondrial genes in yeast *(S. cerevisiae)*

The mitochondrial genome of yeast is large (120 kb) and contains introns. It contains genes for the tRNAs, for the respiratory chain (cytochrome oxidase 1, 2, and 3; cytochrome *b*), for 15S and 21S rRNA, and for subunits 6, 8, and 9 of the ATPase system. The yeast mitochondrial genome is remarkable because its ribosomal RNA genes are separated. The gene for 21S rRNA contains an intron. About 25% of the mitochondrial genome of yeast contains AT-rich DNA without a coding function.

The genetic code of the mitochondrial genome differs from the universal code in nuclear DNA with respect to usage of some codons. The nuclear stop codon UGA codes for tryptophan in mitochondria, while the nuclear codons for arginine (AGA and AGG) function as stop codons in mammalian mitochondria.

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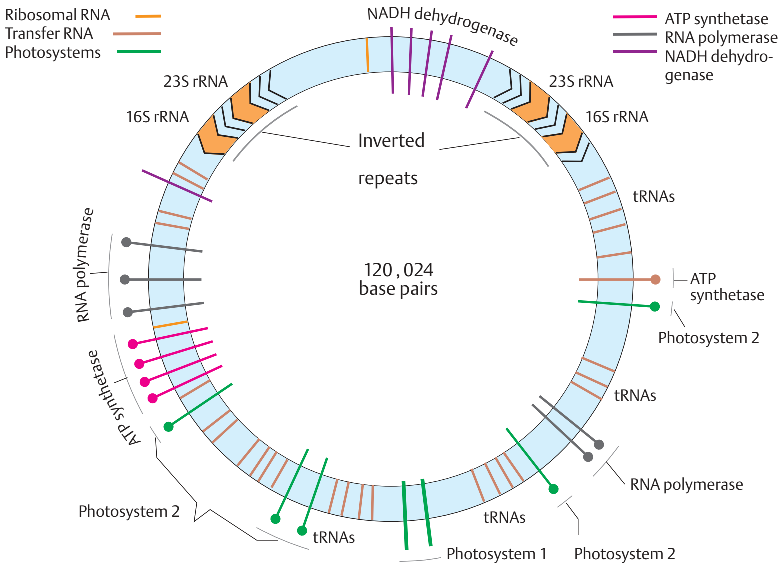
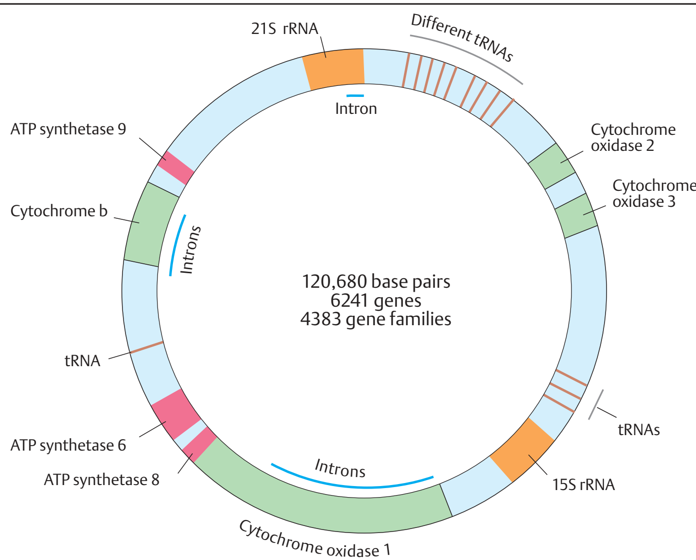
Codon	Nuclear DNA	Mitochondrial genome			
		Mammals	Drosophila	Yeast	Plants
UGA	Stop	Trp	Trp	Trp	Stop
AUA	Ile	Met	Met	Met	Ile
CUA	Leu	Leu	Leu	Thr	Leu
AGA, AGG	Arg	Stop	Ser	Arg	Arg

Differences between the genetic code of the mitochondrial genome and the universal code of nuclear DNA (From B. Alberts et al., 1994).



A. Genes in chloroplasts of a moss (*Marchantia polymorpha*)



B. Mitochondrial genes in yeast (*S. cerevisiae*)

The Mitochondrial Genome of Man

The mitchondrial genome in mammals is small and compact. It contains no introns, and in some regions the genes overlap, so that practically every base pair belongs to a gene. The mitochondrial genomes of humans and mice have been sequenced and contain extensive homologies. Each consists of about 16.5 kb, i.e., is considerably smaller than a yeast mitochondrial genome.

A. Mitochondrial genes in man

Each mitochondrion contains 2 – 10 DNA molecules. The mitochondrial genome of humans contains 13 protein-coding regions: genes for the cytochrome *c* oxidase complex (subunits 1, 2, and 3), for cytochrome *b*, and for subunits 6 and 8 of the ATPase complex. Unlike that of yeast, mammalian mitochondrial DNA contains seven subunits for NADH dehydrogenase (ND1, ND2, ND3, ND4L, ND4, ND5, and ND6). Of the mitochondrial coding capacity, 60% is taken by the seven subunits of NADH reductase (ND). A heavy (H) and a light (L) single strand can be differentiated by a density gradient. Most genes are found on the H strand. The L strand codes for a protein (ND subunit 6) and 8 tRNAs. From the H strand, two RNAs are transcribed, a short one for the rRNAs and a long one for mRNA and 14 tRNAs. A single transcript is made from the L strand. A 7S RNA is transcribed in a counterclockwise manner close to the origin of replication (ORI) (not shown).

B. Cooperation between mitochondrial and nuclear genome

Some mitochondrial proteins are aggregates of gene products of nuclear and mitochondrial genes. These gene products are transported into the mitochondria after nuclear transcription and cytoplasmic translation. In the mitochondria, they form functional proteins from subunits of mitochondrial and nuclear gene products. This explains why a number of mitochondrial genetic disorders show Mendelian inheritance, while purely mitochondrially determined disorders show exclusively maternal inheritance.

C. Evolutionary relationship of mitochondrial genomes

Mitochondrial DNA has a mutation rate tentimes higher than that of nuclear DNA. Mutations are generated during OXPHOS (p. 124) through pathways involving reactive oxygen molecules. Mutations accumulate because effective DNA repair and protective histones are lacking. At birth most mtDNA molecules are identical (homoplasmy); later they differ as a result of mutations accumulated in different mitochondria (heteroplasmy).

Mitochondria probably evolved from independent organisms that were integrated into cells. They replicate, transcribe, and translate their DNA independently of the nuclear DNA. Since mitochondria are present in oocytes only, and not spermatozoa, their genes are exclusively inherited through the mother. The maternal inheritance can be used effectively in studies of evolutionary relationships. An evolutionary tree of related populations can be reconstructed by comparing mtDNA variants, called haplotypes, based on sequence differences. A phylogenetic tree reconstructed from 147 humans of different geographic regions thoughout the world suggested that an ancestral sequence existed about 200 000 years ago (dubbed "Eve"). Mitochondrial DNA can be studied from ancient biological specimens such as Neandertals (see p. 258).

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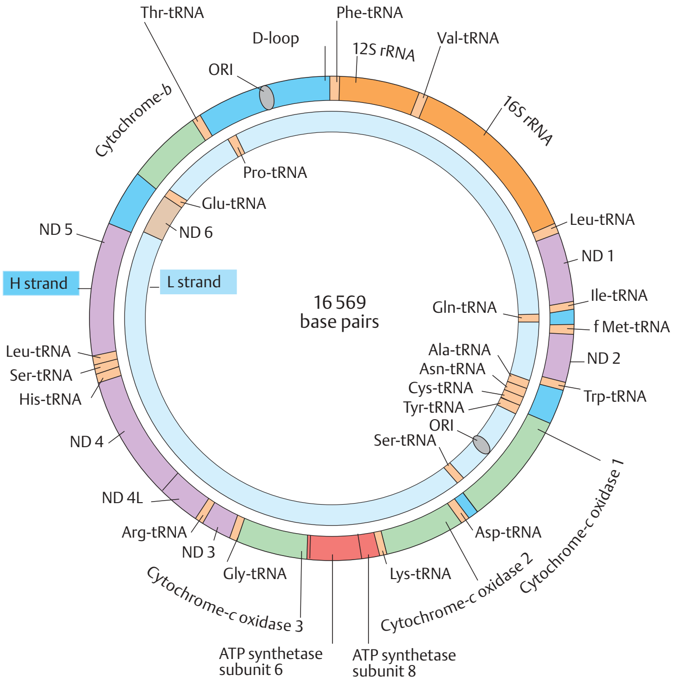
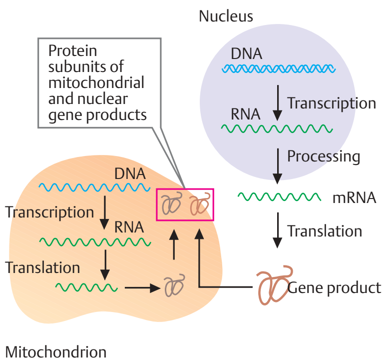
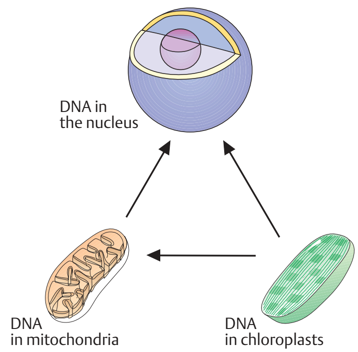


A. Mitochondrial genes in man



B. Cooperation between mitochondrial and nuclear genome



C. Evolutionary relationship of mitochondrial genomes

Mitochondrial Diseases

Mitochondrial function can be disrupted by genetic changes involving one or several of the respiratory chain proteins or the tRNAs or rRNAs. In addition, the interaction of mitochondrial and nuclear genes can be disturbed in many ways. The clinical spectrum and age of onset of mitochondrial diseases is extremely wide. Organs with high energy requirements are particularly vulnerable: the brain, heart, skeletal muscle, eye, ear, liver, pancreas, and kindey.

A. Mutations and deletions in mitochondrial DNA in man

Both deletions and point mutations are causes of mitochondrial genetic disorders. Some are characteristic and recur in different, unrelated patients. Panel A and the table show examples of important mutations and deletions, and mitochondrial diseases. (Figure adapted from Wallace, 1999; MITOMAP; and Marie T. Lott and D. C. Wallace, personal communication).

B. Maternal inheritance of a mitochondrial disease

Hereditary mitochondrial diseases are transmitted only through the maternal line, since spermatozoa contain hardly any mitochondria. Thus the disease will not be transmitted from an affected man to his children.

C. Heteroplasmy for mitochondrial mutations

Mutations or deletions in mitochondria are more frequently limited to a single tissue (mitochondrial cytopathy) than are germline mutations. In such cases, the cells contain different proportions of affected mitochondria (heteroplasmy). The proportion of defective mitochondria varies after repeated cell divisions. This contributes to the considerable variability of mitochondrial diseases.

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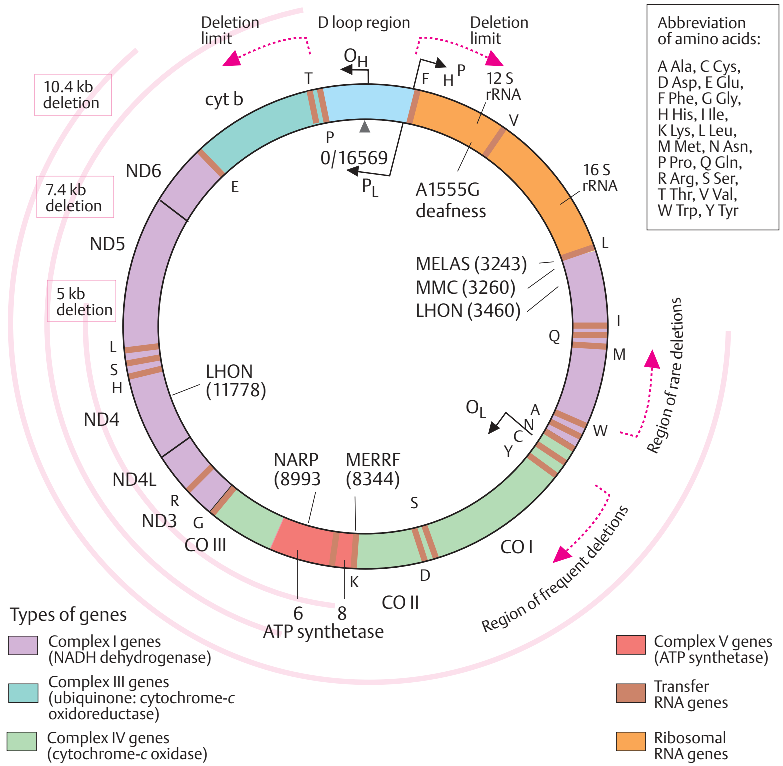
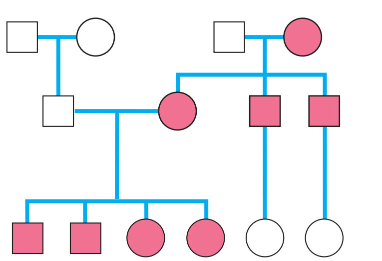
Examples of diseases due to mutations or deletions in mitochondrial DNA

Abbreviation	McKusick No.	Name
KSS	530000	Kearns – Sayre syndrome (ophthalmoplegia, pigmentary degeneration of the retina, and cardiomyopathy)
LHON	535000	Leber hereditary optic neuropathy
MELAS	540000	Encephalomyopathy, Lactic-acidosis with strokelike symptoms
MERRF	545000	Myoclonic epilepsy and ragged red fibers
MMC	590050	Mitochondrial myopathy and cardiomyopathy
NARP	551500	Neurogenic muscle weakness with ataxia and retinitis pigmentosa
CEOP	555000	Chronic external ophthalmoplegia
MNGIE	550900	Myoneurogastrointestinal pseudoobstruction, Encephalopathy
PEAR	557000	Pearson syndrome (bone marrow and pancreas failure)
ADMIMY	157640 550000	Autosomal dominant mitochondrial myopathy with mitochondrial deletions in the D-loop (type Zeviani)
None	515000	Chloramphenicol induced toxicity
None	580000	Streptomycin-induced ototoxicity (A1555G mutation)
None	520000	Diabetes mellitus—deafness

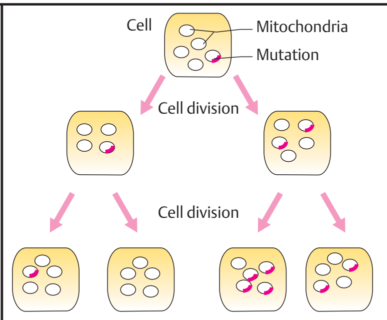
MIM: Mendelian Inheritance in Man, McKusick's Catalog of Human Genes and Genetic Disorders. 12th ed., Johns Hopkins University Press, Baltimore, 1998; MITOMAP; and *http://www.mips.biochem.mpg.de/proj/medgen/mitop/*



A. Mutations and deletions in human mitochondrial DNA



B. Maternal inheritance of a mitochondrial disease



C. Heteroplasmy for mitochondrial mutations

Formal Genetics

The Mendelian Traits

In 1865, the Augustinian monk Gregor Mendel published some remarkable observations in the *Berichte der Naturgeschichtlichen Vereinigung von Brünn* (Proceedings of the Natural History Society of Brno), which received little attention at the time. In this work, entitled "Experiments on Hybrid Plants," Mendel observed that certain traits in garden peas *(Pisum sativum)* are inherited independently of one another. Moreover, Mendel described certain regularities in the pattern of occurrence of individual traits in consecutive generations. His experimental system, the plants, and the observed traits will be presented here.

A. The pea plant (*Pisum sativum*)

The plant consists of a stem, leaves, blossoms, and seedpods. In the blossom, the (female) pistil (comprising the stigma, style, and ovary) and the (male) stamen, comprising the anther and filament, can be differentiated. The garden pea normally reproduces by self-fertilization. Pollen from the anther falls onto the stigma of the same blossom.

However, it is relatively easy to carry out cross-fertilization. To do this, Mendel opened a blossom and removed the anther before pollen could escape and used pollen from another blossom instead. The resulting seedpod contains the seeds from which new plants develop.

B. The observed traits (phenotypes)

Mendel observed a total of seven characteristic traits: (1) height of the plants, (2) location of the blossoms on the stem of the plant, (3) the color of the pods, (4) the form of the pods, (5) the form of the seeds, (6) the color of the seeds, and (7) the color of the seed coat. He observed that each pair of traits was inherited independently from all other pairs of traits. Mendel's main observation was that independent traits are inherited in certain predictable patterns. This was a fundamental new insight into the process of heredity. Since it distinctly deviated from the prevailing concepts about heredity at that time, its significance was not immediately recognized.

Today it is known that genetically determined traits are independently inherited (segregation) only when they are located on different chromosomes or far enough apart on the same chromosome to be separated each time by recombination (i.e., when there is no genetic linkage; see pp. 144). This is true of the genes investigated by Mendel. In recent years, some of these genes have been cloned and their molecular structures have been characterized.

Deviation from the Mendelian pattern of inheritance

Mendelian traits do not always occur in the expected proportions (see p. 134). As a result of the phenomenon of meiotic drive, one trait may occur much more frequently than others. Examples exist in the t complex of the mouse (about 99% instead of 50% of offspring of heterozygous t/+ male mice are also heterozygous) and Segregation Distorter (SD) in *Drosophila.*

In 1993, a mouse population in Siberia was described in which 85% and 65% of offspring were heterozygous for an inversion. Homozygosity for the inversion leads to reduced fitness and is a selective disadvantage. The shift was not due to early embryonal death. Possibly, deviations from Mendelian laws are more frequent than previously assumed. Further deviations occur with genomic imprinting (see p. 226) and germline mosaicism.

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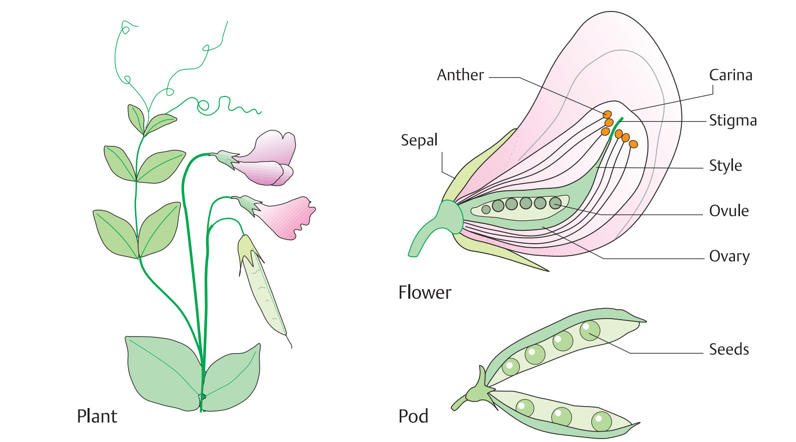
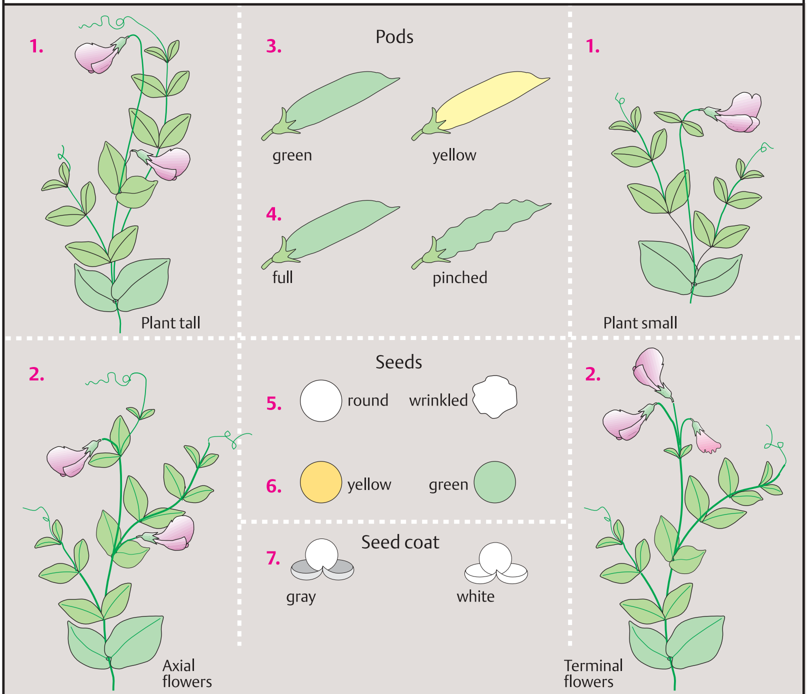


A. The pea plant (*Pisum sativum*)



B. The traits observed (phenotypes)

Distribution (Segregation) of Mendelian Traits

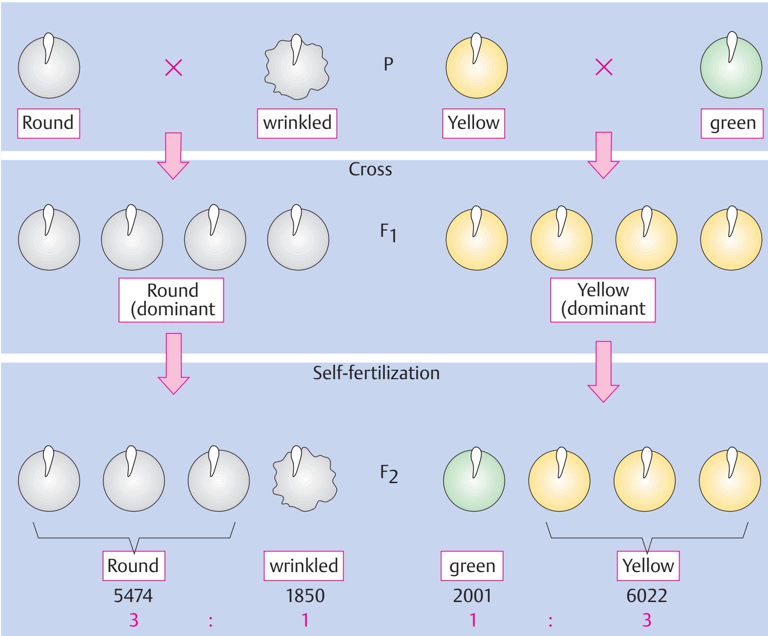
Mendel observed different characteristics of the pea plant and followed their occurrence in consecutive generations. From these observations, certain regularities in their patterns of occurrence (Mendelian laws) became apparent.

A. Segregation of dominant and recessive traits

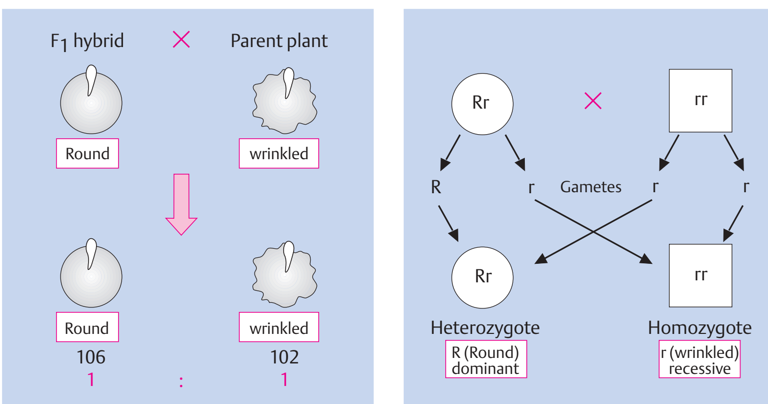
In two different experiments, Mendel observed the shape (smooth or wrinkled) and the color (yellow or green) of the seeds. When he crossed the plants of the parental generation P, i.e., smooth × wrinkled or yellow × green, he observed that in the first filial (daughter) generation, F_1, all seeds were respectively smooth and yellow.
In the next generation, F_2, which arose by the self-fertilization usual for peas, the traits observed in the P generation (smooth and wrinkled, or green and yellow, respectively) reappeared. Among a total of 7324 seeds of one experiment, 5474 were smooth and 1850 were wrinkled. This corresponded to a ratio of 3:1. In the experiment with different colors (green vs. yellow), Mendel observed that in a total of 8023 seeds of the F_2 generation, 6022 were yellow and 2001 green, again corresponding to a ratio of 3:1.
The trait the F_1 generation exclusively showed (round or yellow), Mendel called dominant; the trait that did not appear in the F_1 generation (wrinkled or green) he called recessive. His observation that a dominant and a recessive pair of traits occur (segregate) in the F_2 generation in the ratio 3:1 is known as the first law of Mendel.

B. Backcross of an F_1 hybrid plant with a parent plant

When Mendel backcrossed the F_1 hybrid plant with a parent plant showing the recessive trait (1), both traits occurred in the next generation in a ratio of 1:1 (106 round and 102 wrinkled). This is called the second law of Mendel.
The interpretation of this experiment (2), the backcross of an F_1 hybrid plant with a parent plant, is that different germ cells (gametes) are formed. The F_1 hybrid plant (round) contains two traits, one for round (R, dominant over wrinkled, r) and one for wrinkled (r, recessive to round, R). This plant is a hybrid (heterozygote) and therefore can form two types of gametes (R and r).
In contrast, the other plant is homozygous for wrinkled (r). It can form only one type of gamete (r, wrinkled). Half of the offspring of the heterozygous plant receive the dominant trait (R, round), the other half the recessive trait (r, wrinkled). The resulting distribution of the observed traits is a ratio of 1:1, or 50% each.
The observed trait is called the phenotype (the observed appearance of a particular characteristic). The composition of the two factors (genes) R and r, (Rr) or (rr), is called the genotype. The alternative forms of a trait (here, round and wrinkled) are called alleles. They are the result of different genetic information at one given gene locus.
If the alleles are different, the genotype is heterozygous; if they are the same, it is homozygous (this statement is always in reference to a single, given gene locus).



A. Segregation of dominant and recessive traits



1. Experiment

2. Interpretation

B. Back cross of an F_1-hybrid with a parent plant

Independent Distribution of Two Different Traits

In a further experiment, Mendel observed that two different traits are inherited independently of each other. Each pair of traits shows the same 3:1 distribution of the dominant over the recessive trait in the F_2 generation as he had previously observed. The segregation of two pairs of traits again followed certain patterns.

A. Independent distribution of two traits

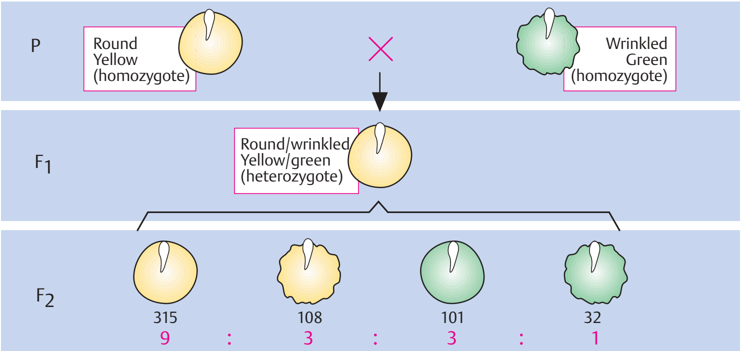
In one experiment, Mendel investigated the crossing of the trait pairs round/wrinkled and yellow/green. When he crossed plants with round and yellow seeds with plants with wrinkled and green seeds, only round and yellow seeds occurred in the F_1 generation. This corresponded with the original experiments, as shown on p. 132. In the F_2 generation, the two pairs of traits occurred in the following distribution: 315 seeds yellow and round, 108 yellow and wrinkled, 101 green and round, 32 green and wrinkled, corresponding to a segregation ratio of 9:3:3:1. This is referred to as the third Mendelian law.

B. Interpretation of the observations

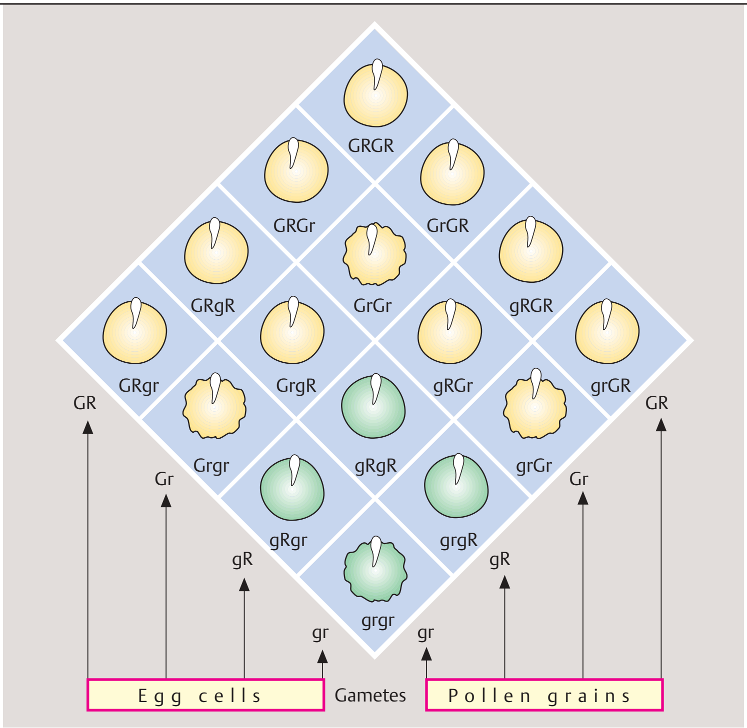
If we assign the capital letter **G** to the dominant gene *yellow,* a lowercase **g** to the recessive gene *green,* the capital letter **R** to the dominant gene *round,* and the lowercase **r** to the recessive gene *wrinkled,* the following nine genotypes of these two traits can occur: **GGRR, GGRr, GgRR, GgRr** (all *yellow* and *round*); **GGrr, Ggrr** (*yellow* and *wrinkled*); **ggRR, ggRr** (*green* and *round*); and **ggrr** (*green* and *wrinkled*). The distribution of the traits shown in A is the result of the formation of gametes of different types, i.e., depending on which of the genes they contain. The ratio of the dominant trait yellow (**G**) to the recessive trait green (**g**) is 12:4, or 3:1.

Also, the ratio of dominant round (**R**) to wrinkled (**r**) seeds was 12:4, i.e., 3:1.

The square (Punnett square, named after an early geneticist) shows the different genotypes that can be formed in the zygote after fertilization. Altogether there are 9/16 yellow round seeds (**GRGR, GRGr, GrGR, GRgR, gRGR, GRgr, GrgR, gRGr, grGR**), 3/16 green round (**gRgR, gRgr, grgR**), 3/16 yellow wrinkled (**GrGr, Grgr, grGr**), and 1/16 green wrinkled seeds (**grgr**). Each of the two traits (dominant yellow versus recessive green or dominant round versus recessive wrinkled) occurs in a 3:1 ratio (dominant vs recessive).



A. Independent segregation of two traits



B. Interpretation of the observation

Phenotype and Genotype

Formal genetic analysis examines the genetic relationship of individuals based on their kinship. These relationships are presented in a pedigree (pedigree analysis). An observed trait is called the phenotype. This could be a disease, a blood group, a protein variant, or any other attribute determined by observation. The phenotype depends to a great degree on the method and accuracy of observation. The term genotype refers to the genetic information on which the phenotype is based.

A. Symbols in a pedigree drawing

The symbols shown here represent a common way of drawing a pedigree. Males are shown as squares, females as circles. Individuals of unknown sex (e.g., because of inadequate information) are shown as diamonds. In medical genetics, the degree of reliability in determining the phenotype, e.g., presence of a disorder, should be stated. In each case it must be stated which phenotype (e.g. which disease) is being dealt with. Established diagnoses (data complete), possible diagnoses (data incomplete), and questionable diagnoses (statements or data doubtable) should be differentiated. False assignment of a phenotype can lead to false conclusions about the mode of inheritance. A number of further symbols are used, e.g., for heterozygous females with X-chromosomal inheritance (see p. 142).

B. Genotype and phenotype

The definitions of genotype and phenotype refer to the genetic information at a given gene locus. The gene locus is the site on a chromosome at which the genetic information for the given trait lies. Different forms of genetic information at a gene locus are called alleles. In diploid organisms — all animals and many plants — there are three possible genotypes with respect to two alleles at any one locus: (1) homozygous for one allele, (2) heterozygous for the two different alleles, and (3) homozygous for the other allele.

Alleles can be differentiated according to whether they can be recognized in the heterozygous state or only in the homozygous state. If they can be recognized in the heterozygous state, they are called dominant. If they can be recognized in the homozygous state only, they are recessive. The concepts dominant and recessive are an attribute of the accuracy in observation and do not apply at the molecular level. If the two alleles can both be recognized in the heterozygous state, they are designated codominant (e.g., the alleles A and B of the blood group system AB0; 0 is recessive to A and B).

If there are more than two alleles at a gene locus, there will be correspondingly more genotypes. With three alleles there are six genotypes; for example, in the AB0 blood group system there are AA, A0 (both phenotype A), BB, B0 (both phenotype B), AB, and 00 (actually, there are more than three alleles in the AB0 system).

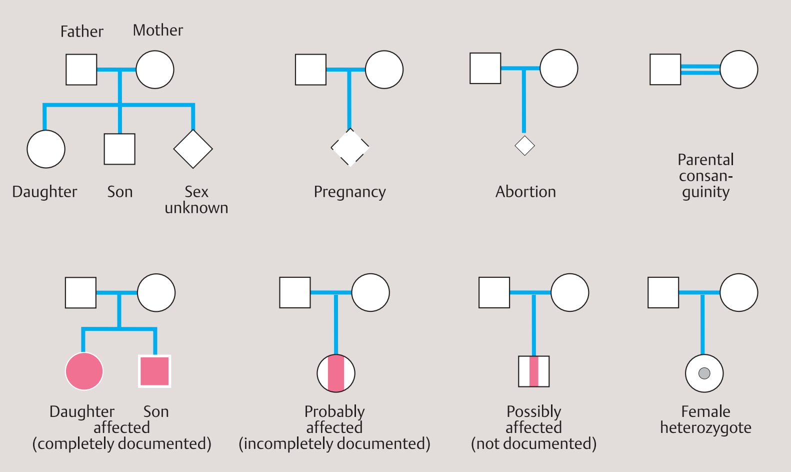
Genetic counseling

Genetic counseling is a communication process relating to the diagnosis and the possible occurrence of a genetically determined disease in a family and in more distant relatives. On the basis of an established diagnosis, the individual risk is determined for the consultand or that person's children. The goal of genetic counseling is to provide comprehensive information, including all possible decisions, course of the disease, medical care, and treatment. Professional confidentiality must be observed. The counselor makes no decisions.

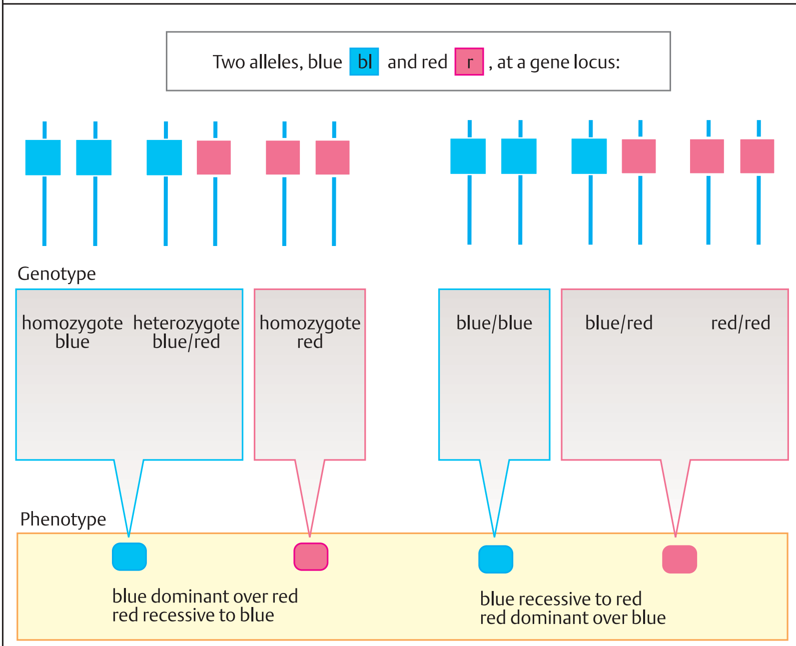
In genetic counseling, one distinguishes the consultand, the person who seeks information, and the patient, since they are very often different persons. The patient whose disease first directed attention to a particular pedigree is called the index patient (or proposita if female and propositus if male). The increasing availability of information about a disease based on a DNA test (predictive DNA testing) prior to disease manifestation calls for the utmost care in establishing whether it is in the interest of a given individual to have a test carried out.

Reference

Harper, P.S.: Practical Genetic Counselling, 5th ed. Butterworth-Heinemann, Oxford, 1998.



A. Symbols used in a pedigree



B. Genotype and phenotype

Segregation of Parental Genotypes

The distribution (segregation) of parental genotypes in the offspring depends on the combination of the alleles in the parents. In each case, they refer to a given gene locus. The Mendelian laws state the expected combination of alleles in the offspring of a parental couple. Depending on the chromosomal location, i.e., a locus on an autosome (see p. 184) or a locus on the X chromosome, one can distinguish autosomal and X-chromosomal inheritance. Depending on the effect of the genotype on the phenotype in the heterozygous state, an allele is classified as dominant or recessive. Hence, there are three basic modes of inheritance: (1) autosomal dominant, (2) autosomal recessive, and (3) X-chromosomal. For genes on the X chromosome it is not important to distinguish dominant and recessive (see below). Since genes on the Y chromosome are always transmitted from the father to all sons and the Y chromosome bears very few disease-causing genes, Y-chromosomal inheritance can be neglected when considering monogenic inheritance.

A. Possible mating combinations of the genotypes for two alleles

For a gene locus with two alleles there are six possible combinations of parental genotypes (1 – 6). Here two alleles, blue (bl) and red (r), are shown, blue being dominant over red. In three of the parental combinations (1, 3, 4) neither of the parents is homozygous for the recessive allele (red). In three parental combinations (2, 5, 6), one or both parents manifest the recessive allele because they are homozygous. The distribution patterns of genotypes and phenotypes in the offspring of the parents are shown in B. In these examples the sex of the parents is interchangable.

B. Distribution pattern in the offspring of parents with two alleles A and a

With three of the parental mating types for the two alleles **A** (dominant over **a**) and **a** (recessive to **A**), there are three combinations that lead to segregation (separation during meiosis) of allelic genes. These correspond to the parental combinations 1, 2, and 3 shown in A. In mating types 1 and 2, one of the parents is a heterozygote (**Aa**) and the other parent is a homozygote (**aa**). The distribution of observed genotypes expected in the offspring is 1: 1; that is 50% (0.50) are **Aa** heterozygotes and 50% (0.50), **aa** homozygotes.

If both parents are heterozygous **Aa** (mating type 3 in A), the proportions of expected genotypes of the offspring (**AA, Aa, aa**) occur in a ratio of 1:2:1. In each case, 25% (0.25) of the offspring will be homozygous **AA,** 50% (0.50) heterozygous **Aa,** and 25% (0.25) homozygous **aa.** If the two parents are homozygotes for different alleles (**AA** and **aa**), all their offspring will be heterozygotes.

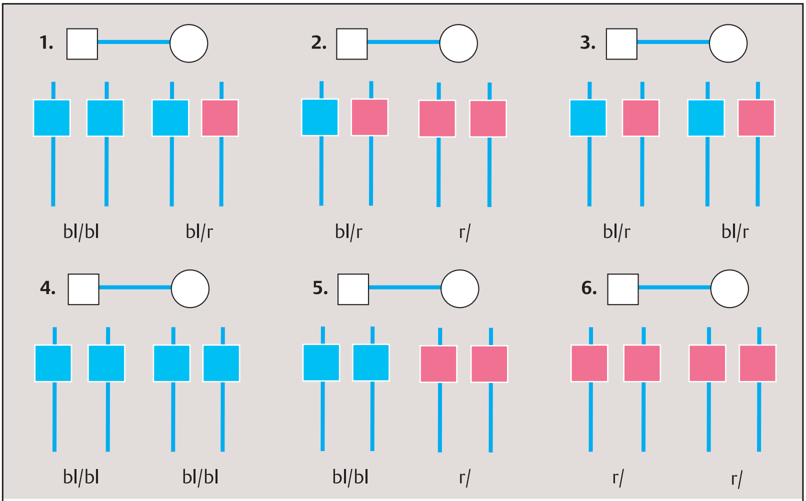
C. Phenotypes and genotypes in the offspring of parents with one dominantly inherited allele

One dominant allele (in the first pedigree, **A**, in the father) is to be expected in 50% of the offspring. If both parents are heterozygous, 25% of the offspring will be homozygous **aa**. If both parents are homozygous, one for the dominant allele **A**, the other for the recessive allele **a**, then all offspring are obligate heterozygotes (i.e., must necessarily be heterozygotes). It should be emphasized that the figures are percentages for expected distributions of the genotypes. The actual distribution may deviate form the expected, especially with small numbers of children.

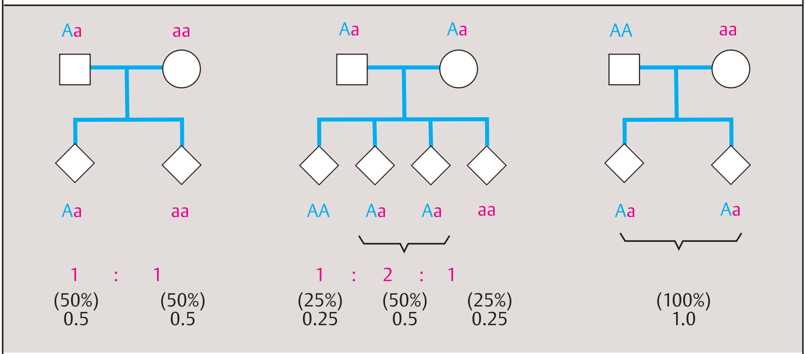
Expected distribution of genotypes for parents with different combinations of genotypes for a dominant allele A and a recessive allele a

Parents	Offspring	Distribution
AA × AA	AA	1
AA × Aa	AA, Aa	1 : 1
Aa × Aa	AA, Aa, aa*	1 : 2 : 1
AA × aa	Aa	1
Aa × aa	Aa, aa	1 : 1
aa × aa	aa	1

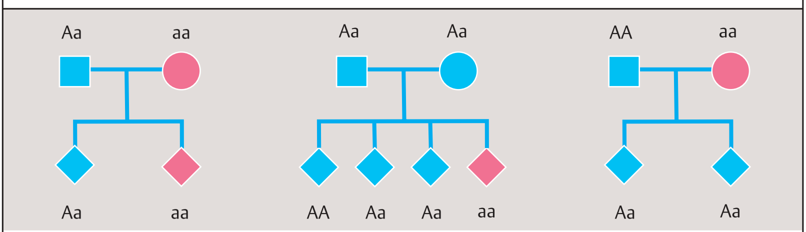
* Dominant phenotype to recessive phenotype 3 : 1



A. Possible mating types of genotypes for two alleles (bl dominant over r)



B. Expected distribution of genotypes in offspring of parents with two alleles, A and a



C. Phenotypes and genotypes in the offspring of parents with a dominant allele A and a recessive allele a

Monogenic Inheritance

Monogenic inheritance refers to the segregation pattern of alleles at a single gene locus when they are transmitted from parents to their offspring. In medical genetics it is customary to assign Roman numerals to consecutive generations. Within a generation, each individual is assigned an Arabic numeral. Individuals can also be assigned nonoverlapping combinations of numbers for computer calculations.

A. Pattern of inheritance in pedigrees with an autosomal dominant trait

Affected individuals are directly related in one or more successive generations. Both males and females are affected in a 1:1 ratio. The expected proportion of affected and unaffected offspring of an affected individual is 50% (0.50) each. An important consideration in autosomal disorders is whether a new mutation is present in a patient without affected parents. Pedigrees 2 and 3 show a new mutation in generation II. In some autosomal dominant disorders a carrier of the mutation does not manifest the disease. This is called nonpenetrance, but it is the exception rather than the rule. The degree of manifestation can vary within a family. This is called variable expressivity.

B. Pattern of inheritance in pedigrees with an autosomal recessive trait

Heterozygous parents have a risk of 25% of affected offspring. The expected segregation of genotypes of children of heterozygous patients is 1:2:1 (25% homozygous normal, 50% heterozygous, 25% homozygous affected). The sexes are affected with equal frequency.
The unaffected parents (II-3 and II-4 of pedigree 1, I-1 and I-2 of pedigree 2, and III-1 and III-2 of pedigree 3) are obligate heterozygotes. In pedigree 3, the homozygosity of the affected child can be traced back to common ancestors of the two parents, who are first cousins. Parental consanguinity (blood relationship) of III-1 and III-2 is indicated by a double line in the pedigree.

C. X-chromosomal inheritance

An X-chromosomal trait usually occurs only in males because they are hemizygous for all genes located on their X chromosome. A female heterozygous for an X-chromosomal mutation has a risk of 50% for an affected son. She also will transmit the X chromosome carrying the mutation to 50% of her daughters, but as heterozygotes they will not be affected. Pedigree 1 shows the distribution of three parental X chromosomes (one from the father, two from the mother) in the offspring. Panel 2 shows the corresponding distribution of the X chromosomes and the Y chromosome from the parents to the offspring.
Since a male has only one X chromosome, his daughter(s) will always inherit this chromosome with its mutation(s). A son will inherit one of his mother's two X chromosomes, but none from his father. The proportion of new mutations is relatively high (3) because males affected with a severe X chromosomal disease cannot reproduce to transmit the mutation to their offspring. One third of males with severe X-chromosomal disorders without family history have a new mutation (Haldane's rule). A new mutation could also occur in a female (4). Typical X-chromosomal inheritance (5) is easy to recognize. Affected males may occur in consecutive generations, but always through a female lineage. Father–son transmission of an X-chromosomal trait is not possible since the son inherits his father's Y, not his X.
Females with an affected son and an affected brother or with two affected sons must be heterozygous. They are called obligate heterozygotes. Those who might be, but possibly are not, heterozygotes are facultative heterozygotes (e.g., III-5 and IV-2). An isolated case of an X-linked disorder poses the question of whether this is due to a new mutation. However, the presence of the mutation in a proportion of germ cells of the mother (germline mosaicism) may increase the apparent risk. The situation can only be resolved on the basis of previous observations of the disease in question.

References

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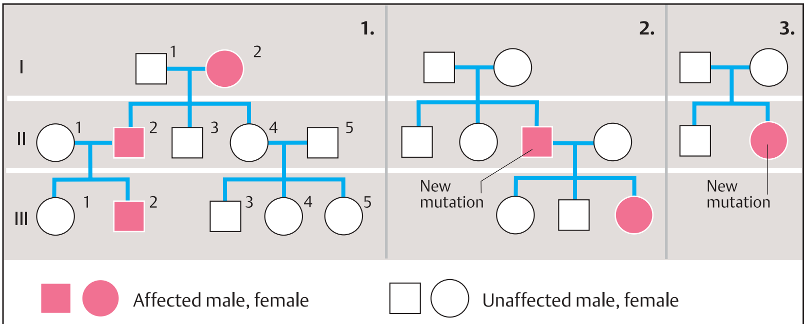
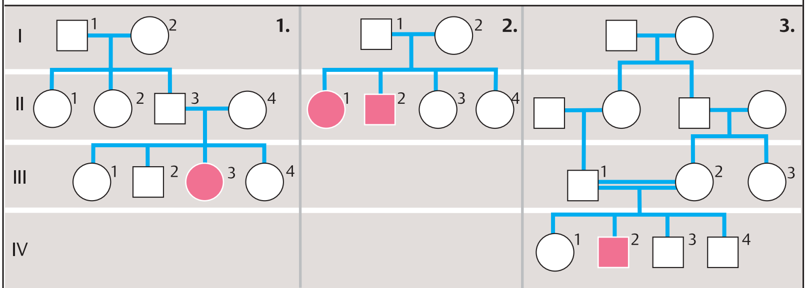
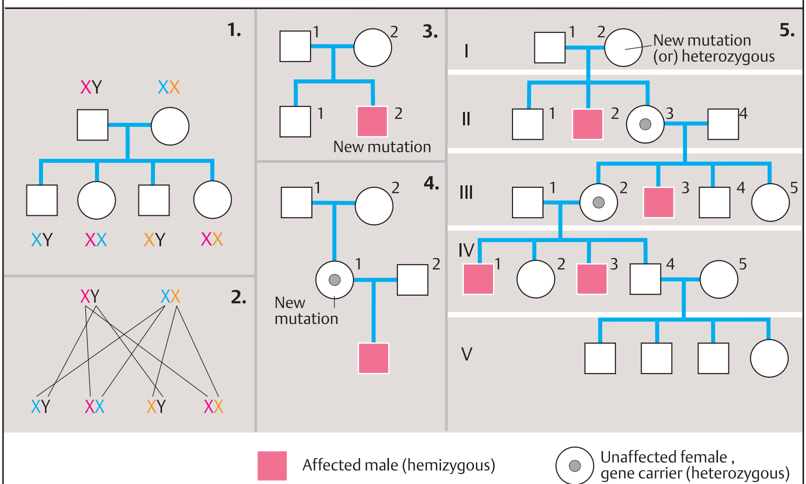
Vogel, F., Motulsky, A.G.: Human Genetics. Problems and Approaches. 3rd ed. Springer Verlag, Heidelberg–New York, 1997.



A. Pattern of inheritance in pedigrees with an autosomal dominant trait



B. Pattern of inheritance in pedigrees with an autosomal recessive trait



C. Pattern of X-chromosomal inheritance

Linkage and Recombination

Linkage refers to two or more genes being inherited together as a result of their location on the same chromosome. This depends on the distance between their loci. The closer they lie next to each other, the more frequently they will be inherited together (linked). Recombination due to crossing-over between the loci (breakage and reunion during meiosis, see p. 116) leads to the formation of a new combination of linked genes. When the loci are very close together, recombination is rare; when they lie farther apart, recombination is more frequent. In fact, the frequency of recombination can be used as a measure of the distance between gene loci. Linkage relates to gene loci, not to specific alleles. Alleles at closely linked gene loci that are inherited together are called a haplotype. If this occurs more frequently or less frequently than expected by the individual frequencies of the alleles involved, it is referred to as linkage disequilibrium (p. 158).

A. Recombination by crossing-over

Whether neighboring genes on the same parental chromosome remain together or become separated depends on the cytological events (1) during meiosis. If there is no crossing-over between the two gene loci **A** and **B,** having the respective alleles **A, a** and **B, b**, then they remain together on the same chromosome (linked). The gamete chromosomes formed during meiosis in this case are not recombinant and correspond to the parental chromosomes. However, if crossing-over occurs between the two gene loci, then the gametes formed are recombinant with reference to these two gene loci. The cytological events (1) are reflected in the genetic result (2). For two neighboring gene loci **A** and **B** on the same chromosome, the genetic result is one of two possibilities: not recombinant (gametes correspond to parental genotype) or recombinant (new combination). The two possibilities can be differentiated only when the parental genotype is informative for both gene loci (**Aa** and **Bb**).

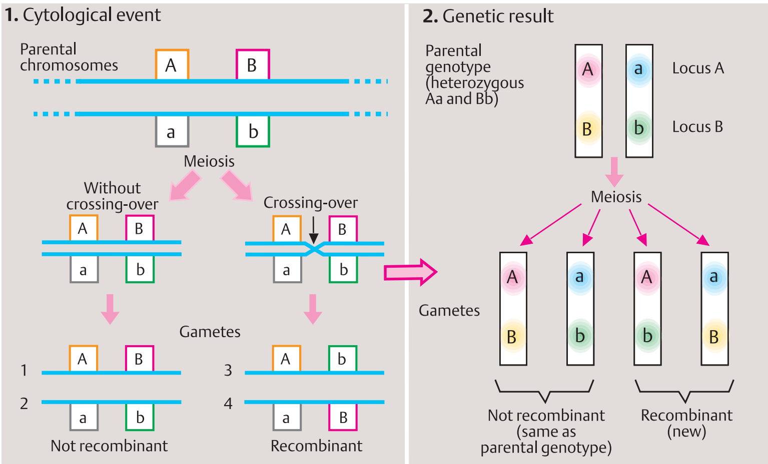
B. Linkage of a gene locus with an autosomal dominant mutation (B) to a marker locus (A)

The segregation of two linked gene loci in a family is shown here. There are two possibilities: 1, no recombination and 2, recombination. One locus (**B**) represents an autosomal dominant mutation that leads to a certain phenotype, e.g., that of an autosomal dominant inherited disorder. The father and three children (red symbols in the pedigree) are affected. The other locus (**A**) is a neighboring marker locus. All three affected children have inherited the mutant allele **B** as well as the marker allele **A** from their father. The three unaffected individuals have inherited the normal allele **b** and the marker allele a from their father. The paternal allele **a** indicates absence of the mutation (i.e., **B** not present). Recombination has not occurred (1).

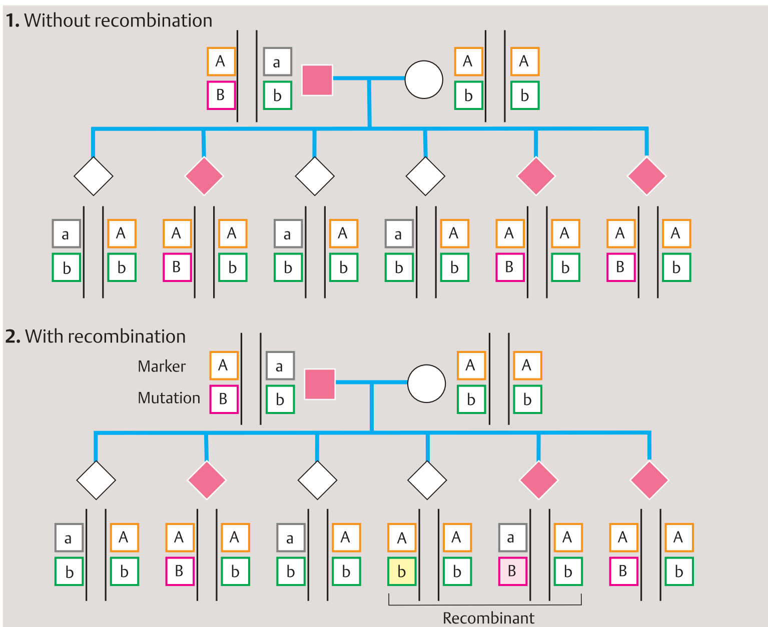
In situation 2, recombination has occurred in two (indicated) persons: An affected individual has inherited alleles **a** and **B** from the father, instead of **A** and **B.** An unaffected individual has inherited allele **A** and allele **b.**

The precondition for differentiating the paternal genotypes is heterozygosity at the father's loci. In the case presented, the alleles **A** and **B** lie on one of the father's chromosomes, and the alleles **a** and **b** on the other (in *cis* position). It would also be possible that allele **A** in the father would lie on one chromosome and allele **B** on the other (in *trans* position). These two possibilities represent two different linkage phases. The recognition of recombination as opposed to nonrecombination assumes knowledge of the parental linkage phase.

Segregation analysis of linked genes is very important in medical genetics because the presence or absence of a disease-causing mutation can be determined without directly knowing the type of mutation (indirect gene analysis). In order to reduce the probability of recombination, closely linked, flanking markers (DNA polymorphisms, see p. 72) are used.



A. Recombination with crossing-over



B. Linkage of a gene locus with autosomal dominant mutation (B) with a marker locus (A)